
Teaching notes: Financial Statistics

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The summaries follow the module's lectures; the explanations, solutions and worked examples are mine.

Def.1 The linear regression model:

The generic form

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_K X_{Ki} + u_i, \quad i = 1, \dots, T. \quad (1)$$

The matrix form

$$Y = X\beta + u, \quad (2)$$

where

$$Y_{T \times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_T \end{bmatrix}, \quad X_{T \times K} = \begin{bmatrix} 1 & X_{12} & X_{13} & \dots & X_{1K} \\ 1 & X_{22} & X_{23} & \dots & X_{2K} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & X_{T2} & X_{T3} & \dots & X_{TK} \end{bmatrix}, \quad \beta_{K \times 1} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}, \quad u_{T \times 1} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_T \end{bmatrix}.$$

The linear regression model consists of a set of assumptions that describe the form of the model and relationships among its parts which are needed for appropriate estimation and inference procedures.

1. **Linearity:** The model specifies a linear relationship between Y and $X_2, \dots, X_K$, that is in (1).
2. **The matrix X is non-stochastic and full rank:** We can treat X_i simply as a matrix of known constants and there must be no exact linear relationship among any of the independent variables in the model.
3. **Homoskedasticity and no autocorrelation:**

$$E[u] = 0 \quad (3)$$

$$\text{Var}(u) = E[uu'] = \sigma^2 I \quad (4)$$

where I is the $T \times T$ identity matrix.

Proof 1. Derive the OLS estimator: We want to find the value of b that minimizes $RSS(b)$

The OLS estimator is the vector $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_K)'$ that minimizes the Residual Sum of Squares (RSS). Let $b \in B$ be any admissible sample estimates of β . Then, for any value of b , we define the vector of residuals as

$$e = Y - Xb. \quad (5)$$

We want to find the value of b that minimizes $RSS(b)$, that is

$$\begin{aligned} \hat{\beta} &= \underset{b \in B}{\text{argmin}} \, RSS(b) \\ &= \underset{b \in B}{\text{argmin}} \, e'e \\ &= \underset{b \in B}{\text{argmin}} \, (Y - Xb)'(Y - Xb) \\ &= \underset{b \in B}{\text{argmin}} \, (Y'Y - 2b'X'Y + b'X'Xb). \end{aligned}$$

Proof. 1. Differentiating with respect to b^a and equating the result to zero, yields the first-order condition (FOC):

$$\left. \frac{\partial RSS(b)}{\partial b} \right|_{b=\hat{\beta}} = -2X'Y + 2X'X\hat{\beta} = 0.^b \quad (6)$$

2. The OLS estimator $\hat{\beta}$ is derived from the FOC (6) as follows

$$\begin{aligned} X'X\hat{\beta} &= X'Y \\ \hat{\beta} &= (X'X)^{-1} X'Y \end{aligned} \quad (7)$$

Notice the importance of the full rank condition which is needed to invert the matrix $X'X$. □

^aRemember the following rules for matrix differentiation:

$$\begin{aligned} \frac{\partial}{\partial x} x' Ay &= Ay, \quad \text{and} \\ \frac{\partial}{\partial x} x' Ax &= (A + A')x \end{aligned}$$

^bYou should remember it and you will use it a lot during the proof. This is nothing but, $(a - b)^2 = a^2 - 2ab + b^2$.

Proof 2. Show the sample covariance between each regressor and the OLS residual e is zero:

To show the sample covariance between each regressor and the OLS residual e is zero, such that

$$\frac{1}{T} \sum_{t=1}^T (X_{it} - \bar{X}_i) (e_t - \bar{e}) = 0, \text{ for } i = 2, \dots, k \quad (8)$$

1. If we rearrange the FOC (6) and remember the definition of residuals in (5), we can also write that

$$\begin{aligned} X'Y - X'X\hat{\beta} &= 0 \\ \Rightarrow X'(Y - X\hat{\beta}) &= 0 \\ \Rightarrow X'e &= 0. \end{aligned} \quad (9)$$

2. Since the first column of X is a column of (1 s), then the first element of (9) implies that

$$\sum_{i=1}^T e_i = 0 \quad (10)$$

that is the least square residuals sum to zero. For the other $K - 1$ entries of (9) we have that

$$X'_k e = 0 \quad \forall k = 2, \dots, K$$

3. We have the following:

$$\frac{1}{T} \sum_{t=1}^T (X_{it} - \bar{X}_i) (e_t - \bar{e}) = \frac{1}{T} \sum_{t=1}^T X_{it} (e_t - \bar{e}) = \frac{1}{T} \sum_{t=1}^T X_{it} e_t = 0. \quad (11)$$

Proof 3. Show we can decompose y into two orthogonal components, $\hat{y}$ and e :

We define the estimate of the conditional mean of Y given X equal to

$$\hat{y} = X\hat{\beta}.$$

To show we can decompose y into $\hat{y}$ and e ,

$$y = \hat{y} + e, \quad (12)$$

such that $\hat{y}$ and e are orthogonal

$$\hat{y}'e = 0. \quad (13)$$

Proof. We have

$$\hat{y}'e = (X\hat{\beta})'e = \hat{\beta}'X'e = 0. \quad (14)$$

This can be easily derived from (9). In the context of OLS, $X'e = 0$ implies that the residuals are orthogonal to the predicted values and to the columns of X . This is a fundamental property of OLS estimators. $\square$

Proof 4. Show the variance decomposition in the context of Ordinary Least Squares (OLS) :

The variance decomposition in the context of Ordinary Least Squares (OLS) regression is often written as:

$$TSS = ESS + RSS \quad (15)$$

Where:

- TSS stands for Total Sum of Squares.
- ESS stands for Explained Sum of Squares.
- RSS stands for Residual Sum of Squares.

Definitions:

1. Total Sum of Squares (TSS):

$$TSS = \sum_{i=1}^T (y_i - \bar{y})^2 \quad (16)$$

where y_i are the actual observations, $\bar{y}$ is the mean of y , and T is the number of observations.

2. Explained Sum of Squares (ESS):

$$ESS = \sum_{i=1}^T (\hat{y}_i - \bar{y})^2 \quad (17)$$

where $\hat{y}_i$ are the predicted values of y .

3. Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^T (y_i - \hat{y}_i)^2 \quad (18)$$

which is also the sum of the squared residuals, where $e_i = y_i - \hat{y}_i$.

Why $\bar{y} = \bar{\hat{y}}$?

Ans:

Sum on both sides of (12)

and in (10) we show if we adding a constant term in regressor

we have sum of errors equals 0

Proof that $TSS = ESS + RSS$:

Proof. Let's decompose TSS into two parts:

$$TSS = \sum_{i=1}^T (y_i - \bar{y})^2 \quad (19)$$

$$= \sum_{i=1}^T ((y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}))^2 \quad (20)$$

Now, let's expand the squared term and then separate the summation:

$$= \sum_{i=1}^T ((y_i - \hat{y}_i)^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + (\hat{y}_i - \bar{y})^2) \quad (21)$$

$$= \sum_{i=1}^T (y_i - \hat{y}_i)^2 + 2 \sum_{i=1}^T (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + \sum_{i=1}^T (\hat{y}_i - \bar{y})^2 \quad (22)$$

By properties of OLS, the residuals ($e_i = y_i - \hat{y}_i$) are orthogonal to the predicted values ($\hat{y}_i$) (See proof 3), which means the second term is zero:

$$= \sum_{i=1}^T (y_i - \hat{y}_i)^2 + \sum_{i=1}^T (\hat{y}_i - \bar{y})^2 \quad (23)$$

So, finally, we have:

$$TSS = RSS + ESS \quad (24)$$

□

Conclusion:

The total variability in the dependent variable y (TSS) is accounted for by the variability explained by the regression model (ESS) plus the variability left unexplained by the model (RSS). This foundational relationship in regression analysis allows for the decomposition of the variance into parts that are explained and unexplained by the model.

Question 1. The basics OLS

Define the linear regression model. Derive the FOC for the OLS estimator using matrix algebra differentiation. What do these imply for the OLS residual $e = y - X\hat{\beta}$? Then derive the estimator itself. Discuss the assumptions that one requires.

Proof. The 'Summary of Lectures 1 and 2' PDF provides a comprehensive coverage of the proof. □

Question 2. Practice on differentiation

For the model

$$Y_i = \alpha + \beta X_i + u_i,$$

derive the covariance matrix of the OLS estimators $\hat{\alpha}, \hat{\beta}$. Are the two estimators uncorrelated?

Proof: Let us first derive the OLS estimators $\hat{\alpha}$ and $\hat{\beta}$. As in the previous exercise, we obtain the OLS estimators by minimizing the sum of squared residuals (RSS). Let $a \in A$ and $b \in B$ be any admissible sample estimates of α and β , respectively, and define the residuals of the model as $e = Y_i - a - bX_i$. The OLS estimator of α is:

$$\begin{aligned}\hat{\alpha} &= \underset{a \in A}{\operatorname{argmin}} \text{RSS} \\ &= \underset{a \in A}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - a - bX_i)^2\end{aligned}$$

The first order condition (FOC) is obtained by differentiating with respect to α and equating the result to zero:

$$\begin{aligned}\left. \frac{\partial \text{RSS}}{\partial a} \right|_{a=\hat{\alpha}} &= -2 \sum_{i=1}^n (Y_i - \hat{\alpha} - bX_i) = 0 \\ 0 &= \sum_{i=1}^n Y_i - n\hat{\alpha} - b \sum_{i=1}^n X_i \\ \hat{\alpha} &= \frac{\sum_{i=1}^n Y_i}{n} - b \frac{\sum_{i=1}^n X_i}{n} \\ \hat{\alpha} &= \bar{Y} - b\bar{X}\end{aligned} \tag{25}$$

where we used the notation $\bar{Y}$ and $\bar{X}$ for the sample means (i.e. $\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$ and $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$). The OLS estimator for β is obtained similarly as:

$$\begin{aligned}\hat{\beta} &= \underset{b \in B}{\operatorname{argmin}} \text{RSS} \\ &= \underset{b \in B}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - \hat{\alpha} - bX_i)^2 \\ \left. \frac{\partial \text{RSS}}{\partial b} \right|_{b=\hat{\beta}} &= -2 \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i) X_i = 0 \\ \Rightarrow 0 &= \sum_{i=1}^n Y_i X_i - \hat{\alpha} \sum_{i=1}^n X_i - \hat{\beta} \sum_{i=1}^n X_i^2\end{aligned}$$

Replacing (25) into this equation becomes:

$$\begin{aligned}0 &= \sum_{i=1}^n Y_i X_i - (\bar{Y} - \hat{\beta}\bar{X}) \sum_{i=1}^n X_i - \hat{\beta} \sum_{i=1}^n X_i^2 \\ 0 &= \sum_{i=1}^n Y_i X_i - \bar{Y} \sum_{i=1}^n X_i + \hat{\beta}\bar{X} \sum_{i=1}^n X_i - \hat{\beta} \sum_{i=1}^n X_i^2 \\ 0 &= \sum_{i=1}^n Y_i X_i - \bar{Y} \sum_{i=1}^n X_i - \hat{\beta} \left(\sum_{i=1}^n X_i^2 - \bar{X} \sum_{i=1}^n X_i \right)\end{aligned} \tag{26}$$

Let us now use the property that $\sum_{i=1}^n X_i = n\bar{X}$, apply it to (26) and solve for $\hat{\beta}$:

$$\begin{aligned}\hat{\beta} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right) &= \sum_{i=1}^n Y_i X_i - n\bar{X}\bar{Y} \\ \hat{\beta} &= \frac{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^n X_i^2 - n\bar{X}^2}\end{aligned} \tag{27}$$

See next page for more proof.

Question 2. Practice on differentiation

Let us now use a couple more tricks. By using simple algebra we can derive that

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}, \quad \text{and}$$

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$$

Substituting these two results in (27) gets:

$$\hat{\beta} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\text{cov}(X, Y)}{\text{var}(X)}$$

Substituting this result back into (25) we obtain the following result for $\hat{\alpha}$:

$$\hat{\alpha} = \bar{Y} - \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \bar{X}.$$

To derive the covariance matrix of the model parameters it is convenient to aggregate the estimators into a vector $\beta = (\alpha, \beta)'$. In this way, we can use matrix notation

$$Y = X\beta + u$$

where

$$Y_{n \times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad X_{n \times 2} = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}, \quad \beta_{2 \times 1} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad u_{n \times 1} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad (28)$$

and it is straightforward to apply the definition in the lecture notes for the variance of the OLS estimator, namely:

$$\text{var}(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

To solve for $\text{var}(\hat{\beta})$ we first have to derive the expression for $(X'X)^{-1}$. Given the definition of X in (8) we have that:

$$X'X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ X_1 & X_2 & \dots & X_n \end{bmatrix} \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^n X_i \\ \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \end{bmatrix}$$

The inverse of this matrix is then obtained as^a

$$(X'X)^{-1} = \frac{1}{|X'X|} \begin{bmatrix} \sum_{i=1}^n X_i^2 & -\sum_{i=1}^n X_i \\ -\sum_{i=1}^n X_i & n \end{bmatrix}$$

where

$$|X'X| = n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2$$

$$\Rightarrow (X'X)^{-1} = \frac{1}{n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2} \begin{bmatrix} \sum_{i=1}^n X_i^2 & -\sum_{i=1}^n X_i \\ -\sum_{i=1}^n X_i & n \end{bmatrix}$$

See next page for more proof.

^aRecall that the inverse of a matrix A

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

is a matrix A^{-1} , such that $AA^{-1} = I$ and it is given by the formula

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

where $|A|$ is the determinant of matrix A given by $|A| = a_{11}a_{22} - a_{21}a_{12}$.

Question 2. Practice on differentiation

The variance of the OLS estimator $\text{var}(\hat{\beta})$ is thus

$$\begin{aligned} \text{var}(\hat{\beta}) &= \sigma^2 (X'X)^{-1} = \frac{\sigma^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} \begin{bmatrix} \sum_{i=1}^n X_i^2 & -\sum_{i=1}^n X_i \\ -\sum_{i=1}^n X_i & n \end{bmatrix} \\ \text{var}(\hat{\beta}) &= \begin{bmatrix} \frac{\sum_{i=1}^n X_i^2 \sigma^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} & \frac{-\sum_{i=1}^n X_i \sigma^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} \\ \frac{-\sum_{i=1}^n X_i \sigma^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} & \frac{n \sigma^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} \end{bmatrix} \end{aligned} \quad (29)$$

We have thus derived the variance of the OLS estimator $\hat{\beta}$, $\text{var}(\hat{\beta})$ but what we need is the covariance between $\hat{\alpha}$ and $\hat{\beta}$, $\text{cov}(\hat{\alpha}, \hat{\beta})$. It turns out that this is actually an element of the variance matrix 29. It is easy to see this if we recall that $\hat{\beta}$ is defined as the vector $(\hat{\alpha}, \hat{\beta})$ so we can write it in terms of its elements as follows :

$$\text{var}(\hat{\beta}) = \begin{bmatrix} \text{var}(\hat{\alpha}) & \text{cov}(\hat{\alpha}, \hat{\beta}) \\ \text{cov}(\hat{\alpha}, \hat{\beta}) & \text{var}(\hat{\beta}) \end{bmatrix} \quad (30)$$

From (29) and (30) it follows that the covariance matrix of $\hat{\alpha}$ and $\hat{\beta}$ is

$$\text{cov}(\hat{\alpha}, \hat{\beta}) = \frac{-\sum_{i=1}^n X_i \sigma^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2}$$

The two estimators $\hat{\alpha}$ and $\hat{\beta}$ are uncorrelated iff $(-\sum_{i=1}^n X_i) = 0$, as this implies $\text{cov}(\hat{\alpha}, \hat{\beta}) = 0$. Notice again the importance of the full rank condition: if the matrix X does not have full rank then the determinant of $X'X$ is zero and the inverse is not defined.

Question 3. Sample Exam Question

(a) Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

and derive the OLS estimator of β_1 .

(b) Then consider the other linear regression model

$$X_i = \gamma_0 + \gamma_1 Y_i + v_i$$

and derive the OLS estimator of γ_1 .

(c) Comparing the two regression models do we obtain

$$(**) \quad \gamma_1 = \frac{1}{\beta_1}?$$

(d) Under which conditions do we obtain the same relationship (**) between the corresponding OLS estimators $\hat{\beta}_1$ and $\hat{\gamma}_1$?

Proof. We know from Question 2 that for a model of the form

$$Y_i = \alpha + \beta X_i + u_i$$

the OLS estimator of β is

$$\hat{\beta} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

By simply adapting this result to our present problem we have that the OLS estimators of β_1 and γ_1 are

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ \hat{\gamma}_1 &= \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \end{aligned} \quad (31)$$

and thus we have solved points (a) and (b). Solving point (c) requires some manipulation of the regression models. Let us first derive the expression for X_i in terms of Y_i . Starting from the model in (a) we have

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + u_i \\ \Rightarrow \beta_1 X_i &= Y_i - \beta_0 - u_i \\ \Rightarrow X_i &= -\frac{\beta_0}{\beta_1} + \frac{Y_i}{\beta_1} - \frac{u_i}{\beta_1}. \end{aligned} \quad (32)$$

From the model in (b) we also know that

$$X_i = \gamma_0 + \gamma_1 Y_i + v_i \quad (33)$$

Then, by equating (32) and (33), we obtain that

$$\gamma_1 = \frac{1}{\beta_1}$$

In point (d) we need to derive the condition under which $\hat{\gamma}_1 = \frac{1}{\hat{\beta}_1}$. Using the OLS estimators in (31), we have that

$$\begin{aligned} \hat{\gamma}_1 = \frac{1}{\hat{\beta}_1} &\iff \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})} \\ &\iff \left(\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X}) \right)^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2 \sum_{i=1}^n (X_i - \bar{X})^2 \end{aligned}$$

□

Def.2 The orthogonal projection matrix:

Given matrix form

$$Y = X\beta + u,$$

where

$$Y_{T \times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_T \end{bmatrix}, \quad X_{T \times K} = \begin{bmatrix} 1 & X_{12} & X_{13} & \cdots & X_{1K} \\ 1 & X_{22} & X_{23} & \cdots & X_{2K} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & X_{T2} & X_{T3} & \cdots & X_{TK} \end{bmatrix}, \quad \beta_{K \times 1} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}, \quad u_{T \times 1} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_T \end{bmatrix}.$$

The linear regression model consists of a set of assumptions that describe the form of the model and relationships among its parts which are needed for appropriate estimation and inference procedures.

1. **Linearity:** The model specifies a linear relationship between Y and $X_2, \dots, X_k$, that is in (1).
2. **The matrix X is non-stochastic and full rank:** We can treat X_i simply as a matrix of known constants and there must be no exact linear relationship among any of the independent variables in the model.
3. **Homoskedasticity and no autocorrelation:**

$$E[u] = 0$$

$$\text{Var}(u) = E[uu'] = \sigma^2 I$$

where I is the $T \times T$ identity matrix.

Define

$$M = I - X(X'X)^{-1}X'. \quad (34)$$

Note that

$$MX = 0.$$

Thus M and X are orthogonal. We call M an **orthogonal projection matrix**, or more colorfully an **annihilator matrix**.

The orthogonal projection matrix $M = I - X(X'X)^{-1}X'$ for any $T \times k$ X with $T \geq k$ has the following algebraic properties

1. M is symmetric ($M' = M$).
2. M is idempotent ($MM = M$).
3. $\text{tr}^a M = T - k$.
4. $My = Mu = e$ where $e = y - X\hat{\beta}$.

^aThe trace of an $n \times n$ square matrix A is defined as

$$\text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}$$

where a_{ij} denotes the entry on the i th row and j th column of A . The basic properties of trace:

- (a) The trace is a linear mapping such as

$$\begin{aligned} \text{tr}(A + B) &= \text{tr}(A) + \text{tr}(B) \\ \text{tr}(cA) &= c \text{tr}(A) \end{aligned}$$

for all square matrices A and B , and all scalars c .

- (b) A matrix and its transpose have the same trace

$$\text{tr}(A) = \text{tr}(A^T).$$

- (c) More generally, the trace is invariant under circular shifts, that is, $\text{tr}(ABCD) = \text{tr}(BCDA) = \text{tr}(CDAB) = \text{tr}(DABC)$.

Thm.1 The Gauss-Markov Theorem:

The Gauss-Markov theorem is a fundamental result in the theory of linear regression. It states that under certain conditions, the ordinary least squares (OLS) estimator has the smallest possible variance among the class of linear unbiased estimators. It can be shown to be the Best Linear Unbiased Estimator, **BLUE**.

Assumptions:

1. **Linearity:** The model specifies a linear relationship between Y and $X_2, \dots, X_k$, that is in (1).
2. **The matrix X is non-stochastic and full rank:** We can treat X_i simply as a matrix of known constants and there must be no exact linear relationship among any of the independent variables in the model.
3. **Homoskedasticity and no autocorrelation:**

$$E[u] = 0$$

$$\text{Var}(u) = E[uu'] = \sigma^2 I$$

where I is the $T \times T$ identity matrix.

Given the assumptions above, the OLS estimator $\hat{\beta}_{OLS} = (X'X)^{-1} X'Y$ is the Best Linear Unbiased Estimator (BLUE) of β . This means that among all linear estimators that are unbiased, the OLS estimator has the smallest variance. In mathematical terms, if $\hat{\beta}_L$ is any other linear unbiased estimator of β , then:

$$\text{Var}(\hat{\beta}_{OLS}) \leq \text{Var}(\hat{\beta}_L).$$

This theorem highlights the efficiency of OLS in the class of linear and unbiased estimators. It does not, however, compare OLS to nonlinear or biased estimators, which can potentially have lower variance under some circumstances.

Testing linear hypothesis:

We wish to test whether the null hypothesis is supported by the data or not:

$$H_0: R\beta = r,$$

where R is $q \times k$ constant matrix and r is q constant vector.

We assume

$$u \sim N(0, \sigma^2 I).$$

then, since linear combination of normal is also normal,

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1})$$

Likewise, under H_0 ,

$$R\hat{\beta} \sim N(r, \sigma^2 R(X'X)^{-1} R')$$

By standardizing

$$(R\hat{\beta} - r)' [\sigma^2 R(X'X)^{-1} R']^{-1} (R\hat{\beta} - r) \sim \chi_q^2.$$

The F-test statistic is given by

$$F \equiv \frac{(R\hat{\beta} - r)' [R(X'X)^{-1} R']^{-1} (R\hat{\beta} - r) / q}{e'e / (T - k)} \sim F(q, T - k). \quad (35)$$

The P-value for the F test

$$p\text{-value} = \text{Prob}(F(q, T - k) > F). \quad (36)$$

where F is a given number, as shown in the F Critical Values Tables. When the P-value is large, this gives confidence to the null hypothesis and the contrary is it is small.

The previous version
contains a typo
should be 'q'



Proof 5. Delve into the Ordinary Least Squares (OLS) estimator and its properties:

The OLS estimator for β is given by:

$$\hat{\beta}_{OLS} = (X'X)^{-1} X'y$$

We want to show,

- Unbiasedness of $\hat{\beta}_{OLS}$:

$$E(\hat{\beta}_{OLS}) = \beta; \quad (37)$$

- Variance of $\hat{\beta}_{OLS}$:

$$\text{Var}(\hat{\beta}_{OLS}) = \sigma^2 (X'X)^{-1}. \quad (38)$$

Proof. To prove the unbiasedness of $\hat{\beta}_{OLS}$, we will show that the expectation of $\hat{\beta}_{OLS}$ is β .

$$\begin{aligned} E(\hat{\beta}_{OLS}) &= E[(X'X)^{-1} X'y] \\ &= E[(X'X)^{-1} X'(X\beta + u)] \\ &= (X'X)^{-1} X'X\beta + (X'X)^{-1} X'E(u) \end{aligned}$$

Assuming $E(u | X) = 0$ (i.e., errors have mean zero given any values of the independent variables), the expectation of u is zero. Therefore,

$$E(\hat{\beta}_{OLS}) = \beta$$

Thus, $\hat{\beta}_{OLS}$ is unbiased. Next, we'll derive the variance.

$$\text{Var}(\hat{\beta}_{OLS}) = E[(\hat{\beta}_{OLS} - E(\hat{\beta}_{OLS}))(\hat{\beta}_{OLS} - E(\hat{\beta}_{OLS}))']$$

First, note that:

$$\begin{aligned} \hat{\beta}_{OLS} - E(\hat{\beta}_{OLS}) &= (X'X)^{-1} X'y - \beta \\ &= (X'X)^{-1} X'(X\beta + u) - \beta \\ &= (X'X)^{-1} X'u \end{aligned}$$

Therefore, plugging in:

$$\text{Var}(\hat{\beta}_{OLS}) = E[(X'X)^{-1} X'uu'X (X'X)^{-1}]$$

Assuming homoscedasticity (constant variance of errors), $E(uu' | X) = \sigma^2 I$. Thus,

$$\text{Var}(\hat{\beta}_{OLS}) = \sigma^2 (X'X)^{-1}$$

This gives the variance of the OLS estimator, under the classical assumptions of the linear regression model. $\square$

Proof 6. Derive the estimate of σ^2 :

let's derive the OLS estimator for σ^2 , which is the variance of the error term in a linear regression model. The OLS estimator of σ^2 is

$$s^2 = \frac{1}{T-k} e'e. \quad (39)$$

We want to prove that s^2 is an unbiased estimator of σ^2

Given:

1. The error term:

$$e = Mu$$

This implies that e is the orthogonal projection of the disturbance u onto the orthogonal complement of the column space of X .

2. The error term (squared form):

$$e'e = u'Mu$$

3. The trace identity: recognize the $u'Mu$ is scalar, the trace of a scalar is the scalar itself. We have

$$u'Mu = \text{tr}(u'Mu).$$

Furthermore, we have

$$E(e'e) = E(u'Mu) = E(\text{tr}(u'Mu)) = E(\text{tr}Mu u') = \sigma^2 \text{tr}(M) = \sigma^2(T-k)$$

This identity captures the total variability of the error term, u , that is projected onto the orthogonal complement of X .

Proof. Let us complete the proof, we begin with the formula for the OLS estimator of σ^2 :

$$s^2 = \frac{1}{T-k} e'e$$

Substitute for $e'e$ using our earlier relationship:

$$s^2 = \frac{1}{T-k} u'Mu$$

Take the expectation:

$$\begin{aligned} E(s^2) &= E\left(\frac{1}{T-k} u'Mu\right) \\ &= \frac{1}{T-k} E(u'Mu) \end{aligned}$$

Substitute in the trace identity:

$$E(s^2) = \frac{1}{T-k} \sigma^2(T-k)$$

Simplify:

$$E(s^2) = \sigma^2$$

This proves that s^2 is an unbiased estimator of σ^2 . □

Question 1.

For the model

$$Y_i = \alpha + u_i,$$

derive the OLS estimator $\hat{\alpha}$ and derive the R^2 of the regression.

Proof. As before, to obtain the OLS estimator $\hat{\alpha}$ of α , we need to solve the FOC of the RSS objective function. Define the residuals of the model as $e = Y_i - a$, where $a \in A$ is any admissible parameter value for α . Then, the OLS estimator $\hat{\alpha}$ is such that

$$\begin{aligned}\hat{\alpha} &= \underset{a \in A}{\operatorname{argmin}} \operatorname{RSS}(a) \\ &= \underset{a \in A}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - a)^2.\end{aligned}$$

Then the FOC implies that

$$\begin{aligned}\left. \frac{\partial \operatorname{RSS}(a)}{\partial a} \right|_{a=\hat{\alpha}} &= -2 \sum_{i=1}^n (Y_i - \hat{\alpha}) = 0 \\ \Rightarrow \sum_{i=1}^n Y_i - n\hat{\alpha} &= 0 \\ \Rightarrow \hat{\alpha} &= \frac{\sum_{i=1}^n Y_i}{n} = \bar{Y}\end{aligned}$$

We now want to derive the R^2 of the model. Using the definition in the lecture notes

$$\begin{aligned}R^2 &= 1 - \frac{\operatorname{RSS}}{\operatorname{TSS}} \\ &= 1 - \frac{\sum_{i=1}^n (Y_i - \hat{\alpha})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \\ &= 1 - \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = 0.\end{aligned}$$

□

Comment on the constant term:

Without a constant term, this version of R^2 can potentially be negative because the model does not ensure that the predicted values have the same mean as the actual values, which can lead to a model that fits the data worse than a simple horizontal line at the mean of y . A negative R^2 indicates that the model without a constant is inappropriate for the data.

Remember, while this calculation of R^2 can be done, it loses its usual interpretation as the proportion of variance explained by the model. It's also worth noting that many statistical software packages will automatically include an intercept in the model, and you typically have to specify if you want to exclude it. When interpreting an R^2 value from a model without an intercept, be cautious and consider other metrics and graphical analyses to assess the fit of your model.

Question 2.

Show the orthogonal projection matrix $M = I - X(X'X)^{-1}X'$ for any $T \times k$ X with $T \geq k$ has the following algebraic properties

1. M is symmetric ($M' = M$).
2. M is idempotent ($MM = M$).
3. $\text{tr}^a M = T - k$.
4. $My = Mu = e$ where $e = y - X\hat{\beta}$.

^aThe trace of an $n \times n$ square matrix A is defined as

$$\text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}$$

where a_{ij} denotes the entry on the i th row and j th column of A . The basic properties of trace:

- (a) The trace is a linear mapping such as

$$\begin{aligned}\text{tr}(A + B) &= \text{tr}(A) + \text{tr}(B) \\ \text{tr}(cA) &= c \text{tr}(A)\end{aligned}$$

for all square matrices A and B , and all scalars c .

- (b) A matrix and its transpose have the same trace

$$\text{tr}(A) = \text{tr}(A')$$

- (c) More generally, the trace is invariant under circular shifts, that is, $\text{tr}(ABCD) = \text{tr}(BCDA) = \text{tr}(CDAB) = \text{tr}(DABC)$.

Proof:

- (1) Symmetry: We need to show $M' = M$.

$$\begin{aligned}M' &= (I - X(X'X)^{-1}X')' \\ &= I' - (X(X'X)^{-1}X')' \\ &= I - X'((X'X)^{-1})'X\end{aligned}$$

Using the property that the inverse of a transpose is the transpose of the inverse, $(A^{-1})' = A'^{-1}$:

$$M' = I - X'(X'X)^{-1}X = M$$

Thus, M is symmetric.

- (2) Idempotency: We need to show $MM = M$.

$$\begin{aligned}MM &= (I - X(X'X)^{-1}X')(I - X(X'X)^{-1}X') \\ &= I^2 - X(X'X)^{-1}X' - X(X'X)^{-1}X' + X(X'X)^{-1}X'X(X'X)^{-1}X'\end{aligned}$$

This simplifies to $I - X(X'X)^{-1}X'$ which is M . Thus, M is idempotent.

- (3) Trace: We need to show $\text{tr} M = T - k$. Using the linearity property of the trace:

$$\begin{aligned}\text{tr} M &= \text{tr}(I - X(X'X)^{-1}X') \\ &= \text{tr}(I) - \text{tr}(X(X'X)^{-1}X') \\ &= \text{tr}(I) - \text{tr}((X'X)^{-1}X'X) \\ &= \text{tr}(I_T) - \text{tr}(I_k) \\ &= T - k\end{aligned}$$

See next page for more proof.

Question 2.

(4) Projection: We need to show $My = Mu = e$ where $e = y - X\hat{\beta}$. Given $e = y - X\hat{\beta}$, and $\hat{\beta} = (X'X)^{-1}X'y$:

$$e = y - X(X'X)^{-1}X'y$$

Now,

$$My = (I - X(X'X)^{-1}X')y = y - X(X'X)^{-1}X'y = e.$$

Similarly, given $u = y - X\beta$, and $MX\beta = 0$:

$$Mu = My - MX\beta = e.$$

Question 3. Sample Exam Question

Consider the multiple linear regression model

$$Y = X\beta + \varepsilon$$

where

$$\beta' = (\beta_1, \beta_2, \beta_3, \beta_4)$$

How would you test the hypothesis that

$$\beta_4 = \beta_2 \text{ and } \beta_1 = 0?$$

Carefully state any assumption you need, the alternative hypothesis, and the test statistic you would use with its distribution. At a 5% significance level, how would you construct the rejection region? Let

$$(X'X)^{-1} = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix}$$

Proof: First notice that our regression model can be written in the following way:

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + u_i$$

so $k = 4$. Moreover, our null hypothesis can be written as

$$H_0: \begin{matrix} \beta_4 - \beta_2 = 0 \\ \beta_1 = 0 \end{matrix}$$

against the alternative

$$H_1: \begin{matrix} \beta_4 - \beta_2 \neq 0 \\ \beta_1 \neq 0 \end{matrix}$$

Therefore, the number of constraints under H_0 is $q = 2$. The above hypothesis can be tested using the F test. Let us first rewrite the null hypothesis in the general form:

$$\begin{aligned} H_0: R\beta &= r \\ \Rightarrow \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} \beta_2 - \beta_4 \\ \beta_1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

As before, we have to derive the expression for $(R\hat{\beta} - r)$:

$$\begin{aligned} R\hat{\beta} - r &= \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \\ \hat{\beta}_4 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \hat{\beta}_2 - \hat{\beta}_4 \\ \hat{\beta}_1 \end{bmatrix} \end{aligned}$$

Now let us derive the expression for $[R(X'X)^{-1}R']^{-1}$:
given

$$(X'X)^{-1} = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix},$$

See next page for more proof.

Question 3. Sample Exam Question

$$\begin{aligned}
 R(X'X)^{-1}R' &= \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ -1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} (c_{21} - c_{41}) & (c_{22} - c_{42}) & (c_{23} - c_{43}) & c_{24} - c_{44} \\ c_{11} & c_{12} & c_{13} & c_{14} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ -1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} c_{22} - 2c_{24} + c_{44} & c_{21} - c_{41} \\ c_{12} - c_{14} & c_{11} \end{bmatrix} \\
 [R(X'X)^{-1}R']^{-1} &= \frac{1}{|R(X'X)^{-1}R'|} \begin{bmatrix} c_{11} & -c_{12} + c_{14} \\ -c_{21} + c_{41} & c_{22} - 2c_{24} + c_{44} \end{bmatrix} \\
 &= \frac{1}{c_{11}(c_{22} - 2c_{24} + c_{44}) - (c_{12} - c_{14})^2} \begin{bmatrix} c_{11} & -c_{12} + c_{14} \\ -c_{21} + c_{41} & c_{22} - 2c_{24} + c_{44} \end{bmatrix}
 \end{aligned}$$

The F test is therefore

$$\begin{aligned}
 F &= \frac{([\hat{\beta}_2 - \hat{\beta}_4, \hat{\beta}_1] [R(X'X)^{-1}R']^{-1} [\hat{\beta}_2 - \hat{\beta}_4]) / 2}{s^2} \\
 F &= \frac{((\hat{\beta}_2 - \hat{\beta}_4)^2 c_{11} + 2\hat{\beta}_1(\hat{\beta}_2 - \hat{\beta}_4)(c_{14} - c_{12}) + \hat{\beta}_1^2(c_{22} - 2c_{24} + c_{44}))}{2s^2 [c_{11}(c_{22} - 2c_{24} + c_{44}) - (c_{12} - c_{14})^2]} \sim F(2, T - 4)
 \end{aligned}$$

The rejection region is constructed by comparing the test statistic (i.e. F) with the corresponding sampling distribution (i.e. $F(2, T - 4)$) which you can find tabulated in F-distribution tables^a. The null hypothesis is rejected if the test statistic is greater than the tabulated value corresponding to the 5% significance level ($F > F(2, T - 4)$). For instance, if we take $T = 24$, the tabulated value corresponding to the 5% significance level for the F distribution with 2 and 20 degrees of freedom (i.e. $F(2, 24 - 4)$) is 3.4928^b. The p -value corresponding to the F test is obtained as the probability that the tabulated value is greater than the test statistic

$$p\text{-value} = \text{Prob}(F(2, T - 4) > F)$$

The smaller the p -value the higher the evidence against the null. The null hypothesis is rejected at 5% level if p -value is less than 0.05.

^a<https://statisticsbyjim.com/hypothesis-testing/f-table/>

^bIn applications you will know what your T is (is the number of observations), this is just an example

Testing linear hypothesis - Generic Form:

We wish to test whether the null hypothesis is supported by the data or not:

$$H_0 : R\beta = r,$$

where R is $q \times k$ constant matrix and r is q constant vector.

We assume

$$u \sim N(0, \sigma^2 I).$$

then, since the linear combination of normal is also normal,

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1})$$

Likewise, under H_0 ,

$$R\hat{\beta} \sim N(r, \sigma^2 R (X'X)^{-1} R')$$

By standardizing

$$(R\hat{\beta} - r)' [\sigma^2 R (X'X)^{-1} R']^{-1} (R\hat{\beta} - r) \sim \chi_q^2.$$

The residuals, e , are:

$$e = y - \hat{y}$$

Which can be expanded to:

$$e = y - X\hat{b}$$

Now, under the classical linear regression model assumptions, the error term u is assumed to be normally distributed with mean 0 and constant variance σ^2 . This implies that the residuals e are also normally distributed with mean 0. Given these assumptions, the quantity:

$$\frac{e'e}{\sigma^2}$$

is a sum of squared standard normal random variables, which is chi-squared distributed. The degrees of freedom for this chi-squared distribution is the number of observations minus the number of parameters estimated, which in this case is $T - k$ (since we are estimating k parameters in b).

Thus:

$$\frac{e'e}{\sigma^2} \sim \chi_{T-k}^2.$$

The reason for $T - k$ degrees of freedom is because, in regression, every estimated parameter uses up one degree of freedom. The intuition is that degrees of freedom reflect the amount of "free" or "unconstrained" information available for estimation. In this context, every time we estimate a parameter, we "use up" a degree of freedom, leaving $T - k$ degrees of freedom for the residuals.

The F-test statistic is given by

$$F \equiv \frac{(R\hat{\beta} - r)' [R (X'X)^{-1} R']^{-1} (R\hat{\beta} - r) / q}{e'e / (T - k)} \sim F(q, T - k). \quad (40)$$

The P -value for the F test

$$p - \text{value} = \text{Prob}(F(q, T - k) > F). \quad (41)$$

where F is a given number, as shown in the F Critical Values Tables. When the P -value is large, this gives confidence to the null hypothesis and the contrary is it is small.

Testing linear hypothesis - Special Case 1:

When the null hypothesis says that all the regression coefficients (other than β_1) are zero

$$\begin{aligned} H_0 : \beta_2 &= 0 \\ \beta_3 &= 0 \\ &\vdots \\ \beta_k &= 0, \end{aligned}$$

the F test becomes

$$F = \frac{R^2 / (k - 1)}{(1 - R^2) / (T - k)} \sim F(k - 1, T - k) \quad (42)$$

where

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \text{ where always } 0 \leq R^2 \leq 1 \quad (43)$$

Recalling:

1. TSS stands for Total Sum of Squares:

$$TSS = \sum_{i=1}^T (y_i - \bar{y})^2$$

2. ESS stands for Explained Sum of Squares:

$$ESS = \sum_{i=1}^T (\hat{y}_i - \bar{y})^2$$

3. RSS stands for Residual Sum of Squares:

$$RSS = \sum_{i=1}^T (y_i - \hat{y}_i)^2$$

The P-value for the F test

$$p\text{-value} = \text{Prob}(F(k - 1, T - k) > F). \quad (44)$$

where F is a given number, as shown in the F Critical Values Tables. When the P-value is large, this gives confidence to the null hypothesis and the contrary is it is small.

Comments:

1. If regression is poor then $R^2 \approx 0$ and also $F \approx 0$:

- This indicates that the regression model does not explain much of the variability in the dependent variable. The predictors in the model aren't useful.
- $F \approx 0$ implies that the model with the predictors is not doing significantly better than a model with no predictors (i.e., an intercept-only model).
- **How is the P-value now?** A small F statistic leads to a large p-value. If the p-value is larger than the significance level (commonly set at 0.05), then we fail to reject the null hypothesis and conclude that the model with the predictors is not statistically better than the model without them.

2. Instead, as $R^2 \uparrow 1$ then $F \uparrow \infty$:

- This indicates that the regression model explains almost all the variability in the dependent variable. The predictors in the model are highly useful.
- A large F statistic implies that the model with the predictors is significantly better than a model without them.
- **How is the P-value now?** A large F statistic leads to a very small p-value. If the p-value is smaller than the significance level (commonly set at 0.05), then we reject the null hypothesis and conclude that the model with the predictors is statistically better than the model without them.

In summary:

- A poor regression with $R^2 \approx 0$ leads to a high p-value, implying the predictors don't add significant value.
- A strong regression with $R^2 \uparrow 1$ leads to a very low p-value, implying the predictors are highly significant.

Testing linear hypothesis - Special Case 2:

When the null hypothesis says that only a subset of coefficients are zero. Rewrite the general model as

$$y = X\beta + u = X_a\beta_a + X_b\beta_b + u, \text{ that is } X = [X_a : X_b]$$

where X_b has $k_b < k$ columns. It is the same as saying that $\beta = (\beta_a' \beta_b')'$, where β_b is of dimension $k_b \times 1$, and let

$$H_0 : \beta_b = 0. \text{ (The restricted regression, 'little model' : } y = X_a\beta_a + u \text{ holds!)}$$

Then

$$F = \frac{(e^{*'}e^* - e'e)/k_b}{e'e/(T-k)} \sim F(k_b, T-k) \quad (45)$$

Here $e^{*'}e^*$ is the RSS from the restricted regression, that is regressing y on X_a . Instead $e'e$ is the (old) RSS from regressing y on all regressors X_a and X_b (unrestricted regression), such as

1. $e^* = y - X_a\hat{\beta}_a$;
2. $e = y - X\hat{\beta}$

The P-value for the F test

$$p\text{-value} = \text{Prob}(F(k_b, T-k) > F). \quad (46)$$

where F is a given number, as shown in the F Critical Values Tables. When the P-value is large, this gives confidence to the null hypothesis and the contrary is it is small.

Testing linear hypothesis - Special Case 3:

When the null hypothesis says that only one regression coefficient, β_i is β_{i0} ,

$$H_0 : \beta_i = \beta_{i0}$$

the F test becomes

$$F = \frac{(\hat{\beta}_i - \beta_{i0})^2}{s^2 c_{ii}} \sim F(1, T-k) \quad (47)$$

where c_{ii} is (i, i) th element on diagonal of $(X'X)^{-1}$. However, since now $q = 1$ we do not need to use the χ^2 trick and obtain directly the t test

$$t \equiv \frac{\hat{\beta}_i - \beta_{i0}}{s\sqrt{c_{ii}}} \sim t(T-k), \quad (48)$$

distributed like a Student t with $T - k$ degree of freedom, dof.

The P-value for the F test

$$p\text{-value} = \text{Prob}(t(T-k) > T_{\text{value}}). \quad (49)$$

where T_{value} is a given number, as shown in the T Critical Values Tables. When the P-value is large, this gives confidence to the null hypothesis and the contrary is it is small.

The confidence interval: For a single parameter, under H_0 ,

$$\begin{aligned} \alpha &= \Pr\left(\left|\frac{\hat{\beta}_i - \beta_{i0}}{s\sqrt{c_{ii}}}\right| < t_{\alpha/2, T-k}\right) \\ &= \Pr(\hat{\beta}_i - t_{\alpha/2, T-k}s\sqrt{c_{ii}} \leq \beta_{i0} \leq \hat{\beta}_i + t_{\alpha/2, T-k}s\sqrt{c_{ii}}) \end{aligned}$$

Therefore the (confidence) interval

$$[\hat{\beta}_i - t_{\alpha/2, T-k}s\sqrt{c_{ii}}, \hat{\beta}_i + t_{\alpha/2, T-k}s\sqrt{c_{ii}}]$$

contains all values of β_{i0} which will not be rejected with probability $1 - \alpha$ when performing a two-tail t test.

Interpretation of a Confidence Interval:

1. **Definition of a Confidence Interval:** A confidence interval provides a range of values which is likely to contain the true population parameter. The confidence level associated with the CI represents the probability that this interval will capture the true parameter if you were to sample from the population repeatedly.
2. **Level of Confidence:** Common levels of confidence are 90%, 95%, and 99%. A 95% confidence interval, for example, means that if you were to take many samples and build a confidence interval from each of them, we expect about 95% of those intervals to contain the true parameter.
3. **Interpreting the Interval:** Let's say you have a 95% confidence interval for the mean weight of apples as (120g, 150g). This would be interpreted as:

"We are 95% confident that the true average weight of apples in the population is between 120 grams and 150 grams."
4. **What It Doesn't Mean:**
 - It doesn't mean that 95% of the data lies within that interval.
 - It also doesn't mean there's a 95% chance the true parameter is in the interval. The true parameter is either in the interval or it's not. The "95% confidence" relates to the method of generating the interval, not the probability of a specific interval containing the parameter. In other words, the 95% confidence is about the long-run frequency of the method we've used to generate the interval. If we were to repeatedly draw samples and construct 95% confidence intervals from these samples, we would expect about 95% of them to contain the true parameter.
5. **Width of the Interval:** A wider interval might suggest more uncertainty about the parameter, while a narrower interval suggests greater precision. However, this width can also be influenced by sample size, with larger samples often resulting in narrower intervals.
6. **Outside the Interval:** If a specific value lies outside a 95% confidence interval, it suggests that this value is not a plausible value for the population parameter at the 95% confidence level. For instance, if someone claims the average weight of apples is 115g, and this value is outside your 95% CI, then based on your data and at the 95% confidence level, this claim seems implausible.
7. **Applications:** When estimating the expected return of a portfolio or a specific asset, analysts might provide a confidence interval to communicate the range within which the true return is expected to lie with a certain degree of confidence.

In summary, a confidence interval provides a range in which we expect, with a certain level of confidence, the true population parameter to lie. It offers a way to express the precision and uncertainty associated with our sample estimate.

Quick Introduction to Large Sample Asymptotics:

In principle the sampling distribution of $\hat{\beta}$ is a function of the joint distribution of (y_i, x_i) and the sample size T , but in practice this function is extremely complicated so it is not feasible to analytically calculate the exact distribution of $\hat{\beta}$ except in very special cases. Therefore we typically rely on approximation methods.

Assumptions:

1. The T by k matrix X is non-stochastic and full rank.
2. Homoskedasticity and no autocorrelation⁴:

$$E[u] = 0$$

$$\text{Var}(u) = E[uu'] = \sigma^2 I$$

where I is the $T \times T$ identity matrix.

For $T \rightarrow \infty$,

$$\frac{X'X}{T} \rightarrow_p Q$$

where Q is a non-random positive definite k by k matrix.

By the Central Limit Theorem and Delta Method, we have

$$\frac{1}{\sqrt{T}} X'u \rightarrow_d N(0, \sigma^2 Q) \quad (50)$$

Under the above conditions, we may derive

$$\hat{\beta} \rightarrow_p \beta \text{ and } \sqrt{T}(\hat{\beta} - \beta) \rightarrow_d N(0, \sigma^2 Q^{-1}). \quad (51)$$

Comments:

1. The first result says that $\hat{\beta}$ is consistent: as T gets large, its randomness vanishes and it equals the true value.
2. Second result allows us to perform inference: as T gets large, we treat (standartized) $\hat{\beta}$ as if was normally distributed. Then we can apply all previous results (t test, F test, etc).
3. A (consistent) estimate of Q^{-1} will be $T(X'X)^{-1}$.

⁴The Homoskedasticity and no autocorrelation assumption provided lead to the inference that the error terms u_i are independent across observations (because of the no autocorrelation assumption) and have a constant variance σ^2 across all observations (homoskedasticity). The IID assumption (Independent and Identically Distributed) is met if the errors have the same distribution (which is implied by having a constant variance) and are independent of each other. So, based on the provided assumptions, the u_i errors can be considered IID with mean 0 and variance σ^2 . The Homoskedasticity assumption also lead to the inference that $E(u_i^2) < \infty$

The Delta Method in OLS:

An important result in order to obtain the asymptotic distribution of a generic nonlinear function of $\hat{\beta}$.

Given

$$\hat{\beta} \rightarrow_p \beta \text{ and } \sqrt{T}(\hat{\beta} - \beta) \rightarrow_d N(0, \sigma^2 Q^{-1}). \quad (52)$$

Let $f: R^k \rightarrow R^j$ be a twice continuously differentiable function(it could be nonlinear).

Then

$$\sqrt{T}(f(\hat{\beta}) - f(\beta)) \rightarrow_d N(0, \sigma^2 F Q^{-1} F') \quad (53)$$

where

$$F(x) \equiv \frac{\partial f(x)}{\partial x'}, \quad F \equiv F(\beta). \quad (54)$$

Question 1. Testing linear hypothesis - Lecture Notes 1

For the regression

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

Show the F test when testing

$$H_0 : \beta_1 = \beta_2.$$

Next, show the F test for the case

$$H_0 : \beta_2 = \beta_3 = 0.$$

Proof: Recall that in the general hypothesis testing setup, any linear null hypothesis can be written in the general form

$$H_0 : \underset{q \times k_k \times 1}{R} \underset{q \times 1}{\beta} = \underset{q \times 1}{r}$$

and the corresponding F test is given by

$$F \equiv \frac{(R\hat{\beta} - r)' [R(X'X)^{-1}R']^{-1} (R\hat{\beta} - r)/q}{e'e/(T-k)} \sim F(q, T-k). \quad (55)$$

So what we have to do is identify the elements of the F test in (55) that apply to this particular setting. More specifically, for our regression model, we have that $k = 3$. Moreover, the null hypothesis

$$H_0 : \beta_1 = \beta_2 \quad \text{vs} \quad H_1 : \beta_1 \neq \beta_2$$

contains just one restriction (i.e. $\beta_1 - \beta_2 = 0$), so that $q = 1$. Therefore, our null hypothesis can be re-written as

$$H_0 : \beta_1 - \beta_2 = 0$$

or, equivalently,

$$\begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = 0$$

Therefore it is easy to see that

$$R = \begin{bmatrix} 1 & -1 & 0 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}, \quad r = 0.$$

Hence:

$$\begin{aligned} R\hat{\beta} - r &= \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} - 0 \\ R\hat{\beta} - r &= \hat{\beta}_1 - \hat{\beta}_2 \end{aligned} \quad (56)$$

Now let us derive the expression for $[R(X'X)^{-1}R']^{-1}$. Let us denote by c_{ij} the generic (i, j) -th element of the matrix $(X'X)^{-1}$. Then,

$$\begin{aligned} (X'X)^{-1} &= \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \\ \Rightarrow R(X'X)^{-1}R' &= \begin{bmatrix} 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} c_{11} - c_{21}, & c_{12} - c_{22}, & c_{13} - c_{23} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \\ &= c_{11} - 2c_{12} + c_{22} \\ \Rightarrow [R(X'X)^{-1}R']^{-1} &= \frac{1}{c_{11} - 2c_{12} + c_{22}} \end{aligned} \quad (57)$$

See next page for more proof.

Question 1. Testing linear hypothesis - Lecture Notes 1

Plugging results (56) and (57) into the equation for the F test (55) we obtain

$$F = \frac{(\hat{\beta}_1 - \hat{\beta}_2)^2}{e'e/(T-k)(c_{11} - 2c_{12} + c_{22})}$$

$$F = \frac{(\hat{\beta}_1 - \hat{\beta}_2)^2}{s^2(c_{11} - 2c_{12} + c_{22})} \sim F(1, T-3)$$

where we used the fact that $e'e/(T-k)$ is the OLS estimator for σ^2 : $s^2 = \frac{e'e}{(T-k)}$.

For the case in which we test the hypothesis

$$H_0: \beta_2 = \beta_3 = 0$$

we notice that the null hypothesis can be re-written as

$$H_0: \begin{bmatrix} \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

so that $q = 2$. It implies that

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and it is clear to see that in this case

$$R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}, \quad r = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad q = 2.$$

The expression for $(R\hat{\beta} - r)$ is then:

$$\begin{aligned} R\hat{\beta} - r &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} \end{aligned} \tag{58}$$

Let us derive the expression for $[R(X'X)^{-1}R']^{-1}$. Using the same notation as in part (a), we have

$$\begin{aligned} (X'X)^{-1} &= \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \\ \Rightarrow R(X'X)^{-1}R' &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_{22} & c_{23} \\ c_{32} & c_{33} \end{bmatrix} \\ &= \frac{1}{|R(X'X)^{-1}R'|} \begin{bmatrix} c_{33} & -c_{32} \\ -c_{23} & c_{22} \end{bmatrix} \\ &= \frac{1}{c_{22}c_{33} - c_{23}^2} \begin{bmatrix} c_{33} & -c_{32} \\ -c_{23} & c_{22} \end{bmatrix} \end{aligned}$$

See next page for more proof.

Question 1. Testing linear hypothesis - Lecture Notes 1

where $|\cdot|$ denotes the determinant of a matrix. The F test is therefore:

$$F = \frac{\left([\hat{\beta}_2, \hat{\beta}_3] \left[R(X'X)^{-1}R' \right]^{-1} \begin{bmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix} \right) / 2}{s^2}$$

$$F = \frac{\hat{\beta}_2^2 c_{33} - 2\hat{\beta}_2 \hat{\beta}_3 c_{23} + \hat{\beta}_3^2 c_{22}}{2s^2 (c_{22}c_{33} - c_{23}^2)} \sim F(2, T-3)$$

Question 2. The Delta Method in OLS - Lecture Notes 1

For the model

$$Y_t = \alpha + \beta X_t + u_t,$$

what is the asymptotic distribution of $\hat{\alpha}/\hat{\beta}$?

Proof: For a general regression model $Y = X\beta + u$ (in matrix form) we know that when $T \rightarrow \infty$ the OLS estimator $\hat{\beta}$ is consistent and asymptotically normally distributed. That is,

$$\hat{\beta} \rightarrow_p \beta \quad \text{and} \quad \sqrt{T}(\hat{\beta} - \beta) \rightarrow_d \mathcal{N}(0, \sigma^2 Q^{-1}).$$

where the $K \times K$ matrix Q is a symmetric and positive definite matrix such that, as $T \rightarrow \infty$,

$$\frac{X'X}{T} \rightarrow_p Q$$

The asymptotic distribution of any nonlinear transformations of $\hat{\beta}$ can be obtained using the Delta Method. More specifically, for a generic nonlinear function $f(\cdot)$ which is continuous and twice differentiable, we have that

$$\sqrt{T}(f(\hat{\beta}) - f(\beta)) \rightarrow_d \mathcal{N}(0, \sigma^2 F Q^{-1} F')$$

where

$$F \equiv \frac{\partial f(\beta)}{\partial \beta'} \equiv \left[\frac{\partial f(\beta)}{\partial \beta_1}, \frac{\partial f(\beta)}{\partial \beta_2}, \dots, \frac{\partial f(\beta)}{\partial \beta_K} \right]$$

Now let us apply this result to our specific setting. Let us first identify the vector of our parameters, that is $\beta = (\alpha, \beta)'$. Then, we can define a twice continuously differentiable nonlinear function $f(\beta)$ as

$$f(\beta) = \frac{\alpha}{\beta}$$

which essentially specifies the form of the nonlinear relation whose asymptotic distribution we want to derive. Applying the method described above, we have that the asymptotic distribution of this nonlinear function is

$$\sqrt{T} \left(\frac{\hat{\alpha}}{\hat{\beta}} - \frac{\alpha}{\beta} \right) \rightarrow_d \mathcal{N}(0, \sigma^2 F Q^{-1} F')$$

where F in our case takes the following form:

$$F = F(\beta) = \frac{\partial f(\beta)}{\partial \beta'} = \left[\frac{\partial f(\beta)}{\partial \alpha}, \frac{\partial f(\beta)}{\partial \beta} \right] = \left[\frac{1}{\beta}, -\frac{\alpha}{\beta^2} \right].$$

In practice, if you want to estimate the asymptotic covariance matrix, we just need to replace Q^{-1} with $\left(\frac{X'X}{T} \right)^{-1}$ and $F(\beta)$ with $F(\hat{\beta})$ and the OLS estimator of σ^2 is $s^2 = \frac{1}{T-K} e'e$.

Question 3. Testing linear hypothesis - Sample Exam Question

Consider the multiple linear regression model:

$$y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + \beta_4 X_{4t} + \varepsilon_t.$$

If we want to test

$$\frac{\beta_1 + \beta_3 + \beta_4}{\beta_2} = 1 \text{ against the alternative } \frac{\beta_1 + \beta_3 + \beta_4}{\beta_2} \neq 1$$

how would you modify the above result? (Hint: a numerical answer is not possible so simply describe how you would implement the test.)

Proof: The objective is to investigate whether the null hypothesis, articulated as

$$H_0 : \beta_1 + \beta_3 + \beta_4 - \beta_2 = 0,$$

holds given the sample data. This hypothesis seeks to ascertain if the linear combination of coefficients when normalized by β_2 , is statistically equivalent to one.

Hypothesis in Matrix Form

The aforementioned hypothesis can be compactly represented in matrix notation:

$$H_0 : R\beta = r$$

where

$$R = \begin{bmatrix} 1 & -1 & 1 & 1 \end{bmatrix}$$

and

$$r = [0].$$

Distribution of $R\hat{\beta}$

Given the multivariate normality of the OLS estimator,

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1})$$

under the null hypothesis, we have:

$$R\hat{\beta} \sim N(r, \sigma^2 R (X'X)^{-1} R').$$

Test Statistic

To evaluate the veracity of the null hypothesis, we compute the following quadratic form:

$$Q = (R\hat{\beta} - r)' [\sigma^2 R (X'X)^{-1} R']^{-1} (R\hat{\beta} - r)$$

This statistic, under the null hypothesis, adheres to a χ^2 distribution with q degrees of freedom, where q denotes the row dimension of matrix R , in this case, $q = 1$.

F-test Statistic

To facilitate hypothesis testing, we convert the previously computed quadratic form into an F-statistic:

$$F = \frac{Q/q}{e'e/(T-k)}$$

where e represents the residuals from the regression, T the sample size, and k the number of parameters (inclusive of the intercept). This F-statistic follows an $F(q, T - k)$ distribution.

P-value Calculation

The corresponding P-value is then:

$$p\text{-value} = \text{Prob}(F(1, T - 4) > F)$$

where the critical value, F , can be found in standard F-distribution tables. A smaller P-value indicates stronger evidence against the null hypothesis at conventional significance levels.

Remembering Thm.1 The Gauss-Markov Theorem:

The Gauss-Markov theorem is a fundamental result in the theory of linear regression. It states that under certain conditions, the ordinary least squares (OLS) estimator has the smallest possible variance among the class of linear unbiased estimators. It can be shown to be the Best Linear Unbiased Estimator, **BLUE**.

Assumptions:

1. **Linearity:** The model specifies a linear relationship between Y and $X_2, \dots, X_k$, that is in (1).
2. **The matrix X is non-stochastic and full rank:** We can treat X_i simply as a matrix of known constants and there must be no exact linear relationship among any of the independent variables in the model.
3. **Homoskedasticity and no autocorrelation:**

$$E[u] = 0$$

$$\text{Var}(u) = E[uu'] = \sigma^2 I$$

where I is the $T \times T$ identity matrix.

Given the assumptions above, the OLS estimator $\hat{\beta}_{OLS} = (X'X)^{-1} X'Y$ is the Best Linear Unbiased Estimator (BLUE) of β . This means that among all linear estimators that are unbiased, the OLS estimator has the smallest variance. In mathematical terms, if $\hat{\beta}_L$ is any other linear unbiased estimator of β , then:

$$\text{Var}(\hat{\beta}_{OLS}) \leq \text{Var}(\hat{\beta}_L).$$

This theorem highlights the efficiency of OLS in the class of linear and unbiased estimators. It does not, however, compare OLS to nonlinear or biased estimators, which can potentially have lower variance under some circumstances.

Introduction of Generalized Least Squares (GLS):

Why do we need it?

Generalized Least Squares (GLS) is an extension of the ordinary least squares (OLS) method, used to estimate the parameters of a linear regression model. **The primary motivation behind GLS is to deal with violations of OLS assumptions, especially when the error terms are heteroskedastic (i.e., have non-constant variance) or are autocorrelated (i.e., not independent).**

Difference between GLS and OLS:

1. **Assumptions:** OLS assumes that the errors (residuals) are homoskedastic (constant variance) and uncorrelated. If these assumptions are violated, OLS estimates remain unbiased but are no longer the best (in terms of being the minimum variance estimator). GLS, on the other hand, allows for heteroskedasticity and/or autocorrelation in the errors.
2. **Efficiency:** In the presence of heteroskedasticity or autocorrelation, GLS estimates are more efficient (i.e., have lower variances) than OLS estimates.
3. **Transformation:** GLS deals with these violations by transforming the original equation into one where the new error terms are homoskedastic and uncorrelated. This is done using a known matrix of variances and covariances.

Real-life Example in Finance:

Consider a financial analyst who is studying the monthly returns of a stock relative to the returns of the market index (like the S&P 500). They are interested in finding out how the stock reacts to the market movements. The analyst decides to run a regression of the stock returns on the market returns.

However, due to certain financial events, **some months experience higher volatility than others, leading to heteroskedasticity in the errors. Additionally, due to momentum effects, the errors this month might be correlated with the errors in the previous month, leading to autocorrelation.**

In this case, simply using OLS might lead to inefficient estimates. Instead, using GLS would provide more reliable results by accounting for the heteroskedasticity and autocorrelation.

GLS Model Outline:

The generic form

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_K X_{Ki} + u_i, \quad i = 1, \dots, T. \quad (59)$$

The matrix form

$$Y = X\beta + u, \quad (60)$$

where

$$Y_{T \times 1} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_T \end{bmatrix}, \quad X_{T \times K} = \begin{bmatrix} 1 & X_{12} & X_{13} & \dots & X_{1K} \\ 1 & X_{22} & X_{23} & \dots & X_{2K} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & X_{T2} & X_{T3} & \dots & X_{TK} \end{bmatrix}, \quad \beta_{K \times 1} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}, \quad u_{T \times 1} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_T \end{bmatrix}.$$

The linear regression model consists of a set of assumptions that describe the form of the model and relationships among its parts which are needed for appropriate estimation and inference procedures.

1. **Linearity:** The model specifies a linear relationship between Y and $X_2, \dots, X_K$, that is in (1).
2. **The matrix X is non-stochastic and full rank:** We can treat X_i simply as a matrix of known constants and there must be no exact linear relationship among any of the independent variables in the model.
3. **Heteroskedasticity or Autocorrelation or both:**

$$E[u] = 0 \quad (61)$$

$$\text{Var}(u) = Eu u' \neq \sigma^2 I \quad (62)$$

where I is the $T \times T$ identity matrix.

Recalling:

$$Eu u' = \begin{pmatrix} E(u_1^2) & E(u_1 u_2) & \dots & E(u_1 u_T) \\ E(u_2 u_1) & E(u_2^2) & \dots & E(u_2 u_T) \\ \dots & \dots & E(u_i u_j) & \dots \\ \vdots & \vdots & \vdots & \vdots \\ E(u_T u_1) & E(u_T u_2) & \dots & E(u_T^2) \end{pmatrix} \quad (63)$$

1. Case Heteroskedasticity Only^a:

$$\text{var}(u_i) = E(u_i^2) \neq \sigma^2$$

2. Case Autocorrelation Only:

$$\text{cov}(u_i, u_j) = E(u_i u_j) \neq 0 \text{ for } i \neq j$$

3. Case Heteroskedasticity and Autocorrelation:

$$\text{var}(u_i) = E(u_i^2) \neq \sigma^2$$

and

$$\text{cov}(u_i, u_j) = E(u_i u_j) \neq 0 \text{ for } i \neq j.$$

We formalize this writing

$$Eu u' = \sigma^2 \Omega,$$

where Ω is known, symmetric positive definite, $T \times T$ matrix. Here σ^2 is just a scaling parameter, unknown.

^aMore generally, if the variance-covariance matrix of disturbance u_i across i has a nonconstant diagonal, the disturbance is heteroscedastic. The matrices below are covariances when there are just three observations across time. The disturbance in matrix A is homoscedastic; this is the simple case where OLS is the best linear unbiased estimator. The disturbances in matrices B and C are heteroscedastic. In matrix B, the variance is time-varying, increasing steadily across time; in matrix C, the variance depends on the value of x . The disturbance in matrix D is homoscedastic because the diagonal variances are constant, even though the off-diagonal covariances are non-zero, and ordinary least squares are inefficient for a different reason: serial correlation.

$$A = \sigma^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad B = \sigma^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad C = \sigma^2 \begin{bmatrix} x_1 & 0 & 0 \\ 0 & x_2 & 0 \\ 0 & 0 & x_3 \end{bmatrix} \quad D = \sigma^2 \begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix}$$

GLS Results:

To introduce GLS, we need to recall that for any Ω symmetric positive definite, there exists a matrix P such that

$$\Omega^{-1} = P'P. \quad (64)$$

P is a sort of square-root matrix. Then if we premultiply the regression model by P

$$Py = PX\beta + Pu \quad (65)$$

or equivalently

$$y^* = X^*\beta + u^* \quad (66)$$

defining $y^* \equiv Py, X^* \equiv PX, u^* \equiv Pu$, we have

$$\begin{aligned} Eu^* &= EPu = PEu = 0, \\ Eu^*u^{*'} &= \sigma^2 P\Omega P' = \sigma^2 PP^{-1}P^{-1}P' = \sigma^2 I. \end{aligned} \quad (67)$$

After the transformation, we transformed the assumption 3 (61 and 62) back to the homoskedasticity and no autocorrelation assumption. Therefore the transformed model with the starred (*) variables satisfies the Gauss-Markov conditions and so OLS is BLUE!^a

^aThis transformation effectively standardizes the scale of and de-correlates the errors. When OLS is used on data with homoscedastic errors, the GaussMarkov theorem applies, so the GLS estimate is the best linear unbiased estimator for β .

Now re-write $\hat{\beta}^* = (X^{*'}X^*)^{-1}X^{*'}y^*$ in terms of y, X, Ω and we finally get the $\hat{\beta}_{GLS}$:

$$\hat{\beta}_{GLS} \equiv \hat{\beta}^* = (X^{*'}X^*)^{-1}X^{*'}y^* = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y. \quad (68)$$

Unbiased estimate of σ^2 is

$$s_{GLS}^2 \equiv \frac{1}{T-k} (y - X\hat{\beta}_{GLS})' \Omega^{-1} (y - X\hat{\beta}_{GLS}). \quad (69)$$

Under normality of u the F test for $H_0: R\beta = r$ is

$$F \equiv \frac{(R\hat{\beta}_{GLS} - r)' [R(X'\Omega^{-1}X)^{-1}R']^{-1} (R\hat{\beta}_{GLS} - r) / q}{s_{GLS}^2} \sim F_{(q, T-k)}$$

Difference between GLS and Feasible GLS:

Our setup above assumed we knew Ω , but in reality that's not the case. In fact, one needs to estimate Ω leading to the Feasible GLS. The main difference between the two methods lies in their assumptions. GLS assumes we know the exact form of the variance-covariance matrix of the errors, Ω , while FGLS estimates this matrix from the data, we use $\hat{\Omega}$. As a result, FGLS is more commonly used in practice since, in many scenarios, we don't have precise knowledge of the heteroskedasticity or autocorrelation structure but can estimate it.

Nolinear Least Squares (NLS) Model Outline:

Regression model can be written more in general as

$$Y_t = h(\beta; x_t) + u_t \quad (70)$$

for some known yet possibly nonlinear scalar function $h(\cdot)$. Linear regression just a special case when $h(\beta; x_t) = x_t' \beta$.

Assumptions:

1. Let u satisfy the Gauss-Markov assumptions and, in addition:

$$2. \quad E(u_t | h(\beta; x_t)) = 0, \quad (71)$$

and

$$E(u_t^2 | h(\beta; x_t)) = \sigma^2. \quad (72)$$

3. $h(\tilde{\beta})$ is twice continuously differentiable function of $\tilde{\beta}$.

4. $h(\beta; x_t) = h(\tilde{\beta}; x_t)$ if and only if $\beta = \tilde{\beta}$.

The NLS problem is defined by

$$\hat{\beta}_{NLS} = \operatorname{argmin}_{\beta \in B} \sum_{t=1}^T (y_t - h(\beta; x_t))^2, \quad (73)$$

where B is some compact subspace of R^k .

Comments:

Although the FOC will exist, no closed-form expression for $\hat{\beta}_{NLS}$ in general.

Nolinear Least Squares (NLS) Results:

Since $\hat{\beta}_{NLS}$ is nonlinear function of the Y_t , for inference we rely on asymptotics. Under the above conditions, as $T \rightarrow \infty$,

$$\hat{\beta}_{NLS} \rightarrow_p \beta, \sqrt{T}(\hat{\beta}_{NLS} - \beta) \rightarrow_d N(0, \sigma^2 Q^{-1}),$$

where Q is a $k \times k$ matrix such that

$$\frac{1}{T} X_0' X_0 \rightarrow_p Q.$$

where $X_0 = (x_{01} \dots x_{0T})'$ and $x_{0t} = \frac{\partial h(\tilde{\beta}; x_t)}{\partial \tilde{\beta}} \Big|_{\tilde{\beta}=\beta}$. We can to usual stuff (t test, F test, confidence intervals, etc)!

To use the asymptotic distribution, for instance, to construct confidence intervals, we need to estimate the covariance matrix $\sigma^2 Q^{-1}$.

σ^2 consistently estimated by

$$s_{NLS}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - h(\hat{\beta}_{NLS}; x_t))^2,$$

and $\hat{Q} = \frac{1}{T} \hat{X}' \hat{X}$, where $\hat{X}$ is obtained stacking

$$\hat{x}_t = \frac{\partial h(\tilde{\beta}; x_t)}{\partial \tilde{\beta}'} \Big|_{\tilde{\beta}=\hat{\beta}_{NLS}}.$$

Question 1. Generalized Least Squares (GLS) Estimator - Lecture Notes 2

Given the model

$$Y_t = \alpha + \beta X_t + u_t$$

where the standard assumptions hold but $E u_t^2 = \sigma^2 X_t^2$, derive the GLS estimator.

In this exercise we need to extend the multiple regression model to the case of error terms that violate the assumption of homoscedasticity and nonautocorrelation (namely, $E[uu'] = \sigma^2 I$). The generalized linear regression model can be formulated in the following way:

$$\begin{aligned} Y_t &= \alpha + \beta X_t + u_t, \\ E[u | X] &= 0, \\ E[uu' | X] &= \sigma^2 \Omega \end{aligned}$$

where Ω is a $T \times T$ positive definite matrix. We know that when $\Omega \neq I$, the OLS estimator retains only some of their desirable properties in this model. That is, the OLS estimator remains unbiased, consistent and asymptotically normally distributed. It will, however, no longer be efficient and, therefore, the usual inference procedures are no longer appropriate. The Generalized Least Square (GLS) estimator has the following form:

$$\hat{\beta}_{GLS} \equiv \left(X' \Omega^{-1} X \right)^{-1} X' \Omega^{-1} Y.$$

Let us try to derive the expression for Ω^{-1} in our case. Using the information in the text we know that $E u_t^2 = \sigma^2 X_t^2$. Defining the $T \times 1$ vector $u = [u_1, \dots, u_t, \dots, u_T]'$, then the covariance matrix has the following form

$$E[uu'] = \sigma^2 \underbrace{\begin{bmatrix} X_1^2 & 0 & \dots & 0 \\ 0 & X_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & X_T^2 \end{bmatrix}}_{\Omega}$$

and hence

$$\Omega^{-1} = \begin{bmatrix} \frac{1}{X_1^2} & 0 & \dots & 0 \\ 0 & \frac{1}{X_2^2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{X_T^2} \end{bmatrix}$$

Question 2. Nolinear Least Squares (NLS) Estimator - Lecture Notes 2

Would you estimate the following model

$$Y_t = \alpha + e^{X_t \beta} + u_t$$

by OLS?

Remember from the first class that one of the assumptions of the linear regression model was the Linearity Assumption. It implies that the model specifies a linear relationship between Y and X . In this exercise the linearity assumption is clearly violated as the β parameter enters through the exponential function. Hence, the standard OLS estimator cannot be applied. We need to derive a more general methodology which includes the linear regression model as special case.

Regression models can be written in a more general form as

$$Y_t = h(\beta; X_t) + u_t$$

for some known and (potentially) nonlinear function $h(\cdot)$. The linear regression model is just a special case when $h(\beta; X_t) = X_t' \beta$.

In order to estimate the model we need to introduce some assumptions. Formally:

1. $E[Y_t | X_t] = h(\beta; X_t)$, $\forall t = 1, \dots, T$ where $h(\beta; X_t)$ is a twice continuously differentiable function of β ,
2. $E[u_t | h(\beta; X_t)] = 0$,
3. $E[u_t^2 | h(\beta; X_t)] = \sigma^2$, and $E[u_t u_s | h(\beta; X_t), h(\beta; X_s)] = 0$
4. Identifiability of the model parameters: The parameter vector in the model is identified (i.e estimable) if there is no nonzero parameter $\tilde{\beta} \neq \beta$ such that $h(\tilde{\beta}, X_t) = h(\beta, X_t)$.

The Nonlinear Least Squares (NLS) estimator is defined as the minimizer of the sum of square residuals

$$\hat{\beta}_{NLS} = \underset{b \in B}{\operatorname{argmin}} \sum_{t=1}^T (Y_t - h(b; X_t))^2$$

where B is the subspace of all the possible admissible values of β . Let us try to determine the NLS estimator in our case. Let $\beta = (a, b)'$ be the 2×1 vector of our parameters and define $b = (a, b)'$ as any admissible value for β . Then

$$\begin{aligned} \hat{\beta}_{NLS} &= \underset{b \in B}{\operatorname{argmin}} \operatorname{RSS}(b) \\ &= \underset{b \in B}{\operatorname{argmin}} \sum_{t=1}^T (Y_t - h(b; X_t))^2 \\ &= \underset{(a,b) \in B}{\operatorname{argmin}} \sum_{t=1}^T (Y_t - a - e^{X_t b})^2. \end{aligned}$$

The FOC for the expression above implies that

$$\begin{aligned} \left. \frac{\partial \operatorname{RRS}(a, b)}{\partial a} \right|_{a=\hat{a}} &= -2 \sum_{t=1}^T (Y_t - \hat{a} - e^{X_t b}) = 0 \\ \left. \frac{\partial \operatorname{RRS}(a, b)}{\partial b} \right|_{b=\hat{\beta}} &= -2 \sum_{t=1}^T (Y_t - \hat{a} - e^{X_t \hat{\beta}}) e^{X_t \hat{\beta}} X_t = 0. \end{aligned} \quad (74)$$

Equation (74) implies that

$$\begin{aligned} \sum_{t=1}^T Y_t - T \hat{a} - \sum_{t=1}^T e^{X_t \hat{\beta}} &= 0 \\ \Rightarrow \hat{a} &= \bar{Y} - \frac{\sum_{t=1}^T e^{X_t \hat{\beta}}}{T} \end{aligned} \quad (75)$$

where $\bar{Y} = \frac{\sum_{t=1}^T Y_t}{T}$.

See next page for more proof.

Question 2. Nolinear Least Squares (NLS) Estimator - Lecture Notes 2

Replacing (75) in (74) we can now solve for $\hat{\beta}$:

$$\begin{aligned} & -2 \sum_{t=1}^T (Y_t - \hat{\alpha} - e^{X_t \hat{\beta}}) e^{X_t \hat{\beta}} X_t = 0 \\ \Rightarrow & \sum_{t=1}^T Y_t X_t e^{X_t \hat{\beta}} - \hat{\alpha} \sum_{t=1}^T X_t e^{X_t \hat{\beta}} - \sum_{t=1}^T e^{2X_t \hat{\beta}} X_t = 0 \\ \Rightarrow & \sum_{t=1}^T Y_t X_t e^{X_t \hat{\beta}} - \left(\bar{Y} - \frac{\sum_{t=1}^T e^{X_t \hat{\beta}}}{T} \right) \sum_{t=1}^T X_t e^{X_t \hat{\beta}} - \sum_{t=1}^T e^{2X_t \hat{\beta}} X_t = 0 \end{aligned} \quad (76)$$

The above expression is a function of $\hat{\beta}$ only. Hence, in principle we can solve for $\hat{\beta}$ and derive the final expression of our $\hat{\beta}_{NLS}$.

For this exercise (and for the final exam), you do not need to solve (76)! But it is important that you know how the estimator can be derived (i.e. you need to know how to set the FOCs for a generic non-linear function).

Question 3. GLS VS. OLS - Lecture Notes 2

In the general linear regression model, suppose Ω is known. Then derive: (i) the covariance matrix of OLS and GLS; (ii) the covariance matrix of the OLS residual $e = y - X\hat{\beta}$; (iii) the covariance matrix of the GLS residual $e = y - X\hat{\beta}_{GLS}$.

As in the previous exercise, the general linear regression model can be formulated as follows

$$\begin{aligned} Y_t &= \beta_1 + \beta_2 X_t + u_{t+} \\ E[u | X] &= 0, \\ E[uu' | X] &= \sigma^2 \Omega \end{aligned}$$

where Ω is a positive $T \times T$ definite matrix. In the rest of the exercise we will assume that Ω is known so that we do not need to estimate it.

(i) Let us start by deriving the covariance matrix of the OLS estimator. Starting from the expression of the estimator we have that

$$\begin{aligned} \hat{\beta}_{OLS} &= (X'X)^{-1} X'Y \\ &= \beta + (X'X)^{-1} X'u \end{aligned} \quad (77)$$

Hence, the covariance matrix is

$$\begin{aligned} \text{Var}(\hat{\beta}_{OLS}) &= E[(\hat{\beta}_{OLS} - \beta)(\hat{\beta}_{OLS} - \beta)'] \\ &= E[(X'X)^{-1} X'uu'X (X'X)^{-1}] \\ &= (X'X)^{-1} X'E[uu']X (X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1} X'\Omega X (X'X)^{-1} \end{aligned} \quad (78)$$

Consider now the GLS estimator. Following the same steps we know that

$$\begin{aligned} \hat{\beta}_{GLS} &= (X'\Omega^{-1}X)^{-1} X'\Omega^{-1}Y \\ &= \beta + (X'\Omega^{-1}X)^{-1} X'\Omega^{-1}u \end{aligned} \quad (79)$$

From this we derive the covariance matrix of the GLS estimator as

$$\begin{aligned} \text{Var}(\hat{\beta}_{GLS}) &= E[(\hat{\beta}_{GLS} - \beta)(\hat{\beta}_{GLS} - \beta)'] \\ &= E[(X'\Omega^{-1}X)^{-1} X'\Omega^{-1}uu'\Omega^{-1}X (X'\Omega^{-1}X)^{-1}] \\ &= (X'\Omega^{-1}X)^{-1} X'\Omega^{-1}E[uu']\Omega^{-1}X (X'\Omega^{-1}X)^{-1} \\ &= \sigma^2 (X'\Omega^{-1}X)^{-1} X'\Omega^{-1}\Omega\Omega^{-1}X (X'\Omega^{-1}X)^{-1} \\ &= \sigma^2 (X'\Omega^{-1}X)^{-1}. \end{aligned} \quad (80)$$

Compare the results in (78) and (80). The variance of the two estimators will be equal only if $\Omega = I$.

(ii) We need to derive the covariance matrix of the OLS residual. Starting from the definition of the residuals and using (77) we have

$$\begin{aligned} e_{OLS} &= Y - X\hat{\beta}_{OLS} \\ &= Y - X(X'X)^{-1} X'Y \\ &= [I - X(X'X)^{-1} X']Y \\ &= M_{OLS}Y \end{aligned} \quad (81)$$

See next page for more proof.

Question 3. GLS VS. OLS - Lecture Notes 2

where $M_{OLS} = [I - X(X'X)^{-1}X']$ is an idempotent $T \times T$ matrix with the following properties:

$$M_{OLS} = M'_{OLS}, \quad M_{OLS}M'_{OLS} = M_{OLS} \quad \text{and} \quad M_{OLS}X = 0.$$

Then,

$$\begin{aligned} \text{Var}(e_{OLS}) &= \text{Var}(M_{OLS}Y) \\ &= \text{Var}(M_{OLS}(X\beta + u)) \\ &= \text{Var}(M_{OLS}u) \\ &= E[M_{OLS}uu'M_{OLS}] \\ &= M_{OLS}E[uu']M_{OLS} \\ &= \sigma^2 M_{OLS}\Omega M_{OLS}. \end{aligned} \tag{82}$$

(iii) To derive the covariance matrix of the GLS residuals we can follow the same steps as in (ii). More specifically, using (79) we have that

$$\begin{aligned} e_{GLS} &= Y - X\hat{\beta}_{GLS} \\ &= Y - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}Y \\ &= [I - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}]Y \\ &= M_{GLS}Y \end{aligned} \tag{83}$$

where we set $M_{GLS} = [I - X(X'\Omega^{-1}X)^{-1}X'\Omega^{-1}]$. Notice that in this case the matrix M_{GLS} satisfies the two properties:

$$M_{GLS}M'_{GLS} = M_{GLS} \quad \text{and} \quad M_{GLS}X = 0.$$

Then

$$\begin{aligned} \text{Var}(e_{GLS}) &= \text{Var}(M_{GLS}Y) \\ &= \text{Var}(M_{GLS}(X\beta + u)) \\ &= \text{Var}(M_{GLS}u) \\ &= E[M_{GLS}uu'M'_{GLS}] \\ &= M_{GLS}E[uu']M'_{GLS} \\ &= \sigma^2 M_{GLS}\Omega M'_{GLS}. \end{aligned} \tag{84}$$

Notice that, when $\Omega = I$, then $\text{Var}(e_{OLS}) = \text{Var}(e_{GLS})$.

Extension of Assumptions on the Regressor Matrix $\mathbf{X}$:

- In Lectures 1 and 2, for simplicity, we assumed that the $T \times K$ regressor matrix $\mathbf{X}$ is non-stochastic. This implies that all the regressors X_i for $i = 1, \dots, K$ are not random variables.
- However, in the context of **Instrumental Variables (IV)**, we treat $\mathbf{X}$ as stochastic.^a This means that all the regressors X_i for $i = 1, \dots, K$ are considered random variables.
- When $\mathbf{X}$ is stochastic, all the results from Lectures 1 and 2 continue to hold, provided they are conditioned on $\mathbf{X}$.

^aThis is a common assumption in econometrics when the regressors are not controlled by the experimenter but are instead observed values that can vary from sample to sample.

Now we have the following,

3. Homoskedasticity and no autocorrelation:

In matrix form:

$$E[u|\mathbf{X}] = 0 \quad (85)$$

The expected value of the error term u given the regressor matrix $\mathbf{X}$ is zero.

$$\text{Var}(u|\mathbf{X}) = E[uu'|\mathbf{X}] = \sigma^2 I \quad (86)$$

The variance of the error term u , given $\mathbf{X}$, is constant across observations (homoskedastic) and there is no autocorrelation (identity matrix I scaled by σ^2).

The generic form for (85) is given by

$$\forall t = 1, \dots, T, \quad E(u_t | x_{t-s}, s > 0) = 0. \quad (87)$$

Taking the expectation on conditional expectation, by the law of iterated expectations (LIE), we have

$$EX'u = E[E[X'u | \mathbf{X}]] = E[X'E[u | \mathbf{X}]] = 0$$

As the expected value of the error term u given the regressor matrix $\mathbf{X}$ is zero, we derive

$$\text{cov}(\mathbf{X}, u) = EX'u = 0. \quad (88)$$

Comments: The step that might be causing confusion is the treatment of $\mathbf{X}$ as non-random within the conditional expectation $E[\cdot | \mathbf{X}]$, which is correct, but it should be understood that $\mathbf{X}$ is still a random matrix outside of this conditional expectation. To remove $\mathbf{X}$ from the conditioning in a mathematically rigorous way, we would apply the Law of Iterated Expectations (LIE).



Check it.

Endogeneity:

Definition In econometrics, endogeneity broadly refers to situations in which an explanatory variable is correlated with the error term, such that

$$\text{cov}(X, u) = E[X'u] \neq 0 \quad (89)$$

Why should we care about it?

To demonstrate that the OLS estimator $\hat{\beta}$ is biased and inconsistent when $\text{cov}(X, u) = E[X'u] \neq 0$, we start with the formula for the OLS estimator:

$$\hat{\beta} = \beta + \left(\frac{X'X}{T} \right)^{-1} \frac{X'u}{T} \quad (90)$$

We will consider the expectation of $\hat{\beta}$ conditioned on X to check for bias and the limit of this expectation as $T \rightarrow \infty$ to check for consistency.

Step 1: Bias

Bias in the estimator is determined by evaluating $E[\hat{\beta}|X]$. Let's compute the conditional expectation of $\hat{\beta}$:

$$E[\hat{\beta}|X] = E \left[\beta + \left(\frac{X'X}{T} \right)^{-1} \frac{X'u}{T} \middle| X \right]$$

Because β is a constant, $E[\beta|X] = \beta$. Now, let's focus on the conditional expectation of the second term:

$$E \left[\left(\frac{X'X}{T} \right)^{-1} \frac{X'u}{T} \middle| X \right] = \left(\frac{X'X}{T} \right)^{-1} E \left[\frac{X'u}{T} \middle| X \right]$$

Since $\text{cov}(X, u) = E[X'u] \neq 0$, this implies that:

$$E \left[\frac{X'u}{T} \right] \neq 0_{K \times 1}$$

Therefore, the conditional expectation of our estimator becomes:

$$E[\hat{\beta}] = E[E[\hat{\beta}|X]] = \beta + E \left[\left(\frac{X'X}{T} \right)^{-1} \right] E \left[\frac{X'u}{T} \right] \neq \beta$$

This indicates that the estimator does not converge to the true value of β as T increases, hence it is not consistent.

Step 2: Consistency

Given the formula for the OLS estimator:

$$\hat{\beta} = \beta + \left(\frac{X'X}{T} \right)^{-1} \frac{X'u}{T}$$

we want to show that the second term on the right-hand side goes to zero in probability as $T \rightarrow \infty$. Let's break down this second term:

$$\left(\frac{X'X}{T} \right)^{-1} \frac{X'u}{T} = \left(\frac{X'X}{T} \right)^{-1} \left(\frac{X'u}{T} \right)$$

Now, applying the Law of Large Numbers (LLN):

1. $\frac{X'X}{T}$ converges in probability to $E[X'X]$ if the entries of X are independent and identically distributed (i.i.d.) with finite second moments. As $T \rightarrow \infty$, $\left(\frac{X'X}{T} \right)$ becomes non-singular and its inverse exists.
2. $\frac{X'u}{T}$ converges in probability to $E[X'u]$. Under the assumption of endogeneity, where u is correlated with X , we have $E[X'u] \neq 0$.

Thus, if X is full rank (the columns of X are linearly independent), and the regressors are correlated with the error term u , then by the LLN, as $T \rightarrow \infty$:

$$\left(\frac{X'X}{T} \right)^{-1} \left(\frac{X'u}{T} \right) \rightarrow_p (E[X'X])^{-1} E[X'u] \neq 0.$$

This indicates that the estimator does not converge to the true value of β as T increases, hence it is not consistent.

More on Next Page!

Endogeneity:

Step 3: Efficiency

If $\text{cov}(X, u) = E[X'u] \neq 0$, this indicates a violation of the Gauss-Markov assumptions, specifically, that the error terms are uncorrelated with the regressors. This condition implies that the error term u and the independent variables X are correlated, suggesting endogeneity in the model. When endogeneity is present, OLS is no longer unbiased, and thus, it cannot be BLUE because BLUE requires the estimator to be unbiased in the first place.

To summarize, if $\text{cov}(X, u) = E[X'u] \neq 0$, the OLS estimator cannot be efficient according to the Gauss-Markov Theorem because it is not even unbiased. Efficiency in the sense of the Gauss-Markov Theorem cannot be discussed for biased estimators, as the theorem's conditions are violated.

Comments:

1. You need to know (90)!
2. Checking Bias is about Taking expectation.
3. Checking Consistency is about apply the LLN.
4. The efficiency of an estimator in statistics is measured by its variance (or more precisely, by its mean squared error (MSE), which is a combination of its variance and bias).

Additional Reading to help you better understand, not required for this class:

The efficiency of an estimator in statistics is commonly measured by its variance or mean squared error (MSE), particularly within the class of linear unbiased estimators. Here's how efficiency is generally assessed:

1. **Variance:** The variance of an estimator is a direct measure of its efficiency. Among unbiased estimators, the estimator with the smallest variance is deemed the most efficient.
2. **Mean Squared Error (MSE):** The MSE is defined as:

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta}, \theta)]^2 \quad (91)$$

An estimator with a smaller MSE is considered more efficient.

3. **Cramér-Rao Lower Bound (CRLB):** This theoretical lower bound for the variance of an unbiased estimator is given by:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)} \quad (92)$$

where $I(\theta)$ is the Fisher information. An efficient estimator that achieves this bound is said to be 'efficient' in a specific sense.

4. **Relative Efficiency:** The relative efficiency of two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ is:

$$e(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)} \quad (93)$$

A relative efficiency greater than 1 indicates that $\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$.

5. **Best Linear Unbiased Estimator (BLUE):** Under the Gauss-Markov theorem, the OLS estimator is the BLUE when the error terms are homoscedastic and uncorrelated with the regressors, implying it has the smallest variance among all linear unbiased estimators.

In practical applications, efficiency can be estimated using sample data to calculate an estimate of the variance or MSE. In regression analysis, standard errors of regression coefficients, which are square roots of the estimated variances, are often used to assess the efficiency of these estimators.



Please read it for fun!

Example of Endogeneity: The errors-in-variables (EIV) problems

Assume $k = 1$ in linear regression:

$$y = \tilde{x}\beta + u$$

where $\tilde{x}$ denotes the true but unobserved regressor. Rather we observe this value with an error:

$$x = \tilde{x} + v$$

where the measurement error v , the regression error u and the true but unobserved regressor $\tilde{x}$ are mutually independent with finite second moment.

If the y are simply regressed on the x , we have

$$y = x\beta + (u - \beta v) = x\beta + \eta, \quad (94)$$

for some

$$\eta = u - \beta v \quad (95)$$

In this errors-in-variables (EIV) problem, we have $\text{cov}(X, \eta) = E[X'\eta] \neq 0$

Let us show that

$$\hat{\beta} \rightarrow_p \beta \left(\frac{\sigma_{\tilde{x}}^2}{\sigma_{\tilde{x}}^2 + \sigma_v^2} \right) < \beta.$$

Proof.

$$\begin{aligned} \hat{\beta} &= \frac{\sum_t x_t y_t}{\sum_t x_t^2} \\ &= \beta + \frac{\sum_t x_t \eta_t}{\sum_t x_t^2} \\ &= \beta + \frac{\sum_t (\tilde{x}_t + v_t) (u_t - \beta v_t)}{\sum_t x_t^2} \\ &= \beta + \frac{\sum_t \tilde{x}_t u_t - \beta \sum_t \tilde{x}_t v_t + \sum_t v_t u_t - \beta \sum_t v_t^2}{\sum_t (\tilde{x}_t + v_t)^2} \\ &= \beta + \frac{\sum_t \tilde{x}_t u_t - \beta \sum_t \tilde{x}_t v_t + \sum_t v_t u_t - \beta \sum_t v_t^2}{\sum_t \tilde{x}_t^2 + 2 \sum_t \tilde{x}_t v_t + \sum_t v_t^2} \end{aligned}$$

Given the mutually independence of $\tilde{x}_t, u_t$, and v_t :

- $E[\tilde{x}_t u_t] = E[\tilde{x}_t] E[u_t] = 0$ because u_t has an expectation of zero.
- $E[\tilde{x}_t v_t] = E[\tilde{x}_t] E[v_t] = 0$ because v_t has an expectation of zero.
- $E[v_t u_t] = E[v_t] E[u_t] = 0$ because both u_t and v_t have an expectation of zero.

The Law of Large Numbers states that as the sample size goes to infinity, the sample average converges in probability to the expected value:

- $\frac{1}{T} \sum_t \tilde{x}_t u_t \rightarrow_p 0$
- $\frac{1}{T} \sum_t \tilde{x}_t v_t \rightarrow_p 0$
- $\frac{1}{T} \sum_t v_t u_t \rightarrow_p 0$

As the measurement error v , the regression error u and the true but unobserved regressor $\tilde{x}$ with finite second moment, we have

- $\frac{1}{T} \sum_t \tilde{x}_t^2 \rightarrow_p \sigma_{\tilde{x}}^2$
- $\frac{1}{T} \sum_t v_t^2 \rightarrow_p \sigma_v^2$

Combining the results and using **the continuous mapping theorem**, we find that the convergence in probability of $\hat{\beta}$ is given by:

$$\hat{\beta} \rightarrow_p \beta - \beta \frac{\sigma_v^2}{\sigma_{\tilde{x}}^2 + \sigma_v^2} = \beta \left(1 - \frac{\sigma_v^2}{\sigma_{\tilde{x}}^2 + \sigma_v^2} \right) = \beta \left(\frac{\sigma_{\tilde{x}}^2}{\sigma_{\tilde{x}}^2 + \sigma_v^2} \right) < \beta.$$

□

Comments:

There appears to be a typo in the original proof in the lecture notes provided. Given the question setup, the regression model only has one regressor and lacks scalar.

The LLN states that as the sample size T increases, the sample average of a set of independent and identically distributed (i.i.d.) random variables converges in probability to the expected value of those variables.

In this case, each $\tilde{x}_t$ is assumed to be an independent draw from a distribution with a finite variance $\sigma_{\tilde{x}}^2$. The term $\tilde{x}_t^2$ is the square of $\tilde{x}_t$, and $\frac{1}{T} \sum_t \tilde{x}_t^2$ is the sample average of these squared terms. According to the LLN, this sample average will converge in probability to the expected value of $\tilde{x}_t^2$, which is $\sigma_{\tilde{x}}^2$ (assuming that $\tilde{x}_t$ has zero mean, otherwise it would be $E[\tilde{x}_t^2]$, the second moment about the origin).

$$\frac{1}{T} \sum_t \tilde{x}_t^2 \rightarrow_p \sigma_{\tilde{x}}^2$$

This correctly indicates that the sample mean of the squares of the true regressor $\tilde{x}_t$ converges in probability to the variance of the true regressor $\tilde{x}_t$, provided that $\tilde{x}_t$ is drawn from a distribution with mean zero. If the $\tilde{x}_t$ are not zero-mean, then the convergence would be to the second moment $E[\tilde{x}_t^2]$, which includes both the variance and the square of the mean.



"This part's notes - my reaction upon reading them."

I have really, really tried my best to make it clear to you!

Instrumental Variable Estimator (IV):

General theory: A consistent estimator of β for the general model $y = X\beta + u$, when $\text{cov}(X, u) \neq 0$ can be obtained if we could find a matrix of instruments Z of order $T \times l$, with $l \geq k$ (more instruments than variables) such that:

- **R1: Relevance Condition** Variables in Z correlated with those in X and $\text{plim } Z'X/T = \Sigma_{ZX}$ finite and full rank (by column or row).
- **R2: Exogeneity Condition** Variables in Z are uncorrelated with the error term u . The population moment condition for this is that $\mathbb{E}[Z'u] = 0$. This implies that $\text{plim } \frac{Z'u}{T} = 0$

Comments:

- R1 states that the instruments are correlated with the endogenous variables in X . The population moment condition for this is that Σ_{ZX} , the limit of $\frac{Z'X}{T}$, is finite and of full rank. **This ensures that Z can explain some of the variations in X .**
- R2 requires that the instruments are uncorrelated with the error term u . The population moment condition for this is that $\mathbb{E}[Z'u] = 0$. This implies that $\text{plim } \frac{Z'u}{T} = 0$, which is **essential for consistency**.

We use the two-step procedure to get the new estimator, $\hat{\beta}_{IV}$

1. **Step 1** We first regress y on Z as well as X on Z obtaining

$$\hat{y} = Z(Z'Z)^{-1}Z'y = Z\hat{\delta}, \quad (96)$$

$$\hat{X} = Z(Z'Z)^{-1}Z'X = Z\hat{\Gamma} \quad (97)$$

2. **Step 2** We regress $\hat{y}$ on $\hat{X}$ getting

$$\hat{\beta}_{IV} = (\hat{X}'\hat{X})^{-1}\hat{X}'\hat{y} \quad (98)$$

Idea: by projecting (regressing) X on Z , hence creating $\hat{X}$, we are taking away the share of X related to u , making $\hat{\beta}_{IV}$ consistent!

To see why IV works, look at:

$$\hat{\beta}_{IV} = \beta + \left(\frac{\hat{X}'\hat{X}}{T}\right)^{-1} \frac{\hat{X}'\hat{u}}{T}. \quad (99)$$

where

$$\frac{\hat{X}'\hat{u}}{T} = \frac{X'Z(Z'Z)^{-1}Z'u}{T} = \frac{X'Z}{T} \left(\frac{Z'Z}{T}\right)^{-1} \frac{Z'u}{T} \quad (100)$$

I. Consistency:

$\hat{\beta}_{IV}$ converges in probability to β , making the IV estimator consistent.

II. Potential Unbiasedness:

An estimator is unbiased if the expected value of the estimator is equal to the true value of the parameter. Whether the IV estimator is unbiased depends on the finite sample properties. The IV estimator is not generally unbiased in finite samples; **it is consistent, which implies unbiasedness as the sample size approaches infinity (asymptotically unbiased)**. However, with a sufficiently large sample and strong instruments, the bias in finite samples may be negligible.

III. Relationship to BLUE:

The Gauss-Markov theorem states that in a linear regression model where the usual least squares assumptions hold (which include, among others, that the error term has zero mean, constant variance, and is uncorrelated with the regressors), the Ordinary Least Squares (OLS) estimator is the BLUE - the Best Linear Unbiased Estimator. However, when one or more regressors are endogenous (i.e., $\text{cov}(X, u) \neq 0$), these assumptions are violated, and OLS is no longer BLUE because it is biased and inconsistent. In such cases, the IV estimator can be a consistent (and thus potentially unbiased in the limit) alternative to OLS. However, **it does not inherit the BLUE property because the IV estimator does not minimize the variance among all linear unbiased estimators when there is endogeneity**.

Comments: To summarize, the IV estimator is designed to be consistent in the presence of endogeneity, provided that the instruments satisfy the relevance and exogeneity conditions. It does not necessarily have the minimum variance among all unbiased estimators, but when the OLS assumptions are violated due to endogeneity, IV can be a preferred method for consistent estimation.

Comments on l and k :

In the context of Instrumental Variables (IV) estimation, we define the instrumental variable matrix Z as a $T \times l$ matrix and the matrix of explanatory variables X as a $T \times k$ matrix, where T represents the number of observations, l the number of instruments, and k the number of explanatory variables. The relationship between l and k is critical for the identification of the model:

- **Over-Identified Case:** If $l > k$, there are more instruments than are necessary to identify the model. This situation allows for over-identification tests that can be used to assess the validity of the instruments.
- **Exactly-Identified Case:** If $l = k$, there are just enough instruments to identify the model. Each endogenous explanatory variable has a corresponding instrument, and there are no degrees of freedom to test the validity of the instruments without additional assumptions.
- **Under-Identified Case:** If $l < k$, the model is under-identified, meaning there are not enough instruments to identify the model. In this case, the IV estimation cannot be consistently estimated because there is not a unique solution to the instrumental variable equations.

For valid IV estimation, the number of instruments l must be at least equal to the number of endogenous explanatory variables among the k regressors. However, having more instruments than endogenous regressors (over-identification) is often desirable, as it allows for testing the exogeneity of the instruments with respect to the error term, which is a crucial assumption for the validity of an IV estimator.

My thoughts on First Stage Regression, Step 1: Why do we also regress Y on Z ?

Although traditionally the first stage of the Two-Stage Least Squares (2SLS) estimation involves only the endogenous explanatory variables, regressing y on Z can be interpreted as fitting the reduced form of the model, which may prove beneficial in certain analytical frameworks.

In our consideration of econometric methodologies, we occasionally encounter a procedure where the dependent variable y is regressed on an instrumental variable matrix Z . Although this is not a canonical component of the instrumental variables (IV) estimation technique, such a regression can offer a myriad of diagnostic benefits.

To begin with, it elucidates the direct relationship between the instruments and the dependent variable, which could furnish us with preliminary insights into the predictive power that the instrumental variables may wield over the outcome of interest. Next, the significance of the relationship between y and Z can serve as an initial test for the relevance of the instruments. While not a definitive test of instrument validity, which also requires the satisfaction of the exogeneity condition, it is a necessary precursor to further analysis.

Furthermore, such a regression can act as a pretest for the exclusion restriction. If the instruments are intended to affect the dependent variable exclusively through the endogenous explanatory variables, then any observed direct effect may signal a violation of this assumption. The strength of the instruments can be also appraised through the F-statistic derived from the regression. This is paramount since weak instruments could potentially compromise the IV estimation, leading to biased results.

In conclusion, while regressing y on Z diverges from the standard IV estimation approach, it can provide invaluable insights and checks that underpin the robustness and validity of the instrumental variable analysis. Careful interpretation in light of economic theory and the assumptions inherent in IV estimation is, however, imperative.

Question 1. The IV estimator - Lecture Notes 2

Show the instrumental variable estimator and show how it does simplify when the number of instruments equals the number of regressors.

Proof: To derive the IV estimator we start from the classic regression setup

$$y = X\beta + u.$$

and pre-multiply it by $Z(Z'Z)^{-1}Z'$ obtaining

$$Z(Z'Z)^{-1}Z'y = Z(Z'Z)^{-1}Z'X\beta + Z(Z'Z)^{-1}Z'u$$

If we denote

$$\hat{y} = Z(Z'Z)^{-1}Z'y, \quad \hat{X} = Z(Z'Z)^{-1}Z'X \text{ and } \hat{u} = Z(Z'Z)^{-1}Z'u$$

we have that

$$\hat{y} = \hat{X}\beta + \hat{u}.$$

The instrumental variable estimator can be obtained by applying OLS on this modified model:

$$\begin{aligned} \hat{\beta}_{IV} &= (\hat{X}'\hat{X})^{-1}\hat{X}'\hat{y} \\ &= \left[\left(Z(Z'Z)^{-1}Z'X \right)' Z(Z'Z)^{-1}Z'X \right]^{-1} \left(Z(Z'Z)^{-1}Z'X \right)' Z(Z'Z)^{-1}Z'y \\ &= \left[X'Z(Z'Z)^{-1}Z'Z(Z'Z)^{-1}Z'X \right]^{-1} X'Z(Z'Z)^{-1}Z'Z(Z'Z)^{-1}Z'y \\ &= \left[X'Z(Z'Z)^{-1}Z'X \right]^{-1} X'Z(Z'Z)^{-1}Z'y \end{aligned} \quad (101)$$

This is how the IV estimator looks in general. If the number of instruments l equals the number of regressors k (i.e. $l = k$) then the product $Z'X$ is a square matrix of dimension $k \times k$ (or $l \times l$), which is non-singular (i.e. invertible)^a. Recall that for two invertible matrices A and B the following holds

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Therefore, we can re-write the term in square brackets in (98) as

$$\begin{aligned} \hat{\beta}_{IV} &= \left[X'Z(Z'Z)^{-1}Z'X \right]^{-1} X'Z(Z'Z)^{-1}Z'y \\ &= (Z'X)^{-1} \left((Z'Z)^{-1} \right)^{-1} (X'Z)^{-1} X'Z(Z'Z)^{-1}Z'y \\ &= (Z'X)^{-1} (Z'Z) (Z'Z)^{-1}Z'y \\ &= (Z'X)^{-1}Z'y. \end{aligned} \quad (102)$$

This represents the IV estimator when the number of instruments equals the number of regressors.

^aOne of the assumptions underlying IV estimation, R1 is that $\frac{Z'X}{T} \xrightarrow{P} \Sigma$, where Σ is finite and full rank. This plus the fact that the matrix is square ensure invertibility

Question 2. The IV estimator - Sample Exam Question

Consider the following generalised linear regression model with stochastic regressors:

$$\begin{aligned}y &= X\beta + \varepsilon \\E(\varepsilon | X) &\neq 0 \\E(\varepsilon | Z) &= 0 \\\text{var}(\varepsilon | X) &= \sigma^2 I \\\text{rank}(X) &= k\end{aligned}$$

where X has dimension $n \times k$ and Z has dimension $n \times l$.

- Assume that $l = k$ and derive the IV estimator of β .
- Making any assumption you need, show that this IV estimator is consistent.
- Is it also unbiased?

Proof:

- As shown in the previous exercise the IV estimator of β when the number of instruments l equals the number of regressors k is

$$\hat{\beta}_{IV} = (Z'X)^{-1} Z'y \quad (103)$$

- To check whether the estimator is consistent, let us start from the IV estimator in (10) and substitute $y = X\beta + \varepsilon$

$$\begin{aligned}\hat{\beta}_{IV} &= (Z'X)^{-1} Z'(X\beta + \varepsilon) \\&= (Z'X)^{-1} Z'X\beta + (Z'X)^{-1} Z'\varepsilon \\&= \beta + (Z'X)^{-1} Z'\varepsilon \\&= \beta + \left(\frac{Z'X}{T}\right)^{-1} \frac{Z'\varepsilon}{T}\end{aligned} \quad (104)$$

In order to obtain a consistent estimator for β we need to impose some assumptions. In particular

- R1: The instrumental variables (Z) should be correlated (as much as possible) with the regressors i.e. $\frac{Z'X}{T} \xrightarrow{p} \Sigma_{XZ}$, where Σ_{XZ} is finite and full rank.
- R2: The instrumental variables (Z) should be uncorrelated with the error terms i.e. $\frac{Z'\varepsilon}{T} \xrightarrow{p} 0$.

The above assumptions imply that the IV estimator for β is consistent i.e.

$$\hat{\beta}_{IV} \xrightarrow{p} \beta.$$

- To check whether the estimator is unbiased let us take expectation of (101) and re-write it as

$$\begin{aligned}E(\hat{\beta}_{IV}) &= E\left[\beta + (Z'X)^{-1} Z'\varepsilon\right] \\&= \beta + E\left[(Z'X)^{-1} Z'\varepsilon\right]\end{aligned} \quad (105)$$

Remember that both X and Z are now stochastic, which means that we cannot just take them out from the expectation in (102). Using the law of iterated expectation, we can write

$$\begin{aligned}E(\hat{\beta}_{IV}) &= \beta + E\left[(Z'X)^{-1} Z'\varepsilon\right] \\&= \beta + E\left[E\left[(Z'X)^{-1} Z'\varepsilon \mid X, Z\right]\right] \\&= \beta + E\left[(Z'X)^{-1} Z'E[\varepsilon \mid X, Z]\right]\end{aligned} \quad (106)$$

From (103) it is clear that the estimator $\hat{\beta}_{IV}$ will be unbiased only if $E[\varepsilon \mid X, Z] = 0$. The problem is that, in general, we do not know what $E[\varepsilon \mid X, Z]$ is equal to. In practice, it could be different from zero, making the estimator potentially biased in finite sample.

See next page for more proof.

Question 2. The IV estimator - Sample Exam Question

The only thing we can conclude is that the estimator is asymptotically unbiased. Indeed, taking the limit and recognize that, as $T \rightarrow \infty$,

$$\frac{Z'X}{T} \xrightarrow{p} \Sigma_{XZ} \quad \text{which is a constant full rank (not random) matrix}$$

$$\frac{Z'\varepsilon}{T} \xrightarrow{p} 0$$

equation in (102) can be written as

$$\begin{aligned} E(\hat{\beta}_{IV}) &= \beta + E\left[\left(\frac{Z'X}{T}\right)^{-1} \frac{Z'\varepsilon}{T}\right] \\ &\xrightarrow{p} \beta + E\left[\Sigma_{XZ}^{-1} \cdot 0\right] \\ &\xrightarrow{p} \beta \end{aligned}$$

So the estimator is asymptotically unbiased, that is $\lim_{T \rightarrow \infty} E[\hat{\beta}_{IV}] = \beta$.

Question 3. Measurement error - Sample Exam Question

Consider the simple regression model:

$$y_t = \alpha + \beta x_t + u_t$$

and assume that all the Gauss Markov assumptions are satisfied. Assume furthermore that $u_t \sim iid(0, \sigma_u^2)$. Unfortunately we can't observe y_t but instead we observe $\tilde{y}_t$ where

$$\tilde{y}_t = y_t + v_t$$

and where $v_t \sim iid(0, \sigma_v^2)$ and it is also independent from u_t and x_t . Provide the definition of consistency of the OLS estimator of β . Is the estimator of β in the above model consistent? Carefully state any extra assumption you need in your proof. What can you conclude about the presence of measurement error in the dependent variable?

Comments:

From your lecture notes you have already seen what happens to the OLS estimator when one of your dependent variable(s) in the regression model is measured with error (see Lecture Notes 2, p. 31). In this case we are going to see a different case of the errors in variable problem, namely when the measurement error is in the dependent variable, y_t .

Proof:

Let us first provide the definition of consistency of the OLS estimator of β . We will say that the OLS estimator $\hat{\beta}$ is a consistent estimator of β if it will converge in probability to the true value of the parameter:

$$\hat{\beta} \xrightarrow{p} \beta.$$

Now we want to verify if this property is still satisfied in the case of measurement error in the dependent variable. Let us start from the true model and rewrite it in terms of the observable value $\tilde{y}_t$. We have

$$\begin{aligned} y_t &= \alpha + \beta x_t + u_t \\ \Rightarrow (\tilde{y}_t - v_t) &= \alpha + \beta x_t + u_t \\ \Rightarrow \tilde{y}_t &= \alpha + \beta x_t + \underbrace{(u_t + v_t)}_{\eta_t} \\ \Rightarrow \tilde{y}_t &= \alpha + \beta x_t + \eta_t \end{aligned} \quad (107)$$

Let us now estimate β in (104) using OLS. Do you remember what the expression for $\hat{\beta}_{OLS}$ is in the case of a simple regression model with just two parameters? We derived it in our previous class:

$$\begin{aligned} \hat{\beta}_{OLS} &= \frac{\sum_{t=1}^T (x_t - \bar{x})(\tilde{y}_t - \bar{\tilde{y}})}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= \frac{\sum_{t=1}^T (x_t - \bar{x})((\beta(x_t - \bar{x}) + \eta_t))}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= \frac{\beta \sum_{t=1}^T (x_t - \bar{x})^2 + \sum_{t=1}^T (x_t - \bar{x}) \eta_t}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= \beta + \frac{\sum_{t=1}^T (x_t - \bar{x}) \eta_t}{\sum_{t=1}^T (x_t - \bar{x})^2} \\ &= \beta + \frac{\sum_{t=1}^T (x_t - \bar{x}) u_t}{\sum_{t=1}^T (x_t - \bar{x})^2} + \frac{\sum_{t=1}^T (x_t - \bar{x}) v_t}{\sum_{t=1}^T (x_t - \bar{x})^2} \end{aligned} \quad (108)$$

Now, if we take the limit of (105) as $T \rightarrow \infty$, we have

$$\hat{\beta}_{OLS} \xrightarrow{p} \beta + \frac{\text{Cov}(x_t, u_t)}{\text{Var}(x_t)} + \frac{\text{Cov}(x_t, v_t)}{\text{Var}(x_t)} \quad (109)$$

We can now use the assumption that v_t is independent of x_t , which implies that $\text{Cov}(x_t, v_t) = 0$. We also know from the exercise that all the Gauss Markov assumptions are satisfied. That is, we also know that x_t is uncorrelated with the error term u_t , which implies that $\text{Cov}(x_t, u_t) = 0$. Then, using these two assumptions, (106) reduces to

$$\hat{\beta}_{OLS} \xrightarrow{p} \beta$$

which implies that the OLS estimator of β is still consistent. Hence, we can conclude that when the errors-in-variable problem affects the dependent variable y_t and the Gauss Markov assumptions hold, the OLS estimator will still retain the property of consistency.

More proof on the next page!

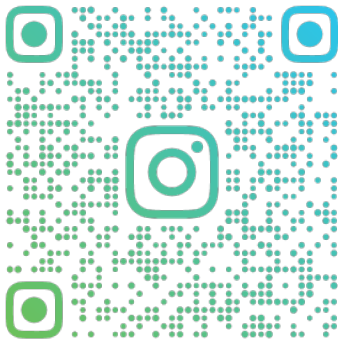
Question 3. Measurement error - Sample Exam Question

Comments on (108) Step 2: Now we know that:

$$\begin{aligned}\tilde{y}_t &= y_t + v_t = \alpha + \beta x_t + u_t + v_t \\ \tilde{y}_t - \bar{\tilde{y}} &= \beta (x_t - \bar{x}) + (u_t + v_t) - \overline{(u + v)}\end{aligned}$$

Since $\overline{(u + v)}$ is the mean of the error terms and is zero under the assumptions, the equation simplifies to:

$$\tilde{y}_t - \bar{\tilde{y}} = \beta (x_t - \bar{x}) + (u_t + v_t)$$



REEL SHARED ON 10月 4, 2023
BY @COLLEGEFESSING

Sharing these funny memes with you all.

Life wouldn't be easy, but we are in this together!

Cheers!

The Maximum Likelihood Estimation Method:

Idea: Maximum Likelihood Estimation (MLE) is a statistical method used for estimating the parameters, θ of a statistical model. The core idea behind MLE is to choose parameter values for the model that make the observed data most probable. In other words, MLE finds the parameter values that maximize the likelihood function, which is a function of the parameters given the data.

Question Setup: Let $y = (y_1, \dots, y_T)$ be sample of dimension T of a rv with known pdf $f(y; \theta)$ that depends on unknown parameter. θ , of finite dimension $k \times 1$ is called the true parameter vector. We want to apply MLE to estimate the $\hat{\theta}_{MLE}$.

Assumptions:

1. Well-Specified Model: This includes having a clear understanding of the distribution that the data is assumed to follow (e.g., normal, binomial, Poisson) and the relationship between the variables. (The following is given to us :)

$$f(y; \theta) = f(y_1; \theta) f(y_2 | y_1; \theta) \cdots f(y_T | y_{T-1} \dots y_1; \theta) \quad (110)$$

where $f(y_t | y_{t-1} \dots y_1; \theta_0)$ is the conditional pdf.

2. Independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$: For simplicity, the Classic Assumption is used. We may further simplify (110) to

$$f(y; \theta) = f(y_1; \theta) f(y_2; \theta) \cdots f(y_T; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (111)$$

How MLE Works:

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (112)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{t=1}^T \log f(y_t; \theta) \quad (113)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta) \quad (114)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (115)$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (116)$$

$$H = \begin{bmatrix} \frac{\partial^2 l}{\partial \theta_1^2} & \frac{\partial^2 l}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 l}{\partial \theta_1 \partial \theta_k} \\ \frac{\partial^2 l}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 l}{\partial \theta_2^2} & \cdots & \frac{\partial^2 l}{\partial \theta_2 \partial \theta_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 l}{\partial \theta_k \partial \theta_1} & \frac{\partial^2 l}{\partial \theta_k \partial \theta_2} & \cdots & \frac{\partial^2 l}{\partial \theta_k^2} \end{bmatrix} \quad (117)$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (118)$$

Comments: The Fisher Information for a parameter in a statistical model gives an idea of how much information the data carries about that parameter. When we have a sample of independent and identically distributed (iid) observations, the Fisher Information is the sum of the information provided by each individual observation. This scaling by the number of observations is a general property of the Fisher Information for iid samples: the information accumulates linearly with the number of observations. Hence, more observations provide more information about the parameter, which is reflected in the larger Fisher Information.

Comments:

$$L(\theta) \equiv f(y; \theta)$$

This is given by the lecture notes, **it is not a typo** as y is the T by 1 vector which collects all the iid observations of random variable. Thus $f(y; \theta)$ is the joint distribution of all the y .

The properties of MLE:

Under regularity conditions, as $T \rightarrow \infty$,

1. Consistency of MLE:

$$\hat{\theta}_{MLE} \rightarrow_p \theta \quad (119)$$

2. Asymptotic normality of MLE

$$\sqrt{T} (\hat{\theta}_{MLE} - \theta) \rightarrow_d N(0, \mathcal{I}^{-1}(\theta)) \quad (120)$$

where $\mathcal{I}(\theta) = \text{plim} \frac{I(\theta)}{T}$.

3. $\hat{\theta}_{MLE}$ is asymptotically efficient and invariant:

- Asymptotic efficiency means that for any other estimator with acm (asymptotically covariance matrix) V then the difference $V - \mathcal{I}^{-1}(\theta)$ is 'non-negative' matrix (more formally, it is positive semi-definite). MLE is more precise than anything else! This is known as the Cramer-Rao result.
- Invariance means that the MLE of $g(\theta)$, for any continuously differentiable function $g(\cdot)$ is $g(\hat{\theta}_{MLE})$. As a result, there is no need to perform another numerical optimization!

Example 1 of MLE: Normal Distribution

Assumptions:

Suppose we have independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$ and the probability density function (PDF) for a single observation y_t is given by

$$f(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right). \quad (121)$$

Let us derive the MLE.

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (122)$$

$$L(\mu, \sigma^2 | y_1, y_2, \dots, y_T) = \prod_{t=1}^T \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_t - \mu)^2}{2\sigma^2}\right) \quad (123)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{t=1}^T \log f(y_t; \theta) \quad (124)$$

$$\log L(\mu, \sigma^2 | y_1, y_2, \dots, y_T) = -\frac{T}{2} \log(2\pi) - T \log(\sigma) - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \mu)^2 \quad (125)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta) \quad (126)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

$$\hat{\mu} = \frac{1}{T} \sum_{i=1}^T y_i, \quad \hat{\sigma}^2 = \frac{1}{T} \sum_{i=1}^T (y_i - \hat{\mu})^2 \quad (127)$$

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (128)$$

$$\frac{\partial \log L}{\partial \mu} = \frac{1}{\sigma^2} \sum_{t=1}^T (y_t - \mu), \quad \frac{\partial \log L}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{t=1}^T (y_t - \mu)^2 \quad (129)$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (130)$$

$$H(\mu, \sigma^2) = \begin{pmatrix} -\frac{T}{\sigma^2} & 0 \\ 0 & \frac{T}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{t=1}^T (y_t - \mu)^2 \end{pmatrix} \quad (131)$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (132)$$

$$I(\mu, \sigma^2) = \begin{pmatrix} \frac{T}{\sigma^2} & 0 \\ 0 & \frac{T}{2\sigma^4} \end{pmatrix} \quad (133)$$

Comments on the Hessian of the Gaussian likelihood: no cross-derivative effect between mean and variance

The Hessian of the Gaussian likelihood is a matrix of second derivatives of the log-likelihood function with respect to the parameters of a Gaussian distribution. In the context of Gaussian distributions, the parameters are typically the mean (μ) and the variance (σ^2).

For a Gaussian distribution, the probability density function (PDF) is given by:

$$p(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The log-likelihood ($\mathcal{L}$) of observing a set of data points $\{x_1, x_2, \dots, x_n\}$ under this distribution is:

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

The Hessian matrix $\mathbf{H}$ of this log-likelihood with respect to the parameters μ and σ^2 is a 2×2 matrix where each element is a second partial derivative of $\mathcal{L}$ with respect to the parameters:

$$\begin{bmatrix} \frac{\partial^2 \mathcal{L}}{\partial \mu^2} & \frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2} \\ \frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu} & \frac{\partial^2 \mathcal{L}}{\partial \sigma^4} \end{bmatrix}$$

The specific elements of this Hessian matrix are:

1. $\frac{\partial^2 \mathcal{L}}{\partial \mu^2} = -\frac{n}{\sigma^2}$ - This is the second derivative with respect to the mean, indicating how the curvature of the log-likelihood changes as the mean changes.
2. $\frac{\partial^2 \mathcal{L}}{\partial \sigma^4} = -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6} \sum_{i=1}^n (x_i - \mu)^2$ - This is the second derivative with respect to the variance, reflecting the curvature of the log-likelihood as the variance changes.
3. The off-diagonal terms $\frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2}$ and $\frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu}$ are typically zero for the Gaussian likelihood, indicating no cross-derivative effect between mean and variance.

To understand why the cross-derivative terms are zero in the Hessian matrix of the Gaussian likelihood, let's delve into the derivation of these terms. The cross-derivative terms we're interested in are $\frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2}$ and $\frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu}$. By the symmetry of mixed partial derivatives (Clairaut's theorem), these two should be equal, so we only need to calculate one of them. We start with the log-likelihood function for a Gaussian distribution:

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

1. First, compute the first partial derivative with respect to μ :

$$\frac{\partial \mathcal{L}}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

2. Next, compute the first partial derivative of this expression with respect to σ^2 :

$$\frac{\partial}{\partial \sigma^2} \left(\frac{\partial \mathcal{L}}{\partial \mu} \right) = \frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \right)$$

To perform this derivative, note that the sum $\sum_{i=1}^n (x_i - \mu)$ does not depend on σ^2 , so it can be treated as a constant with respect to σ^2 . Therefore, we are essentially differentiating a constant multiplied by $\frac{1}{\sigma^2}$, which gives:

$$\frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \right) = -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)$$

3. Observation: The key point here is that this derivative is zero under the assumption of the Gaussian distribution where the data is centered around the mean (μ). That is, if you sum up $x_i - \mu$ over all i , you essentially get a sum that tends to zero, especially as the number of data points increases, because the mean μ is the point where this sum is minimized (by definition of the mean in a Gaussian distribution).

Therefore, the cross-derivative term $\frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2}$ (and consequently $\frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu}$) is effectively zero for a Gaussian distribution. This result is particularly pronounced when dealing with a large number of data points, as the sum $\sum_{i=1}^n (x_i - \mu)$ becomes increasingly close to zero, reinforcing the assumption that these cross-derivative terms do not contribute to the curvature of the likelihood function in the parameter space.

Example 2 of MLE: Exponential Distribution

Assumptions:

Suppose we have independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$ and the probability density function (PDF) for a single observation y_t is given by

$$f(y|\lambda) = \lambda \exp(-\lambda y) \quad (134)$$

Let us derive the MLE.

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (135)$$

$$L(\lambda|y_1, y_2, \dots, y_T) = \prod_{t=1}^T \lambda \exp(-\lambda y_t) \quad (136)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{t=1}^T \log f(y_t; \theta) \quad (137)$$

$$\log L(\lambda|y_1, y_2, \dots, y_T) = T \log(\lambda) - \lambda \sum_{t=1}^T y_t \quad (138)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta) \quad (139)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

$$\hat{\lambda} = \frac{T}{\sum_{i=1}^T y_i} \quad (140)$$

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (141)$$

$$\frac{\partial \log L}{\partial \lambda} = \frac{T}{\lambda} - \sum_{t=1}^T y_t \quad (142)$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (143)$$

$$H(\lambda) = -\frac{T}{\lambda^2} \quad (144)$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (145)$$

$$I(\lambda) = \frac{T}{\lambda^2} \quad (146)$$

Comments: The Fisher Information for a parameter in a statistical model gives an idea of how much information the data carries about that parameter. In the case of the exponential distribution, the Fisher Information for a single observation for the parameter λ is indeed $\frac{1}{\lambda^2}$. However, when we have a sample of independent and identically distributed (iid) observations, the Fisher Information is the sum of the information provided by each individual observation.

For T iid observations from an exponential distribution, the Fisher Information for the parameter λ is the sum of the Fisher Information for each observation. Since each observation contributes $\frac{1}{\lambda^2}$ to the information, the total Fisher Information for T observations is $T \times \frac{1}{\lambda^2}$, which is $\frac{T}{\lambda^2}$.

This scaling by the number of observations is a general property of the Fisher Information for iid samples: the information accumulates linearly with the number of observations. Hence, more observations provide more information about the parameter, which is reflected in the larger Fisher Information.

Example 3 of MLE: Binomial Distribution

Assumptions:

Suppose we have independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$ and the probability mass function (PMF) for a single observation y_t is given by

$$P(y|n, p) = \binom{n}{y} p^y (1-p)^{n-y} \quad (147)$$

Let us derive the MLE.

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (148)$$

$$L(p|y_1, y_2, \dots, y_T) = \prod_{t=1}^T \binom{n}{y_t} p^{y_t} (1-p)^{n-y_t} \quad (149)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{t=1}^T \log f(y_t; \theta) \quad (150)$$

$$\log L(p|y_1, y_2, \dots, y_T) = \sum_{t=1}^T \left[\log \binom{n}{y_t} + y_t \log(p) + (n - y_t) \log(1-p) \right] \quad (151)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta) \quad (152)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

$$\hat{p} = \frac{\sum_{i=1}^T y_i}{nT} \quad (153)$$

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (154)$$

$$\frac{\partial \log L}{\partial p} = \sum_{t=1}^T \left[\frac{y_t}{p} - \frac{n - y_t}{1-p} \right] \quad (155)$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (156)$$

$$H(p) = - \sum_{t=1}^T \left[\frac{y_t}{p^2} + \frac{n - y_t}{(1-p)^2} \right] \quad (157)$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (158)$$

$$I(p) = \frac{nT}{p(1-p)} \quad (159)$$

Example 4 of MLE: Bernoulli Distribution

Assumptions:

Suppose we have independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$ and the probability mass function (PMF) for a single observation y_i is given by

$$P(y|p) = p^y(1-p)^{1-y} \quad (160)$$

Let us derive the MLE.

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{i=1}^T f(y_i; \theta) \quad (161)$$

$$L(p|y_1, y_2, \dots, y_T) = \prod_{i=1}^T p^{y_i} (1-p)^{1-y_i} \quad (162)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{i=1}^T \log f(y_i; \theta) \quad (163)$$

$$\log L(p|y_1, y_2, \dots, y_T) = \sum_{i=1}^T [y_i \log(p) + (1-y_i) \log(1-p)] \quad (164)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta) \quad (165)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

$$\hat{p} = \frac{1}{T} \sum_{i=1}^T y_i \quad (166)$$

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (167)$$

$$\frac{\partial \log L}{\partial p} = \sum_{i=1}^T \left[\frac{y_i}{p} - \frac{1-y_i}{1-p} \right] \quad (168)$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (169)$$

$$H(p) = - \sum_{i=1}^T \left[\frac{y_i}{p^2} + \frac{1-y_i}{(1-p)^2} \right] \quad (170)$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (171)$$

$$I(p) = \frac{T}{p(1-p)} \quad (172)$$

Example 5 of MLE: Uniform Distribution

Assumptions:

Suppose we have independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$ and the probability mass function (PMF) for a single observation y_i is given by

$$f(y|a, b) = \frac{1}{b-a} \text{ for } a \leq y \leq b \quad (173)$$

For a uniform distribution with parameters a and b , we assume a and b are independent. Let us derive the MLE.

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (174)$$

$$L(a, b|y_1, y_2, \dots, y_T) = \prod_{t=1}^T \frac{1}{b-a} \quad (175)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{t=1}^T \log f(y_t; \theta) \quad (176)$$

$$\log L(a, b|y_1, y_2, \dots, y_T) = -T \log(b-a) \quad (177)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta) \quad (178)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

In this case we cannot solve the problem in the standard way (i.e by setting the FOCs equal to zero), since we have a constraint on θ : $a \leq y_t \leq b \quad \forall t = 1, \dots, T$. However, there is an easy way to derive the MLE for θ_1 and θ_2 in this case. Look at the expression in (178), maximizing (178) is also equivalent to minimizing

$$(b-a) \quad a \leq y_t \leq b$$

Then, given the constraint that $a \leq y_t \leq b \quad \forall t = 1, \dots, T$, the above expression is minimized when

$$\hat{a} = \min\{y_1, y_2, \dots, y_T\}, \quad \hat{b} = \max\{y_1, y_2, \dots, y_T\} \quad (179)$$

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (180)$$

$$\frac{\partial \log L}{\partial a} = \frac{T}{b-a}, \quad \frac{\partial \log L}{\partial b} = -\frac{T}{b-a} \quad (181)$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (182)$$

$$H(a, b) = \begin{pmatrix} \frac{T}{(b-a)^2} & -\frac{T}{(b-a)^2} \\ -\frac{T}{(b-a)^2} & \frac{T}{(b-a)^2} \end{pmatrix} \quad (183)$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (184)$$

$$I(a, b) = \begin{pmatrix} -\frac{T}{(b-a)^2} & \frac{T}{(b-a)^2} \\ \frac{T}{(b-a)^2} & -\frac{T}{(b-a)^2} \end{pmatrix} \quad (185)$$

Example 6 Maximum Likelihood Estimation: regression example

Assumptions:

Suppose we have independent and identically distributed (iid) observations $y_1, y_2, \dots, y_T$ and store them into a T by 1 vector, y . Consider linear regression of model

$$y = X\beta_0 + u, u \sim N(0, \sigma_0^2 I),$$

where now we denote by β_0 and σ_0^2 the true values. Assume that Gauss-Markov conditions^a hold. Let us derive the MLE.

^a

1. **Linearity:** The model specifies a linear relationship between Y and $X_2, \dots, X_k$, that is in (1).
2. **The matrix X is non-stochastic and full rank:** We can treat X_i simply as a matrix of known constants and there must be no exact linear relationship among any of the independent variables in the model.

3. **Homoskedasticity and no autocorrelation:**

$$E[u] = 0 \quad (186)$$

$$\text{Var}(u) = E[uu'] = \sigma^2 I \quad (187)$$

where I is the $T \times T$ identity matrix.

Given $u \sim N(0, \sigma_0^2 I)$, then y is also multi-normal with mean and variance

$$E(y | X) = X\beta_0, \quad \text{var}(y | X) = \sigma_0^2 I.$$

Therefore,

$$y | X \sim N(X\beta_0, \sigma_0^2 I).$$

The pdf for a multivariate normal distribution $N(\mu, \Sigma)$ where μ is the mean vector and Σ is the covariance matrix, for a vector x is:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)' \Sigma^{-1} (x - \mu)\right) \quad (188)$$

In our case, $\mu = X\beta_0$ and $\Sigma = \sigma_0^2 I$, and $x = y$. Where T is the number of observations.

Step 1: Set up the likelihood function

$$L(\theta) \equiv f(y; \theta) = \prod_{t=1}^T f(y_t; \theta) \quad (189)$$

So the likelihood function $L(\theta; y)$ for arbitrary $\theta = (\sigma^2, \beta')'$ is:

$$L(\theta; y) = \frac{1}{(2\pi\sigma^2)^{T/2}} \exp\left(-\frac{1}{2\sigma^2} (y - X\beta)' (y - X\beta)\right) \quad (190)$$

Step 2: Take logarithm and get Log likelihood function:

$$l(\theta) = \log L(\theta) = \log f(y_1; \theta) + \log f(y_2; \theta) + \dots + \log f(y_T; \theta) = \sum_{t=1}^T \log f(y_t; \theta) \quad (191)$$

$$l(\theta) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (y - X\beta)' (y - X\beta) \quad (192)$$

Step 3: Solve the MLE Problem:

The MLE is defined by

$$\hat{\theta}_{MLE} = \arg\max_{\theta \in \Theta} l(\theta) \quad (193)$$

where Θ included in R^k given θ is a k by 1 true parameter vector.

$$\hat{\beta} = (X'X)^{-1} X'y, \quad \hat{\sigma}^2 = (y - X\hat{\beta})' (y - X\hat{\beta}) / T. \quad (194)$$

Check out the next page!

Example 6 Maximum Likelihood Estimation: regression example

Assuming $l(\cdot)$ is twice differentiable, we have the following

1. The score vector:

$$s(\theta) \equiv \frac{\partial l(\theta)}{\partial \theta} \quad (195)$$

$$\frac{\partial l(\hat{\theta})}{\partial \beta} = -\frac{1}{\hat{\sigma}^2} (-X'y + X'X\hat{\beta}) \quad (196)$$

$$\frac{\partial l(\hat{\theta})}{\partial \sigma^2} = -\frac{T}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4} (y - X\hat{\beta})'(y - X\hat{\beta}),$$

2. The Hessian matrix:

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \quad (197)$$

$$\frac{\partial^2 l(\theta)}{\partial \beta \partial \beta'} = -\frac{X'X}{\sigma^2},$$

$$\frac{\partial^2 l(\theta)}{\partial \beta \partial \sigma^2} = -\frac{X'u}{\sigma^4}, \quad (198)$$

$$\frac{\partial^2 l(\theta)}{\partial^2 \sigma^2} = \frac{T}{2\sigma^4} - \frac{u'u}{\sigma^6}.$$

3. The Fisher information matrix:

$$I(\theta) \equiv -EH(\theta) \quad (199)$$

$$E\left(\frac{\partial^2 l(\theta)}{\partial \beta \partial \beta'} \mid X\right) = -\frac{X'X}{\sigma^2}$$

$$E\left(\frac{\partial^2 l(\theta)}{\partial \beta \partial \sigma^2} \mid X\right) = -\frac{E(X'u \mid X)}{\sigma^4} \quad (200)$$

$$E\left(\frac{\partial^2 l(\theta)}{\partial^2 \sigma^2} \mid X\right) = \frac{T}{2\sigma^4} - \frac{E(u'u \mid X)}{\sigma^6}.$$

Since $E(X'u \mid X) = X'E(u) = 0$, $E(u'u \mid X) = T\sigma^2$ substituting and changing sign gives

$$I(\theta) = \frac{1}{\sigma^2} \begin{pmatrix} X'X & 0 \\ 0 & \frac{T}{2\sigma^2} \end{pmatrix} \quad (201)$$

Dividing by T and taking the limit

$$\frac{I(\theta)}{T} = \frac{1}{\sigma^2} \begin{pmatrix} \frac{X'X}{T} & 0 \\ 0 & \frac{1}{2\sigma^2} \end{pmatrix} \rightarrow_p \mathcal{I}(\theta_0) = \frac{1}{\sigma^2} \begin{pmatrix} Q & 0 \\ 0 & \frac{1}{2\sigma^2} \end{pmatrix}.$$

It gives the asymptotic covariance matrix (the precision!) of the MLE:

$$\mathcal{I}^{-1}(\theta_0) = \sigma^2 \begin{pmatrix} Q^{-1} & 0 \\ 0 & 2\sigma^2 \end{pmatrix}.$$

Under regularity conditions, as $T \rightarrow \infty$,

1. Consistency of MLE:

$$\hat{\theta}_{MLE} \rightarrow_p \theta \quad (202)$$

2. Asymptotic normality of MLE

$$\sqrt{T}(\hat{\theta}_{MLE} - \theta) \rightarrow_d N(0, \mathcal{I}^{-1}(\theta)) \quad (203)$$

where $\mathcal{I}(\theta) = \text{plim} \frac{I(\theta)}{T}$.

3. $\hat{\theta}_{MLE}$ is asymptotically efficient and invariant:

- Asymptotic efficiency means that for any other estimator with acm (asymptotically covariance matrix) V then the difference $V - \mathcal{I}^{-1}(\theta)$ is 'non-negative' matrix (more formally, it is positive semi-definite). MLE is more precise than anything else! This is known as the Cramer-Rao result.
- Invariance means that the MLE of $g(\theta)$, for any continuously differentiable function $g(\cdot)$ is $g(\hat{\theta}_{MLE})$. As a result, there is no need to perform another numerical optimization!

Define the Generalized Method of Moment estimation (GMM) principle:

Assumption:

Often, financial and economic theory leads to orthogonality conditions describing some aspects of a given economic model, such as:

$$Eg(y_t, X_t, \theta_0) = 0. \quad (204)$$

Here y and X are the endogenous and exogenous variables respectively, θ_0 a k -dimensional vector of parameters and $g(\cdot)$ a L -dimensional vector, with $L \geq k$. **Setup:**

The GMM estimator is defined as:

$$\hat{\theta}_{GMM} \equiv \operatorname{argmin}_{\theta \in \Theta} \hat{g}'(y, X, \theta) W_T \hat{g}(y, X, \theta), \quad (205)$$

where

$$\hat{g}(y, X, \theta) \equiv \frac{1}{T} \sum_{t=1}^T g(y_t, X_t, \theta) \quad (206)$$

and W_T is a $L \times L$ matrix (independent of θ), such that

$$W_T \rightarrow_p W > 0. \quad (207)$$

Under regularity conditions, we have

1. $\hat{\theta}_{GMM} \rightarrow_p \theta_0$
2. $\sqrt{T}(\hat{\theta}_{GMM} - \theta_0) \rightarrow_d N\left(0, (G'WG)^{-1} G'W\Omega WG (G'WG)^{-1}\right)$

setting

$$G = E \left. \frac{\partial g(y_t, X_t, \theta)}{\partial \theta'} \right|_{\theta=\theta_0}, \quad (208)$$

$$\Omega = E g(y_t, X_t, \theta_0) g(y_t, X_t, \theta_0)'$$

Comments: Make sure you know these two steps. The idea behind the GMM principle is very simple and it is basically based on the Law of Large Numbers. Let θ be the parameter of interest. To derive the GMM estimator of θ , $\hat{\theta}_{GMM}$:

1. The first thing to do is identifying a set of equations which relate the population moment (e.g, the expected value, the variance, etc.) to the parameter of interest θ .
2. We then replace the population moments by the sample counterparts (e.g, sample mean, sample variance, etc.) and derive expressions for the parameter θ as functions of the sample moments.

Comments:

(Not required by this Class)

Some of you have expressed interest in understanding how the Method of Moments (MM) can be practically applied in the real world. To provide you some intuition, you may go through the survey, 'Applications of Generalized Method of Moments Estimation' written by Prof. Jeffrey M. Wooldridge which has been uploaded to the Insendi File folder.

The PE teacher when he had to
watch over a math exam



Question 1. The MLE - Lecture Notes 4

Consider the Bernoulli rv y , which takes value 1 with probability θ and zero with probability $1 - \theta$. Verify that $E(y) = \theta$ and $\text{var}(y) = \theta(1 - \theta)$. Find the MLE of θ when a random sample is available. Find the acm of the MLE.

Proof: To verify the expressions for the mean and variance of the random variable y it is convenient to summarize its possible realizations as

$$y = \begin{cases} 1 & \text{with prob. } \theta \\ 0 & \text{with prob. } 1 - \theta \end{cases}$$

The expected value of the variable is then easily estimated as the probability weighted average of all its possible realizations. Hence, we have that the expected value of y is given by

$$E(y) = 1 \cdot \theta + 0 \cdot (1 - \theta) = \theta$$

The variance is obtained similarly as

$$\begin{aligned} \text{var}(y) &= E(y^2) - E(y)^2 \\ &= (1^2 \cdot \theta + 0^2 \cdot (1 - \theta)) - \theta^2 \\ &= \theta - \theta^2 \\ &= \theta(1 - \theta). \end{aligned}$$

We have thus shown that $E(y) = \theta$ and $\text{var}(y) = \theta(1 - \theta)$.

To derive the MLE we need to know the probability distribution function (pdf) of the random variable. In case of our random variable the probability distribution function (actually called probability mass function because we are dealing with a discrete random variable) is given by:

$$f(y_i, \theta) = \theta^{y_i} (1 - \theta)^{1 - y_i} \quad \text{for } y_i \in \{0, 1\}$$

Since we are dealing with a random sample $y = (y_1, \dots, y_T)$, the observations are independent and the likelihood function is

$$\begin{aligned} L(\theta) &\equiv f(y; \theta) = f(y_1; \theta) f(y_2; \theta) \dots f(y_T; \theta) = \prod_{i=1}^T f(y_i; \theta) \\ &= \prod_{i=1}^T \theta^{y_i} (1 - \theta)^{1 - y_i} \\ &= \theta^{\sum_{i=1}^T y_i} (1 - \theta)^{\sum_{i=1}^T (1 - y_i)} \\ &= \theta^{\sum_{i=1}^T y_i} (1 - \theta)^{T - \sum_{i=1}^T y_i} \end{aligned}$$

Now it becomes clear why we said that it is often convenient to work with the logarithm of the likelihood function, the log-likelihood function $l(\theta)$:

$$l(\theta) = \log(L(\theta)) = \sum_{i=1}^T y_i \log(\theta) + \left(T - \sum_{i=1}^T y_i \right) \log(1 - \theta).$$

The MLE of θ_0 is obtained by solving the maximization problem

$$\hat{\theta}_{ML} = \underset{\theta \in \Theta}{\operatorname{argmax}} l(\theta).$$

This maximization problem is solved by differentiating with respect to θ and equating to zero, which yields the first order condition

$$\begin{aligned} \frac{\partial l(\theta)}{\partial \theta} \bigg|_{\theta = \hat{\theta}_{ML}} &= \frac{\sum_{i=1}^T y_i}{\hat{\theta}_{ML}} - \frac{T - \sum_{i=1}^T y_i}{1 - \hat{\theta}_{ML}} = 0 \\ &\Rightarrow (1 - \hat{\theta}_{ML}) \sum_{i=1}^T y_i - \hat{\theta}_{ML} \left(T - \sum_{i=1}^T y_i \right) = 0 \\ &\Rightarrow \hat{\theta}_{ML} = \frac{\sum_{i=1}^T y_i}{T} \end{aligned}$$

More proof on the next page!

Question 1. The MLE - Lecture Notes 4

To find the asymptotic covariance matrix of the MLE, let us first recall that the MLE is asymptotically normal

$$\sqrt{T} (\hat{\theta}_{ML} - \theta_0) \sim \mathcal{N} (0, \mathcal{I}^{-1}(\theta_0))$$

where $\mathcal{I}(\theta_0) = \text{plim} \frac{I(\theta_0)}{T}$. We see that in order to obtain the asymptotic covariance matrix we first have to derive the information matrix

$$I(\theta) \equiv -EH(\theta)$$

which in turn depends on the Hessian matrix, $H(\theta)$, defined as

$$H(\theta) \equiv \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'}$$

Let us first derive the Hessian matrix

$$\begin{aligned} H(\theta) &= \frac{\partial}{\partial \theta} \left[\frac{\sum_{i=1}^T y_i}{\theta} - \frac{T - \sum_{i=1}^T y_i}{1 - \theta} \right] \\ &= -\frac{\sum_{i=1}^T y_i}{\theta^2} - \frac{T - \sum_{i=1}^T y_i}{(1 - \theta)^2} \end{aligned}$$

Hence the information matrix is

$$\begin{aligned} I(\theta) &= -EH(\theta) \\ &= E \left[\frac{\sum_{i=1}^T y_i}{\theta^2} + \frac{T - \sum_{i=1}^T y_i}{(1 - \theta)^2} \right] \\ &= \frac{T}{\theta} + \frac{T}{(1 - \theta)} \\ &= \frac{T}{\theta(1 - \theta)} \end{aligned}$$

where we used the fact that $\forall i, y_i$ is iid (random sampling) and $E[y_i] = \theta$. Now we can obtain $\mathcal{I}(\theta_0)$

$$\mathcal{I}(\theta_0) = \text{plim} \frac{I(\theta_0)}{T} = \text{plim} \frac{1}{\theta(1 - \theta)} = \frac{1}{\theta(1 - \theta)}$$

and by inverting this result we obtain the asymptotic covariance matrix

$$\mathcal{I}^{-1}(\theta_0) = \theta_0(1 - \theta_0).$$

Comments:

As you can see, we derive the MLE based on the knowledge of Pdf (or Pmf) following the **3 Steps**, later on you will see the MLE for exponential distribution is also the sample mean but it is just a coincidence. **MLE may not always be the sample mean!**



Question 2. The MLE - Lecture Notes 4

A sample of three values (1, 2, 3) is drawn from the exponential distribution with pdf

$$f(x) = \frac{1}{\theta} e^{-x/\theta}$$

Derive the MLE of θ and the ML estimate for the above sample. Derive the acm of the MLE and derive an estimate of it based on the ML estimate.

Let us first re-write the pdf as a function of parameters as

$$f(x; \theta) = \frac{1}{\theta} e^{-x/\theta}$$

Then, using the definition of the likelihood function, $L(\theta)$, for the case of independent observations we have that

$$\begin{aligned} L(\theta) &= f(x; \theta) = \prod_{i=1}^T f(x_i; \theta) \\ &= \prod_{i=1}^T \frac{1}{\theta} e^{-x_i/\theta} \\ &= \frac{1}{\theta^T} e^{-\sum_{i=1}^T x_i/\theta} \end{aligned}$$

The log-likelihood function $l(\theta)$ can then be obtained as

$$\begin{aligned} l(\theta) &= \log(L(\theta)) = -\sum_{i=1}^T x_i/\theta + \log \theta^{-T} \\ &= -\frac{\sum_{i=1}^T x_i}{\theta} - T \log \theta \end{aligned}$$

The MLE is obtained as before by taking the derivative with respect to θ and equating to zero

$$\begin{aligned} \frac{\partial l(\theta)}{\partial \theta} \Big|_{\theta=\hat{\theta}_{ML}} &= \frac{\sum_{i=1}^T x_i}{\hat{\theta}_{ML}^2} - \frac{T}{\hat{\theta}_{ML}} = 0 \\ \Rightarrow \hat{\theta}_{ML} &= \frac{\sum_{i=1}^T x_i}{T} \end{aligned}$$

The ML estimate for the sample (1, 2, 3) is then

$$\hat{\theta}_{ML} = \frac{1+2+3}{3} = 2.$$

To derive the asymptotic covariance matrix we follow the same steps as in the previous exercise. Namely, let us first derive the Hessian matrix

$$\begin{aligned} H(\theta) &= \frac{\partial^2 l(\theta)}{\partial \theta \partial \theta'} \\ &= \frac{\partial}{\partial \theta} \left[\frac{\sum_{i=1}^T x_i}{\theta^2} - \frac{T}{\theta} \right] \\ &= -\frac{2 \sum_{i=1}^T x_i}{\theta^3} + \frac{T}{\theta^2} \end{aligned}$$

The information matrix is thus

$$\begin{aligned} I(\theta) &= -EH(\theta) \\ &= E \left[\frac{2 \sum_{i=1}^T x_i}{\theta^3} - \frac{T}{\theta^2} \right] \\ &= \frac{2E \left[\sum_{i=1}^T x_i \right]}{\theta^3} - \frac{T}{\theta^2} \end{aligned}$$

Check the Next Page!

Question 2. The MLE - Lecture Notes 4

Using the fact a generic exponentially distributed variable with pdf $f(x) = \lambda e^{-\lambda x}$ has an expected value given by $E(x) = 1/\lambda$, we have that our information matrix can be re-written as

$$I(\theta) = \frac{2T\theta}{\theta^3} - \frac{T}{\theta^2} = \frac{T}{\theta^2}.$$

Therefore $\mathcal{I}(\theta_0)$ is

$$\mathcal{I}(\theta_0) = \text{plim} \frac{I(\theta_0)}{T} = \frac{1}{\theta_0^2}$$

and the asymptotic covariance matrix is

$$\mathcal{I}(\theta_0)^{-1} = \theta_0^2$$

The ML estimate of the asymptotic covariance matrix is

$$\mathcal{I}^{-1}(\hat{\theta}_{ML}) = \hat{\theta}_{ML}^2 = 2^2 = 4.$$

Question 3. The GMM - Lecture Notes 4

Assume we have a random sample from the distribution

$$f(x) = \theta - 0.5\theta^2 x \text{ for } 0 \leq x \leq 2/\theta$$

and zero everywhere else. Derive $E(X)$ and derive the MM estimate of θ .

Let us re-write the distribution as

$$f(x) = \begin{cases} \theta - 0.5\theta^2 x & \text{for } x \in [0, 2/\theta] \\ 0 & \text{otherwise} \end{cases}$$

Recall that the expected value of a continuous variable X defined over the sample space Ω is

$$E(X) = \int_{\Omega} x f(x) dx$$

In our case we have that

$$\begin{aligned} E(X) &= \int_0^{2/\theta} x f(x) dx \\ &= \int_0^{2/\theta} x (\theta - 0.5\theta^2 x) dx \\ &= \int_0^{2/\theta} (\theta x - 0.5\theta^2 x^2) dx \\ &= \left(\frac{1}{2}\theta x^2 - \frac{1}{6}\theta^2 x^3 \right) \Big|_{x=0}^{2/\theta} \\ &= \frac{1}{2}\theta \left(\frac{2}{\theta} \right)^2 - \frac{1}{6}\theta^2 \left(\frac{2}{\theta} \right)^3 \\ &= \frac{2}{\theta} - \frac{4}{3\theta} = \frac{2}{3\theta}. \end{aligned}$$

The Method-of-Moment estimator $\hat{\theta}_{MM}$ is obtained by replacing the population moment with the sample moment. Hence

$$\begin{aligned} E(X) &= \frac{2}{3\theta} \\ \bar{X} &= \frac{2}{3\hat{\theta}_{MM}} \end{aligned}$$

where we substituted the population mean $E(X)$ with the sample mean $\bar{X} = \frac{\sum_{i=1}^T x_i}{T}$ and then it is straightforward to solve for $\hat{\theta}_{MM}$:

$$\hat{\theta}_{MM} = \frac{2}{3\bar{X}}$$

Comments on the Hessian of the Gaussian likelihood: no cross-derivative effect between mean and variance

The Hessian of the Gaussian likelihood is a matrix of second derivatives of the log-likelihood function with respect to the parameters of a Gaussian distribution. In the context of Gaussian distributions, the parameters are typically the mean (μ) and the variance (σ^2).

For a Gaussian distribution, the probability density function (PDF) is given by:

$$p(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The log-likelihood ($\mathcal{L}$) of observing a set of data points $\{x_1, x_2, \dots, x_n\}$ under this distribution is:

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

The Hessian matrix $\mathbf{H}$ of this log-likelihood with respect to the parameters μ and σ^2 is a 2×2 matrix where each element is a second partial derivative of $\mathcal{L}$ with respect to the parameters:

$$\begin{bmatrix} \frac{\partial^2 \mathcal{L}}{\partial \mu^2} & \frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2} \\ \frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu} & \frac{\partial^2 \mathcal{L}}{\partial \sigma^4} \end{bmatrix}$$

The specific elements of this Hessian matrix are:

1. $\frac{\partial^2 \mathcal{L}}{\partial \mu^2} = -\frac{n}{\sigma^2}$ - This is the second derivative with respect to the mean, indicating how the curvature of the log-likelihood changes as the mean changes.
2. $\frac{\partial^2 \mathcal{L}}{\partial \sigma^4} = -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6} \sum_{i=1}^n (x_i - \mu)^2$ - This is the second derivative with respect to the variance, reflecting the curvature of the log-likelihood as the variance changes.
3. The off-diagonal terms $\frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2}$ and $\frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu}$ are typically zero for the Gaussian likelihood, indicating no cross-derivative effect between mean and variance.

To understand why the cross-derivative terms are zero in the Hessian matrix of the Gaussian likelihood, let's delve into the derivation of these terms. The cross-derivative terms we're interested in are $\frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2}$ and $\frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu}$. By the symmetry of mixed partial derivatives (Clairaut's theorem), these two should be equal, so we only need to calculate one of them. We start with the log-likelihood function for a Gaussian distribution:

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

1. First, compute the first partial derivative with respect to μ :

$$\frac{\partial \mathcal{L}}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

2. Next, compute the first partial derivative of this expression with respect to σ^2 :

$$\frac{\partial}{\partial \sigma^2} \left(\frac{\partial \mathcal{L}}{\partial \mu} \right) = \frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \right)$$

To perform this derivative, note that the sum $\sum_{i=1}^n (x_i - \mu)$ does not depend on σ^2 , so it can be treated as a constant with respect to σ^2 . Therefore, we are essentially differentiating a constant multiplied by $\frac{1}{\sigma^2}$, which gives:

$$\frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \right) = -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)$$

3. Observation: The key point here is that this derivative is zero under the assumption of the Gaussian distribution where the data is centered around the mean (μ). That is, if you sum up $x_i - \mu$ over all i , you essentially get a sum that tends to zero, especially as the number of data points increases, because the mean μ is the point where this sum is minimized (by definition of the mean in a Gaussian distribution).

Therefore, the cross-derivative term $\frac{\partial^2 \mathcal{L}}{\partial \mu \partial \sigma^2}$ (and consequently $\frac{\partial^2 \mathcal{L}}{\partial \sigma^2 \partial \mu}$) is effectively zero for a Gaussian distribution. This result is particularly pronounced when dealing with a large number of data points, as the sum $\sum_{i=1}^n (x_i - \mu)$ becomes increasingly close to zero, reinforcing the assumption that these cross-derivative terms do not contribute to the curvature of the likelihood function in the parameter space.

Keep it up, everyone! 🍷

When you feel it's tough and want to give up, remember that I'm still organizing your notes late into the night at 4 am 🌙! May all our efforts be rewarded! Cheers! 🎉



Testing linear hypotheses: preliminaries

1. Define type I and type II error:

When we do a hypothesis test, two types of errors are possible: type I and type II.

1. If we reject the null hypothesis H_0 when it is actually true, then we are making a Type I error. We will denote with

$$\alpha = \Pr(\text{Type I error}) = \Pr(\text{Reject } H_0 \mid H_0 \text{ true})$$

Size of a test: is the probability of a type I error, conventionally denoted by α or the significance level.

2. Type II error arises when we do not reject the null hypothesis H_0 even when H_0 is false. We will define

$$\begin{aligned}\beta &= \Pr(\text{Type II error}) = \Pr(\text{Accept } H_0 \mid H_0 \text{ false}) \\ &= 1 - \Pr(\text{Reject } H_0 \mid H_0 \text{ false})\end{aligned}$$

Comments: In general, for every test we conduct, we would like to

- Minimize the probability α of committing a Type I error (typically a significance level of $\alpha \leq 0.10$ is desirable).
- Minimize the probability β , aiming for a type II error that is as small as possible.

Hence, ideally, we would like to minimize both Type I and Type II errors. Unfortunately, for any given sample size, it is not possible to minimize both errors simultaneously: if we try to reduce the probability of rejecting the true hypothesis, at the same time we increase the probability of accepting the false hypothesis and vice versa. There is a trade-off involved between these two types of errors. What econometricians generally do in practice is to fix the value of α at 1 or 5 (at most 10) percent level and then choose a test statistic that would make β as small as possible.

2. Define the power of a test:

The power of a test is defined as the probability of rejecting the null hypothesis H_0 when it is false. Since $\beta = 1 - \Pr(\text{Reject } H_0 \mid H_0 \text{ false})$, then the power of the test will be denoted by

$$\text{power} = 1 - \beta = \Pr(\text{Reject } H_0 \mid H_0 \text{ false})$$

3. The null hypothesis

Data generated by pdf $f(y; \theta_0)$ and we wish to test the linear hypothesis:

$$H_0 : R\theta_0 = r,$$

where R, r are respectively a known constant matrix of dimension $q \times k$, with $q < k$, and a known constant vector of dimension $q \times 1$.

Testing linear hypotheses: preliminaries (Not likely to appear on exam.)

- Sometimes able to construct uniformly most powerful (UMP) tests: have greater power (small β) than any other tests for given α .
- Consistent test: test is consistent when its power goes to one as the sample size goes to infinity.

Comments Whether a test is an UMP test depends on the testing situation and assumptions made about the population. UMP tests are more commonly identified in simpler testing scenarios with specific parameter distributions and well-defined null and alternative hypotheses.

The consistency of tests assumes that the models they are applied to are correctly specified and that certain regularity conditions are met. In practice, this often translates to having a sufficiently large sample size and ensuring that the necessary assumptions for each test (such as normality, independence, etc.) are reasonably satisfied in the data being analyzed.

Likelihood Ratio (LR) test

The null hypothesis

Data generated by pdf $f(y; \theta_0)$ and we wish to test the linear hypothesis:

$$H_0 : R\theta_0 = r,$$

where R, r are respectively a known constant matrix of dimension $q \times k$, with $q < k$, and a known constant vector of dimension $q \times 1$.

Procedures:

1. Estimate MLE $\hat{\theta}$

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta),$$

and get $l(\hat{\theta})$.

2. Apply the Lagrangian to estimate the restricted MLE $\tilde{\theta}$

$$\tilde{\theta} = \operatorname{argmax}_{\theta \in \Theta} (l(\theta) - \mu'(R\theta - r)),$$

and get $l(\tilde{\theta})$. Here μ is the $q \times 1$ vector of Lagrange multipliers.

3. Construct the LR test:

$$LR \equiv 2(l(\hat{\theta}) - l(\tilde{\theta})) \rightarrow_d \chi_q^2 \quad (209)$$

Special Case:

For linear regression model, alternative expression for LR is:

$$LR = T \log \left(1 + \frac{\tilde{e}'\tilde{e} - e'e}{e'e} \right) \rightarrow_d \chi_q^2 \quad (210)$$

where

1. $\tilde{e} = y - X\tilde{\beta}$ and $e = y - X\hat{\beta}$
2. $\hat{\beta} = (X'X)^{-1} X'y$
3. $\tilde{\beta} = \hat{\beta} + (X'X)^{-1} R' (R(X'X)^{-1} R')^{-1} (r - R\hat{\beta})$

Thing to Know:

$LR \geq 0$: The global maximum is not smaller than the local maximum.

Comments on the Wald test (W test) and the F test:

- The Wald test is more general and can be used for a wider range of hypotheses, including non-linear relationships.
- The F test is specifically designed for testing hypotheses about linear restrictions in nested models.

In certain cases, particularly in linear regression with linear hypotheses, the Wald test and the F test can yield equivalent results. This is because, under certain conditions, the Wald test statistic can be algebraically transformed into an F-test statistic. However, this equivalence does not extend to all scenarios, especially when dealing with non-linear models or non-linear hypotheses.

Constrained Least Squares: $\tilde{\beta}$

Given the linear regression model:

$$y = X\beta_0 + u, \quad u \sim N(0, \sigma_0^2 I)$$

where:

1. y is a $T \times 1$ vector of observations.
2. X is a $T \times k$ matrix of explanatory variables.
3. β_0 is a $k \times 1$ vector of true coefficients.
4. u is a $T \times 1$ vector of error terms, follows Gauss-Markov conditions.

The null hypothesis H_0 is:

$$H_0 : R\beta_0 = r$$

where R is a $q \times k$ matrix and r is a $q \times 1$ vector.

Derive Restricted MLE.

Under H_0 , we seek to find the MLE of β_0 subject to the restriction $R\beta_0 = r$. We can use the method of Lagrange multipliers to solve this. The Lagrangian can be written as

$$\mathcal{L}(\beta, \lambda) = (y - X\beta)'(y - X\beta) + 2\lambda'(R\beta - r)$$

where λ is a q by 1 vector of Lagrange multipliers.

Taking the derivative of $\mathcal{L}$ with respect to β and setting it equal to zero gives

$$-2X'y + 2X'X\beta + 2R'\lambda = 0$$

Rearranging, we get

$$X'X\beta = X'y - R'\lambda$$

Multiplying both sides by $(X'X)^{-1}$, we have

$$\beta = (X'X)^{-1}X'y - (X'X)^{-1}R'\lambda.$$

Using $\hat{\beta}$, this becomes

$$\beta = \hat{\beta} - (X'X)^{-1}R'\lambda$$

Now, we need to find λ such that $R\beta = r$. Substituting the expression for β , we have

$$R(\hat{\beta} - (X'X)^{-1}R'\lambda) = r.$$

Rearranging and solving for λ , we get

$$R\hat{\beta} - R(X'X)^{-1}R'\lambda = r$$

$$\lambda = (R(X'X)^{-1}R')^{-1}(R\hat{\beta} - r).$$

Substituting λ back into the expression for β , we have

$$\tilde{\beta} = \hat{\beta} - (X'X)^{-1}R' \left[(R(X'X)^{-1}R')^{-1}(R\hat{\beta} - r) \right]$$

Simplifying, we obtain

$$\tilde{\beta} = \hat{\beta} + (X'X)^{-1}R' \left[(R(X'X)^{-1}R')^{-1}(r - R\hat{\beta}) \right].$$

This is the restricted MLE under the linear hypothesis $H_0 : R\beta_0 = r$.

Wald (W) test

The null hypothesis

Data generated by pdf $f(y; \theta_0)$ and we wish to test the linear hypothesis:

$$H_0 : R\theta_0 = r,$$

where R, r are respectively a known constant matrix of dimension $q \times k$, with $q < k$, and a known constant vector of dimension $q \times 1$.

Procedures:

1. Estimate MLE $\hat{\theta}$

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} l(\theta),$$

and get $l(\hat{\theta})$ and $I(\hat{\theta})$.

2. Apply the asymptotic normality of MLE $\hat{\theta}$

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow_d N(0, \mathcal{I}^{-1}(\theta_0))$$

and

$$\mathcal{I}^{-1}(\theta_0) \approx (I(\hat{\theta})/T)^{-1} = TI^{-1}(\hat{\theta})$$

3. Construct the Wald (W) test:

$$W = (R\hat{\theta} - r)' (RI^{-1}(\hat{\theta})R')^{-1} (R\hat{\theta} - r) \rightarrow_d \chi_q^2 \quad (211)$$

Special Case:

For linear regression model, $H_0 : R\beta_0 = r$. We have seen that

$$\sqrt{T}(R\hat{\beta} - r) \rightarrow_d N(0, \sigma^2 RQ^{-1}R')$$

where $Q = E[X'X]$. As a result, we have W for the linear regression model, which is

$$W \equiv \frac{(R\hat{\beta} - r)' (R(X'X)^{-1}R')^{-1} (R\hat{\beta} - r)}{\hat{\sigma}^2} \rightarrow_d \chi_q^2, \quad (212)$$

where the MLE sample estimator of the variance $\hat{\sigma}^2 = \frac{e'e}{T}$.

An alternative expression for W is:

$$W = T \frac{\tilde{e}'\tilde{e} - e'e}{e'e} \rightarrow_d \chi_q^2. \quad (213)$$

Thing to Know:

$W \geq 0$: The global minimum is not greater than the local minimum^b.

^aChi-squared distribution definitions: If $Z_1, \dots, Z_q$ are independent, standard normal random variables, then the sum of their squares,

$$Q = \sum_{i=1}^q Z_i^2,$$

is distributed according to the chi-squared distribution with q degrees of freedom. This is usually denoted as

$$Q \sim \chi^2(q) \text{ or } Q \sim \chi_q^2.$$

The chi-squared distribution has one parameter: a positive integer q that specifies the number of degrees of freedom (the number of random variables being summed, Z_i 's).

^bRemember the definition of Residual Sum of Squares(RSS)

$$\begin{aligned} \hat{\beta} &= \operatorname{argmin}_{b \in B} \text{RSS}(b) \\ &= \operatorname{argmin}_{b \in B} e'e \end{aligned}$$

The Wald test statistic: comments on its derivation (211)

The Wald test is a statistical test used to determine whether a set of parameters in a model is significant. It's often used in the context of econometrics or other statistical modeling. The idea behind the Wald test is to see whether a certain linear combination of the estimated parameters significantly differs from a specific value under the null hypothesis.

Given a null hypothesis $H_0 : R\theta = r$, where R is a known matrix and r is a known vector, the Wald test assesses whether this hypothesis is plausible given the estimated parameters $\hat{\theta}$ from your model. The asymptotic distribution of $\hat{\theta}$ under H_0 is important for this test.

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow_d N(0, \mathcal{I}^{-1}(\theta_0))$$

indicates that, as the sample size T grows, the distribution of the scaled difference between the estimated parameters $\hat{\theta}$ and the true parameters θ_0 converges in distribution to a multivariate normal distribution with mean 0 and variance-covariance matrix $\mathcal{I}^{-1}(\theta_0)$, where $\mathcal{I}(\theta_0)$ is the Fisher Information matrix evaluated at θ_0 .

Now, let's derive the Wald test statistic:

1. Start with the Null Hypothesis: Under the null hypothesis $H_0 : R\theta = r$, we are interested in testing whether $R\hat{\theta}$ is close to r .
2. Use the Asymptotic Distribution: We know that $\sqrt{T}(\hat{\theta} - \theta_0)$ converges in distribution to a normal distribution. So, we can standardize $R\hat{\theta} - r$ using the variance of $R\hat{\theta}$.
3. Calculate the Variance of $R\hat{\theta}$: The variance of $R\hat{\theta}$ can be derived using the Delta method or directly, given the asymptotic variance of $\hat{\theta}$ is $\mathcal{I}^{-1}(\theta_0)$. The variance of $R\hat{\theta}$ is $R\mathcal{I}^{-1}(\theta_0)R'$.
4. Formulate the Test Statistic: The Wald test statistic is then:

$$W = T(R\hat{\theta} - r)' [R\mathcal{I}^{-1}(\theta_0)R']^{-1} (R\hat{\theta} - r)$$

5. Approximate the Inverse Fisher Information Matrix, $\mathcal{I}^{-1}(\theta_0)$:

We have $\mathcal{I}^{-1}(\theta_0) \approx (I(\theta_0)/T)^{-1} = TI^{-1}(\theta_0)$, where $I(\theta_0)$ is the Fisher Information matrix evaluated at θ .

This approximation is based on the fact that the inverse Fisher Information matrix, $\mathcal{I}^{-1}(\theta_0)$, can be estimated by the inverse of the observed information divided by the sample size, T . This is a standard approach in large sample theory. The Wald test statistic is:

$$W = (R\hat{\theta} - r)' (RI^{-1}(\hat{\theta})R')^{-1} (R\hat{\theta} - r)$$

Here, $I^{-1}(\hat{\theta})$ is used as an estimate for the asymptotic variance-covariance matrix of $\hat{\theta}$. This substitution is common in practical applications, where θ_0 is unknown and is replaced by its ML estimate $\hat{\theta}$.

It turns out that I made a mistake in my teaching, and Paolo's lecture notes are correct.



Lagrange multiplier (LM) test (a.k.a the Score Test)

The null hypothesis

Data generated by pdf $f(y; \theta_0)$ and we wish to test the linear hypothesis:

$$H_0 : R\theta_0 = r,$$

where R, r are respectively a known constant matrix of dimension $q \times k$, with $q < k$, and a known constant vector of dimension $q \times 1$.

Procedures:

1. Apply the Lagrangian to estimate the restricted MLE $\tilde{\theta}$

$$\tilde{\theta} = \operatorname{argmax}_{\theta \in \Theta} (l(\theta) - \mu'(R\theta - r)),$$

Here μ is the $q \times 1$ vector of Lagrange multipliers.

2. Calculate the Score Vector: Consider the Lagrangian,

$$\tilde{l}(\theta, \mu) = l(\theta) - \mu'(R\theta - r), \quad (214)$$

we have

$$s(\tilde{\theta}) = \left. \frac{\partial \tilde{l}(\theta, \mu)}{\partial \theta} \right|_{\theta=\tilde{\theta}} \quad (215)$$

and we get $I(\tilde{\theta})$.

3. Construct the **LM** test:

$$LM = s'(\tilde{\theta}) I^{-1}(\tilde{\theta}) s(\tilde{\theta}) \rightarrow_d \chi_q^2 \quad (216)$$

Special Case:

For linear regression model, $H_0 : R\beta_0 = r$. We have seen that

$$s(\tilde{\theta}) = \begin{pmatrix} \tilde{\sigma}^{-2} X' \tilde{e} \\ 0 \end{pmatrix},$$

and

$$I^{-1}(\tilde{\theta}) = \begin{pmatrix} \tilde{\sigma}^2 (X'X)^{-1} & 0 \\ 0 & T^{-1} 2\tilde{\sigma}^4 \end{pmatrix}.$$

For linear regression model, alternative expression for LM is:

$$LM = T \frac{\tilde{e}' X (X'X)^{-1} X' \tilde{e}}{\tilde{e}' \tilde{e}} \rightarrow_d \chi_{q'}^2 \quad (217)$$

where

1. $\tilde{e} = y - X\hat{\beta}$
2. $\hat{\beta} = (X'X)^{-1} X'y$
3. $\tilde{\beta} = \hat{\beta} + (X'X)^{-1} R' \left(R (X'X)^{-1} R' \right)^{-1} (r - R\hat{\beta})$

Thing to Know:

The LM test measures the closeness to zero of the score evaluated at restricted and unrestricted MLE. Whenever $\hat{\theta}, \tilde{\theta}$ are close one another, we have

$$s(\tilde{\theta}) = R' \tilde{\mu} \approx 0 = s(\hat{\theta}).$$

In other words, if the restriction truly holds in the data, imposing the restriction is redundant and so the estimated $\tilde{\mu}$ is insignificant.

An important relationship

All three tests are equivalent asymptotically (they have the same asymptotic distribution). In finite samples, however, different conclusions can be drawn depending on the type of test used. In particular, **for linear regression models**, the following relation holds

$$W \geq LR \geq LM. \quad (218)$$

In linear regression models, Wald tests are rejected more often and are considered conservative.

Testing non-nested hypotheses

For not nested models^a above theory does not apply. Consider the two rival linear models:

$$M_1 : \mathbf{y} = \mathbf{X}\beta + \mathbf{u}_1, \quad \mathbf{u}_1 \sim N(0, \sigma_1^2 \mathbf{I})$$

$$M_2 : \mathbf{y} = \mathbf{Z}\gamma + \mathbf{u}_2, \quad \mathbf{u}_2 \sim N(0, \sigma_2^2 \mathbf{I})$$

where $\mathbf{X}$ is $T \times k$ and $\mathbf{Z}$ is $T \times l$. There may be some regressors in common, yielding

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_* \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{Z}_* \end{bmatrix}.$$

The goal is to determine which model is statistically superior.

Procedures:

1. We write the composite model:

$$M_3 : y = (1 - \alpha)X\beta + \alpha(Z\gamma) + u,$$

where α scalar parameter such that, for $\alpha = 0$ then M_3 reduces to M_1 whereas for $\alpha = 1$ M_3 reduces to M_2 .

2. Assume that M_1 is being tested.

Then estimate M_2 by OLS and replace $Z\gamma$ in M_3 by $Z\hat{\gamma} = Z(Z'Z)^{-1}Z'y \equiv \hat{y}_2$.

3. Set $\delta = (1 - \alpha)\beta$ and estimate model

$$y = X\delta + \alpha\hat{y}_2 + u.$$

4. Then t test of $\hat{\alpha}$ will tell if M_1 supported by data.

^aThe models were nested when the model under H_0 was a restricted version of an unrestricted model.

Question 1. The Tests - Lecture Notes 5

For the linear regression model

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + u_t$$

derive the LR, W and LM test for the hypothesis

$$H_0 : \beta_3 = -1$$

Proof: Let us first re-write our null hypothesis in such a way as to relate it to the generic hypothesis testing framework

$$H_0 : R\beta = r \quad \text{v.s.} \quad R\beta \neq r.$$

We do this by expressing our null hypothesis

$$H_0 : \beta_3 = -1$$

in matrix form as follows

$$H_0 : \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = -1$$

Hence we can identify

$$R = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \quad \text{and} \quad r = -1.$$

The Likelihoods Ratio test is

$$LR \equiv T \log \left(1 + \frac{\tilde{e}'\tilde{e} - e'e}{e'e} \right) = T \log \left(\frac{\tilde{e}'\tilde{e}}{e'e} \right) \rightarrow_d \chi_q^2$$

where $\tilde{e} = y - X\tilde{\beta}$ and $e = y - X\hat{\beta}$. We know from the lecture notes that for a linear regression $y = X\beta + u$ with null hypothesis $R\beta = r$, the restricted MLE $\tilde{\beta}$ is given by

$$\tilde{\beta} = \hat{\beta} + (X'X)^{-1} R' \left(R (X'X)^{-1} R' \right)^{-1} (r - R\hat{\beta})$$

The Wald test is defined as

$$W \equiv \frac{(R\hat{\beta} - r)' \left(R (X'X)^{-1} R' \right)^{-1} (R\hat{\beta} - r)}{\hat{\sigma}^2} \rightarrow_d \chi_q^2$$

where the MLE sample estimator of the variance $\hat{\sigma}^2 = \frac{e'e}{T}$.
The Lagrange Multiplier test is defined as

$$LM \equiv T \frac{\tilde{e}' X (X'X)^{-1} X' \tilde{e}}{\tilde{e}' \tilde{e}} \rightarrow_d \chi_q^2.$$

Question 2. The Tests - Sample Exam Question

1. Define type one and type two errors.
2. Define the power of a test.
3. Define the Lagrange multiplier test, the Wald test and the Likelihood ratio test. Discuss the asymptotic and finite sample relationship between these tests.

See 'Summary of Lectures 13 and 14' Page 1.

Question 3. The MME - Sample Exam Question

Define the Method of Moment estimation (MME) principle. In the multiple linear regression model:

$$Y = X\beta + \epsilon$$

- i) Let X be a $T \times k$ matrix and define Z as a $T \times l$ matrix. Assume that $E(Z'\epsilon) = 0$ and $E(X'\epsilon) \neq 0$. Then explain why you can't estimate β by OLS, proving your claim.
- ii) Write down the IV estimator. How many instruments do you need for this estimator to be defined?
- iii) Assume now $l = k$. Show that the IV estimator is actually the MM estimator based on the orthogonality condition $E(Z'\epsilon) = 0$.

Proof:

(i) Let us now consider the linear regression model $Y = X\beta + \epsilon$ and assume that $E(Z'\epsilon) = 0$, but $E(X'\epsilon) \neq 0$. In this case, we cannot apply OLS to estimate β , since one of the Gauss Markov assumption is violated. As a consequence, $\hat{\beta}_{OLS}$ will not be consistent anymore. To show why this happens, suppose we estimated our β by simply applying OLS to the original model. Then, assuming that

$$\text{plim}_{T \rightarrow \infty} \frac{X'X}{T} = Q$$

where Q is a finite and full rank matrix of dimension $k \times k$, we have

$$\begin{aligned}\hat{\beta}_{OLS} &= (X'X)^{-1} X'Y \\ &= \beta + (X'X)^{-1} X'\epsilon \\ &= \beta + \left(\frac{X'X}{T} \right)^{-1} \frac{X'\epsilon}{T} \\ &\xrightarrow{p} \beta + Q^{-1} E(X'\epsilon)\end{aligned}$$

which implies that $\hat{\beta}_{OLS} \xrightarrow{p} \beta$ unless $E(X'\epsilon) = 0$.

(ii) The IV estimator is derived by using a two-step procedure. In the first step we regress Y on Z and X on Z , obtaining

$$\begin{aligned}\hat{Y} &= Z(Z'Z)^{-1} Z'Y \\ \hat{X} &= Z(Z'Z)^{-1} Z'X.\end{aligned}$$

Then, in the second step we regress $\hat{Y}$ on $\hat{X}$, obtaining the IV estimator

$$\begin{aligned}\hat{\beta}_{IV} &= (\hat{X}'\hat{X})^{-1} \hat{X}'\hat{Y} \\ &= (X'Z(Z'Z)^{-1} Z'X)^{-1} X'Z(Z'Z)^{-1} Z'Y\end{aligned}$$

In order for this estimator to be identified we need that $l \geq k$, i.e we need at least k instruments.

(iii) Let us now assume that $l = k$, so that the model is just identified. In this case we have proved (check Week 5 - Question 2.i) that the IV estimator reduces to

$$\hat{\beta}_{IV} = (Z'X)^{-1} Z'Y$$

since $(X'Z(Z'Z)^{-1} Z'X)^{-1}$ can be written as $(Z'X)^{-1} (Z'Z) (X'Z)^{-1}$. We now need to apply MM to the linear regression model using $E(Z'\epsilon) = 0$ as moment conditions (or orthogonality conditions) and check if the resulting MM estimator for β coincides with IV. Let us start by the (population) moment condition:

$$E(Z'\epsilon) = 0 \Rightarrow E(Z'(Y - X\beta)) = 0.$$

Then, replace the population moment with the sample analogue:

$$\frac{1}{T} (Z'(Y - X\beta)) = 0$$

and finally solve (12) for β , obtaining the MM estimator

$$\hat{\beta}_{MM} = (Z'X)^{-1} Z'Y$$

which coincides precisely with IV. Therefore, we can conclude that IV can be viewed as a special case of MM when we use $E(Z'\epsilon) = 0$ as moment conditions.

Take a deep breath — there's no need to memorize every detail from this summary. My goal is to provide you with my intuitive understanding, so you'll always find your way, even in complex situations.

The white noise:

Definition

The process ϵ_t is white noise if $\mathbb{E}(\epsilon_t) = 0$, $\mathbb{E}(\epsilon_t^2) = \sigma^2 < \infty$, and $\text{cov}(\epsilon_t, \epsilon_{t-k}) = 0$ for $k \neq 0$.

Comments:

White noise in the context of time series analysis refers to a sequence of random variables where each value has a mean of zero, constant finite variance, and no serial correlation. In other words, white noise is a random signal that is equally distributed across different frequencies with no predictable pattern or trend.

Why it matters?

1. **Benchmark for Predictability:** White noise serves as a baseline for determining the predictability of a time series. If a financial time series is found to be white noise, it suggests that the series is purely random and cannot be predicted. This has implications for trading strategies, risk management, and portfolio allocation.
2. **Model Evaluation:** When developing econometric models, such as ARIMA (Autoregressive Integrated Moving Average) models, the residuals (errors) of the model are expected to be white noise. If the residuals are not white noise, it may indicate that the model has not fully captured the underlying process, and there may be more information to extract for better predictions.
3. **Risk Management:** Understanding whether a financial time series behaves like white noise can help in risk management. For instance, if returns on an asset are white noise, it suggests that past movements do not provide any information about future movements, which can influence hedging strategies and the assessment of financial risk.
4. **Efficient Markets Hypothesis:** The concept of white noise is closely related to the Efficient Markets Hypothesis (EMH), which posits that asset prices fully reflect all available information. Under the strong form of EMH, price movements would be unpredictable and resemble a white noise process, as all known and unknown information is already incorporated into prices, leaving only random, new information to affect prices going forward.

The covariance or weakly stationary stochastic process:

When observing $(y_1, \dots, y_T)$ we are now considering y_t as part of a stochastic process, which means a collection of rv , one for each t .

Definition

$\{y_t, t = 0, \pm 1, \pm 2, \dots\}$ is covariance or weakly stationary if the mean and variance matrix are independent of any t , and the autocovariances are independent of t for all s

$$E y_t \equiv \mu_t = \mu < \infty$$

$$\text{var}(y_t) = E(y_t - \mu)^2 \equiv \gamma(0) < \infty$$

$$\text{cov}(y_t, y_s) = E(y_t - \mu)(y_{t+s} - \mu) \equiv \gamma(s) \text{ with } |\gamma(s)| < \infty.$$

Graph of $(s, \gamma(s))$ defines autocovariance function, hereafter ACF.

Often one uses the autocorrelation function (scale free): $\rho(s) = \frac{\gamma(s)}{\gamma(0)}$, $s = 0, \pm 1, \dots$ where $-1 \leq \rho(s) \leq 1$

Comments:

A weakly stationary (or covariance stationary) stochastic process is one whose mean, variance, and autocovariance structure do not change over time. The autocovariance at any given lag is the same regardless of the time at which it is measured. This implies that the process has a constant mean, a constant variance, and a consistent autocorrelation structure that only depends on the lag between observations, not on the time at which the autocorrelation is computed.

Why it matters?

1. Modeling and Forecasting: Many time series models, such as ARIMA models, assume stationarity as a prerequisite. Stationarity implies that the model's parameters are constant over time, which simplifies both the model estimation process and the interpretation of the results. Non-stationary data can lead to spurious results and unreliable forecasts.
2. Mean-Reversion: In a stationary process, the mean-reverting property implies that values tend to return to the mean, which can inform trading strategies. For example, if an asset price deviates significantly from its historical mean, one might expect it to eventually revert back, presenting a potential trading opportunity.
3. Financial Econometrics: Many econometric tests, such as those for the presence of unit roots (e.g., the Augmented Dickey-Fuller test), are used to determine whether a financial time series is stationary. If a time series is found to be non-stationary, it may need to be differenced or transformed to achieve stationarity before further analysis.
4. Economic Theory: Many economic theories and financial models assume equilibrium behaviors that are consistent with stationarity. For example, the consumption-based Capital Asset Pricing Model (CAPM) assumes that consumption growth rates are stationary.
5. Integration and Cointegration: In finance, especially in pairs trading or arbitrage strategies, the concept of cointegration relies on the stationarity of the linear combination of non-stationary financial time series. If two or more non-stationary series are cointegrated, it suggests that there is a stable, long-term relationship between them, which can be exploited for profit.

The strictly stationary stochastic process:

When observing $(y_1, \dots, y_T)$ we are now considering y_t as part of a stochastic process, which means a collection of rv , one for each t .

Definition

$\{y_t\}$ is strictly stationary if the joint distribution of $(y_t, \dots, y_{t+\ell})$ is independent of t for all ℓ .

Comments:

All strictly stationary processes with a defined mean and variance are also weakly stationary, but not all weakly stationary processes are strictly stationary. Weak stationarity is a less stringent condition than strict stationarity because it only concerns the first two moments of the distribution (mean and variance) and the autocovariance function, not the entire distribution. In finance, this is often sufficient for many practical purposes, such as forecasting with ARIMA models, since it ensures a stable mean and variance, which are often of primary interest.

Theorem

If y_t is i.i.d., then it strictly stationary.

The Moving Average (MA) Processes:

The first-order moving average process, denoted MA(1), is

$$y_t = \mu + \epsilon_t + \theta\epsilon_{t-1}$$

where ϵ_t is i.i.d of white noise process. The model is called a "moving average" because y_t is a weighted average of the shocks ϵ_t and ϵ_{t-1} .

It is straightforward to calculate that a MA(1) has the following moments.

1. $E(y_t) = \mu$
2. $\text{var}(y_t) = (1 + \theta^2) \sigma^2$
3. $\gamma(1) = \theta\sigma^2$
4. $\gamma(s) = \rho(s) = 0, \quad s \geq 2$
5. $\rho(1) = \frac{\theta}{1+\theta^2}$

The q^{th} -order moving average process, denoted MA(q), is

$$y_t = \mu + \theta_0\epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \cdots + \theta_q\epsilon_{t-q}$$

where $\theta_0 = 1$. It is straightforward to calculate that a MA(q) has the following moments.

1. $E(y_t) = \mu$
2. $\text{var}(y_t) = \left(\sum_{j=0}^q \theta_j^2\right) \sigma^2$
3. $\gamma(s) = \left(\sum_{j=0}^{q-s} \theta_{j+s}\theta_j\right) \sigma^2, \quad s \leq q$
4. $\gamma(s) = \rho(s) = 0, \quad s > q$
5. $\rho(s) = \frac{\sum_{j=0}^{q-s} \theta_{j+s}\theta_j}{\sum_{j=0}^q \theta_j^2}$

An infinite-order moving average process, denoted MA(∞), also known as a linear process, is

$$y_t = \mu + \sum_{j=0}^{\infty} \theta_j \epsilon_{t-j}$$

where ϵ_t is a iid white noise process, and

$$\sum_{j=0}^{\infty} \theta_j^2 < \infty.$$

1. $E(y_t) = \mu$
2. $\text{var}(y_t) = \left(\sum_{j=0}^{\infty} \theta_j^2\right) \sigma^2$
3. $\gamma(s) = \left(\sum_{j=0}^{\infty} \theta_{j+s}\theta_j\right) \sigma^2$
4. $\rho(s) = \frac{\sum_{j=0}^{\infty} \theta_{j+s}\theta_j}{\sum_{j=0}^{\infty} \theta_j^2}$

The Moving Average (MA) Processes:

Things to know:

1. Thus the MA(1) process has a non-zero first autocorrelation, with the remainder zero.
2. An MA(1) process with $\theta \neq 0$ is serially correlated, with each pair of adjacent observations (y_{t-1}, y_t) correlated. If $\theta > 0$ the pair are positively correlated, while if $\theta < 0$ they are negatively correlated. The serial correlation, however, is limited in that observations separated by multiple periods are mutually independent.
3. A MA(q) has q non-zero autocorrelations, with the remainder zero.
4. As the ACF, $\gamma(s)$ is truncated, MA have little memory.
5. A MA(q) process y_t is strictly stationary.

The MA(q) Process Estimation:

1. In model identification for an MA(q) process, the order (q) is determined by observing where the Autocorrelation Function (ACF) significantly cuts off after lag q.
2. Once the model order (q) is identified, estimation is commonly done using the Method of Moments or Maximum Likelihood Estimation (MLE).
3. Diagnostic checking: the residuals $\hat{\epsilon}_t$ are examined to ensure they behave like white noise and are independently and identically distributed (i.i.d.).

Why it matters?

1. Modeling Short-Term Dependencies: The MA(q) process can model the short-term dependencies in financial time series data. The series y_t is expressed as the sum of the current and past q random shock terms ϵ , which are typically assumed to be independently and identically distributed with a mean of zero. This allows the model to capture the 'shock' effects and their impact over the subsequent q periods.
2. Shock Absorption: The MA(q) model assumes that the impact of a shock to the system will be temporary and will last for q periods. This is consistent with financial markets where a news event or market shock may have an immediate impact on asset prices but the effect diminishes over time.

The Lag Operator:

Definition: The lag operator L satisfies $Ly_t = y_{t-1}$.

1. In general, $L^k y_t = y_{t-k}$.
2. $L^k c = c$ for any constant c .

The Autoregressive (AR) Processes:

The first-order autoregressive process, denoted AR(1), is

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \epsilon_t$$

where ϵ_t is i.i.d of white noise process. To find a linear process representation for y_t we can use backward recursion.

Theorem: If $|\alpha_1| < 1$ then the AR(1) process has the convergent representation

$$y_t = \mu + \sum_{j=0}^{\infty} \alpha_1^j \epsilon_{t-j}$$

where $\mu = \alpha_0 / (1 - \alpha_1)$. The AR(1) process y_t is absolutely convergent, strictly stationary.

For simplicity, we may set $\alpha_0 = 1$.

We can compute the moments of y_t , remember if $|\alpha_1| < 1$ we have stationarity which implies $E[y_t] = E[y_{t-1}] = \mu$ and $\text{var}(y_{t-1}) = \text{var}(y_t) = \gamma(0)$.

1. **Mean of y_t :** The mean of the process is given by $\mu = \frac{\alpha_0}{1 - \alpha_1}$.
2. **Variance of y_t ($\gamma(0)$):** The variance of the process is $\gamma(0) = \frac{\sigma^2}{1 - \alpha_1^2}$.
3. **Autocovariance Function $\gamma(s)$:** The autocovariance function is given by $\gamma(s) = \alpha_1^s \gamma(0)$.
4. **Autocorrelation Function $\rho(s)$:** The autocorrelation function is $\rho(s) = \alpha_1^s$.

The second-order autoregressive process, denoted AR(2), is

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \epsilon_t$$

where ϵ_t is iid of white noise process.

Using the lag operator we can write above as

$$y_t - \alpha_1 L y_t - \alpha_2 L^2 y_t = \alpha_0 + \epsilon_t,$$

or

$$\alpha(L) y_t = \alpha_0 + \epsilon_t$$

where

$$\alpha(L) = 1 - \alpha_1 L - \alpha_2 L^2.$$

We call $\alpha(z)$ the autoregressive polynomial of y_t .

We would like to describe the conditions for the stationarity of y_t . For simplicity set $\alpha_0 = 0$. Factor the autoregressive polynomial as

$$\alpha(z) = 1 - \alpha_1 z - \alpha_2 z^2 = (1 - \rho_1 z)(1 - \rho_2 z)$$

Theorem: If $|\rho_1| < 1$ and $|\rho_2| < 1$ (or equivalently, all roots λ_j of $\alpha(z)$ satisfy $|\lambda_j| > 1$ for $j = 1, 2$) then the AR(2) process is absolutely convergent, strictly stationary.

The p^{th} -order autoregressive process, denoted AR(p), is

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \dots + \alpha_p y_{t-p} + \epsilon_t$$

where ϵ_t is iid of white noise process. Using the lag operator,

$$y_t - \alpha_1 L y_t - \alpha_2 L^2 y_t - \dots - \alpha_p L^p y_t = \alpha_0 + \epsilon_t,$$

or

$$\alpha(L) y_t = \alpha_0 + \epsilon_t$$

where

$$\alpha(L) = 1 - \alpha_1 L - \alpha_2 L^2 - \dots - \alpha_p L^p.$$

We call $\alpha(z)$ the autoregressive polynomial of y_t .

Theorem: If $|\rho_j| < 1$ for $j = 1, \dots, p$ (or equivalently, all roots of $\alpha(z)$ lie outside the unit circle), then the AR(p) process is absolutely convergent, strictly stationary.

Comments: Derivation of Key Moments in AR(1) Time Series Models: A Simplified Approach

To derive the ACF of an AR(1) process, we use the properties of the process and its definition. Let's start by assuming the process is stationary, which implies the mean (μ) and variance ($\gamma(0)$) are constant over time, and the covariance between two observations depends only on the time lag between them.

1. **Mean of y_t :** For a stationary AR(1) process, we have

$$E[y_t] = \alpha_0 + \alpha_1 E[y_{t-1}].$$

Assuming stationarity, $E[y_t] = E[y_{t-1}] = \mu$. Solving $\mu = \alpha_0 + \alpha_1 \mu$ gives

$$\mu = \frac{\alpha_0}{1 - \alpha_1}.$$

2. **Variance of y_t ($\gamma(0)$):** Consider the equation

$$y_t - \mu = \alpha_1 (y_{t-1} - \mu) + \epsilon_t.$$

Taking variance on both sides, we get

$$\gamma(0) = \alpha_1^2 \gamma(0) + \sigma^2.$$

Solving for $\gamma(0)$ gives

$$\gamma(0) = \frac{\sigma^2}{1 - \alpha_1^2}.$$

3. **Autocovariance Function $\gamma(s)$:** For $s > 0$, using the AR(1) model, we have

$$\gamma(s) = \text{cov}(\alpha_1 y_{t-1} + \epsilon_t, y_{t-s}).$$

As ϵ_t is uncorrelated with y_{t-s} and the covariance between two observations depends only on the time lag between them, this simplifies to

$$\gamma(s) = \alpha_1 \gamma(s-1).$$

By recursive substitution,

$$\gamma(s) = \alpha_1^s \gamma(0).$$

4. **ACF $\rho(s)$:** The autocorrelation function $\rho(s)$ is the autocovariance function normalized by the variance, i.e.,

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)}.$$

Therefore,

$$\rho(s) = \alpha_1^s.$$

The Autoregressive (AR) Processes:

Things to know:

For AR(1) Process

1. Thus the AR(1) process with $\alpha_1 \neq 0$ has non-zero autocorrelations of all orders which decay to zero geometrically as s increases.
2. For $\alpha_1 > 0$ the autocorrelations are all positive.
3. For $\alpha_1 < 0$ the autocorrelations alternate in sign.

The AR(p) Process Estimation:

1. In model identification for an AR(p) process, the order (p) is determined by the Akaike Information Criterion (AIC) or the Bayesian Information Criterion (BIC) as they can be used to compare models and select the best fitting one.
2. Once the model order (p) is identified, estimation is commonly done using methods like Ordinary Least Squares (OLS) or Maximum Likelihood Estimation (MLE).
3. Diagnostic checking: the residuals $\hat{\epsilon}_t$ are examined to ensure they behave like white noise and are independently and identically distributed (i.i.d.).

Why it matters?

1. Forecasting Financial Time Series: AR models are fundamental tools for forecasting in finance. They are used to predict future values of financial time series, such as stock prices, exchange rates, interest rates, and economic indicators, based on their own past values.
2. Understanding Market Dynamics: AR models help in understanding the dynamics of financial markets by quantifying the extent to which past values of a time series influence its future values. This understanding can be crucial for investment strategy and risk management.
3. Volatility Clustering: AR models, especially when combined with moving average (MA) components to form ARMA or ARIMA models, can be effective in modeling and forecasting volatility clustering, a common phenomenon in financial time series where periods of high volatility are followed by high volatility and vice versa.

Question 1. The MA(2) - Lecture Notes 6

Derive the ACF for the MA(2) scheme:

$$y_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}$$

Proof: To derive the Autocovariance Function (ACF) of a process we need to derive the variance and the covariance of the process with itself at different time points.

Let us start by deriving the variance of the process. To derive it we first need to derive its mean. Starting from the fact that ϵ_t is a white noise with zero mean and variance equal to σ^2 (that is $\epsilon_t \sim iid(0, \sigma^2)$), we get

$$\begin{aligned} E[y_t] &= E[\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}] \\ &= 0 \end{aligned}$$

and hence

$$\begin{aligned} \text{Var}(y_t) &= \text{Var}(\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}) \\ &= E[\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}]^2 \\ &= \sigma^2 + \theta_1^2 \sigma^2 + \theta_2^2 \sigma^2 \\ &= (1 + \theta_1^2 + \theta_2^2) \sigma^2 \end{aligned}$$

Consider now the covariance between y_t and y_{t-1} . Rewriting

$$y_{t-1} = \epsilon_{t-1} + \theta_1 \epsilon_{t-2} + \theta_2 \epsilon_{t-3}$$

and remembering that $\text{Cov}(\epsilon_t, \epsilon_s) = 0, \forall t \neq s$, we have

$$\begin{aligned} \text{Cov}(y_t, y_{t-1}) &= E[(\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2})(\epsilon_{t-1} + \theta_1 \epsilon_{t-2} + \theta_2 \epsilon_{t-3})] \\ &= E[\epsilon_t \epsilon_{t-1}] + \theta_1 E[\epsilon_t \epsilon_{t-2}] + \theta_2 E[\epsilon_t \epsilon_{t-3}] \\ &\quad + \theta_1 E[\epsilon_{t-1} \epsilon_{t-1}] + \theta_1^2 E[\epsilon_{t-1} \epsilon_{t-2}] + \theta_1 \theta_2 E[\epsilon_{t-1} \epsilon_{t-3}] \\ &\quad + \theta_2 E[\epsilon_{t-2} \epsilon_{t-1}] + \theta_1 \theta_2 E[\epsilon_{t-2} \epsilon_{t-2}] + \theta_2^2 E[\epsilon_{t-2} \epsilon_{t-3}] \\ &= \theta_1 \sigma^2 + \theta_1 \theta_2 \sigma^2 \\ &= \sigma^2 (\theta_1 + \theta_1 \theta_2). \end{aligned}$$

Using the same steps to derive the covariance between y_t and y_{t-2} , we have

$$\begin{aligned} \text{Cov}(y_t, y_{t-2}) &= E[(\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2})(\epsilon_{t-2} + \theta_1 \epsilon_{t-3} + \theta_2 \epsilon_{t-4})] \\ &= \theta_2 \sigma^2 \end{aligned}$$

since all the covariances between the error term at different points in time are zero. What about the covariance between y_t and y_{t-3} ?

$$\begin{aligned} \text{Cov}(y_t, y_{t-3}) &= E[(\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2})(\epsilon_{t-3} + \theta_1 \epsilon_{t-4} + \theta_2 \epsilon_{t-5})] \\ &= 0 \end{aligned}$$

since all the error terms are now uncorrelated to each other. The same is true for all the other lags of order greater than 3. That is

$$\text{Cov}(y_t, y_{t-u}) = 0 \quad \forall |u| \geq 3$$

Hence, remembering that the ACF is a symmetric function, we can summarize it as follows

$$\text{Cov}(y_t, y_{t-u}) = \begin{cases} (1 + \theta_1^2 + \theta_2^2) \sigma^2 & u = 0 \\ \sigma^2 (\theta_1 + \theta_1 \theta_2) & u = \pm 1 \\ \theta_2 \sigma^2 & u = \pm 2 \\ 0 & |u| \geq 3 \end{cases}$$

Using the ACF we can derive the Autocorrelation Function by simply dividing the autocovariance of the process by its variance. More specifically

$$\rho(u) = \frac{\text{Cov}(y_t, y_{t-u})}{\text{Var}(y_t)} = \begin{cases} 1 & u = 0 \\ \frac{(\theta_1 + \theta_1 \theta_2)}{(1 + \theta_1^2 + \theta_2^2)} & u = \pm 1 \\ \frac{\theta_2}{(1 + \theta_1^2 + \theta_2^2)} & u = \pm 2 \\ 0 & |u| \geq 3 \end{cases}$$

Question 2. The AR(1) - Lecture Notes 6

Derive the mean, the variance and ACF of Δy_t where

$$y_t = \delta_0 + \delta_1 t + u_t$$

with

$$u_t = \alpha u_{t-1} + \epsilon_t, \quad 0 < \alpha < 1$$

Proof: Let us first start by deriving the expression for Δy_t .

$$\begin{aligned} \Delta y_t &= y_t - y_{t-1} = \delta_0 + \delta_1 t + u_t - (\delta_0 + \delta_1(t-1) + u_{t-1}) \\ &= \delta_1 + u_t - u_{t-1}. \end{aligned} \quad (219)$$

From (219), it is clear that, if we want to derive the mean and the variance and the ACF of Δy_t , we first need to derive the mean, the variance and the ACF of the u_t process.

Notice that u_t follows an AR(1) process and, by assumption, it is also stationary (since $0 < \alpha < 1$). It implies that its mean, variance and ACF will not vary over time.

For an AR(1) process defined as $u_t = \alpha_0 + \alpha_1 u_{t-1} + \epsilon_t$, where ϵ_t is a white noise process, the following properties hold:

1. **Mean of u_t :** The mean of the process is given by

$$\mu = \frac{\alpha_0}{1 - \alpha_1} = \frac{0}{1 - \alpha} = 0.$$

2. **Variance of u_t ($\gamma(0)$):** The variance of the process is

$$\gamma(0) = \frac{\sigma^2}{1 - \alpha_1^2} = \frac{\sigma^2}{1 - \alpha^2}.$$

3. **Autocovariance Function $\gamma(s)$:** The autocovariance function is given by

$$\gamma(s) = \alpha_1^s \gamma(0) = \alpha^s \gamma(0).$$

4. **Autocorrelation Function $\rho(s)$:** The autocorrelation function is

$$\rho(s) = \alpha_1^s = \alpha^s.$$

Using this result we can now derive the mean, the variance and the ACF of the Δy_t process. Let us start from by deriving the mean. Since the u_t process has zero mean, then

$$\begin{aligned} E[\Delta y_t] &= E[\delta_1 + u_t - u_{t-1}] \\ &= E[\delta_1] + E[u_t] - E[u_{t-1}] \\ &= \delta_1 \end{aligned}$$

and its variance will be

$$\begin{aligned} \text{Var}[\Delta y_t] &= \text{Var}[\delta_1 + u_t - u_{t-1}] \\ &= \text{Var}[u_t - u_{t-1}] \\ &= \text{Var}(u_t) + \text{Var}(u_{t-1}) - 2\text{Cov}(u_t, u_{t-1}) \\ &= \frac{\sigma^2}{(1 - \alpha^2)} + \frac{\sigma^2}{(1 - \alpha^2)} - 2 \frac{\alpha \sigma^2}{(1 - \alpha^2)} \\ &= \frac{2\sigma^2(1 - \alpha)}{(1 - \alpha^2)} = \frac{2\sigma^2}{(1 + \alpha)} \end{aligned}$$

More Proof on Next Page!

Question 2. The AR(1) - Lecture Notes 6

Then, for a general $k = 1, 2, 3 \dots$ we have

$$\begin{aligned}
 \text{Cov} [\Delta y_t, \Delta y_{t-k}] &= E [\Delta y_t \Delta y_{t-k}] - E [\Delta y_t] E [\Delta y_{t-k}] \\
 &= E [\Delta y_t \Delta y_{t-k}] - \delta_1^2 \\
 &= E [(\delta_1 + u_t - u_{t-1})(\delta_1 + u_{t-k} - u_{t-k-1})] - \delta_1^2 \\
 &= \delta_1^2 + E [u_t u_{t-k}] - E [u_t u_{t-k-1}] - E [u_{t-1} u_{t-k}] + E [u_{t-1} u_{t-k-1}] - \delta_1^2 \\
 &= \text{Cov} (u_t, u_{t-k}) - \text{Cov} (u_t, u_{t-k-1}) - \text{Cov} (u_{t-1}, u_{t-k}) + \text{Cov} (u_{t-1}, u_{t-k-1})^a \\
 &= 2 \text{Cov} (u_t, u_{t-k}) - \text{Cov} (u_t, u_{t-k-1}) - \text{Cov} (u_t, u_{t-k+1}) \\
 &= \sigma^2 \frac{(2\alpha^k - \alpha^{k+1} - \alpha^{k-1})}{(1 - \alpha^2)}^b \\
 &= \sigma^2 \frac{\alpha^{k-1} (2\alpha - \alpha^2 - 1)}{(1 - \alpha^2)} \\
 &= -\sigma^2 \frac{\alpha^{k-1} (\alpha - 1)^2}{(1 - \alpha^2)} \\
 &= \sigma^2 \frac{\alpha^{k-1} (\alpha - 1)^2}{(\alpha^2 - 1)} \\
 &= \sigma^2 \frac{\alpha^{k-1} (\alpha - 1)}{(\alpha + 1)}.
 \end{aligned}$$

The Autocorrelation Function is then obtained as

$$\begin{aligned}
 \rho(k) &= \frac{\text{Cov} [\Delta y_t, \Delta y_{t-k}]}{\text{Var} [\Delta y_t]} \\
 &= \frac{\sigma^2 \alpha^{k-1} (\alpha - 1) (1 + \alpha)}{(\alpha + 1) 2\sigma^2} \\
 &= \frac{\alpha^{k-1} (\alpha - 1)}{2}
 \end{aligned}$$

^aNotice that u_t follows an AR(1) process and, by assumption, it is also stationary (since $0 < \alpha < 1$). It implies that its mean, variance, and ACF will not vary over time. The covariance between two observations depends only on the time lag between them.

^bTo save your life, leaving it here is perfectly fine in the exam.

Question 3. Sample Exam Question

A macroeconomic postulates that the log of US real GNP can be represented by

$$A(L)(y_t - \delta_0 - \delta_1 t) = \epsilon_t, \quad A(L) = 1 - \alpha_1 L - \alpha_2 L^2. \quad (220)$$

An OLS fit yields

$$y_t = -0.321 + 0.003t + 1.335y_{t-1} - 0.401y_{t-2} + \hat{\epsilon}_t. \quad (221)$$

- (i) Derive the values of $\alpha_1, \alpha_2, \delta_0, \delta_1$.
- (ii) Compute the roots of the characteristic equation.
- (iii) What is the estimated value of $A(1)$?
- (iv) An alternative specification fitted to the same data yields

$$\Delta y_t = 0.003 + 0.369\Delta y_{t-1} + v_t.$$

what are the roots of this equation?

Proof:

- (i) To derive the value of $\alpha_1, \alpha_2, \delta_0, \delta_1$ we have to re-write the model in (220) in such way that we can directly compare it with the fitted model (221). More specifically, let us re-write (220) as

$$\begin{aligned} (1 - \alpha_1 L - \alpha_2 L^2)(y_t - \delta_0 - \delta_1 t) &= \epsilon_t \\ y_t - \delta_0 - \delta_1 t - \alpha_1 y_{t-1} + \alpha_1 \delta_0 + \alpha_1 \delta_1 (t-1) - \alpha_2 y_{t-2} + \alpha_2 \delta_0 + \alpha_2 \delta_1 (t-2) &= \epsilon_t \\ y_t = (\delta_0 - \alpha_1 \delta_0 + \alpha_1 \delta_1 - \alpha_2 \delta_0 + 2\alpha_2 \delta_1) + (\delta_1 - \alpha_1 \delta_1 - \alpha_2 \delta_1)t + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \epsilon_t \end{aligned}$$

Then, by mapping this result into the fitted model in (221) we have that

$$\begin{aligned} \delta_0 - \alpha_1 \delta_0 - \alpha_2 \delta_0 + \alpha_1 \delta_1 + 2\alpha_2 \delta_1 &= -0.321 \\ \delta_1 - \alpha_1 \delta_1 - \alpha_2 \delta_1 &= 0.003 \\ \alpha_1 &= 1.335 \\ \alpha_2 &= -0.401 \end{aligned}$$

Using results on α_1 and α_2 we can back out δ_1 as

$$\Rightarrow \delta_1 = \frac{0.003}{1 - \alpha_1 - \alpha_2} = \frac{0.003}{1 - 1.335 + 0.401} = 0.045$$

and then we can back out δ_0

$$\begin{aligned} \Rightarrow \delta_0 &= \frac{-0.321 - \alpha_1 \delta_1 - 2\alpha_2 \delta_1}{1 - \alpha_1 - \alpha_2} \\ &= \frac{-0.321 - 1.335 * 0.045 - 2 * (-0.401) * 0.045}{1 - 1.335 + 0.401} \\ &= -5.22 \end{aligned}$$

- (ii) Let us recall that the characteristic equation is essentially given by the lag polynomial

$$A(L) = 1 - \alpha_1 L - \alpha_2 L^2$$

More specifically, the characteristic equation is obtained by replacing the lag operator L by a variable, say z , and set the resulting polynomial equal to zero

$$1 - \alpha_1 z - \alpha_2 z^2 = 0.$$

Given $\alpha_1 = 1.335$ and $\alpha_2 = -0.401$ we have

$$1 - 1.335z + 0.401z^2 = 0$$

The general formula for finding the roots of a quadratic equation of the form $ax^2 + bx + c = 0$ is given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The roots of the equation $1 - 1.335z + 0.401z^2 = 0$ are approximately 1.138 and 2.191.

More Proof on Next Page!

Question 3. Sample Exam Question

(iii) The estimated value of $A(1)$ is obtained by starting from the general expression for $A(L)$ in (220) and then plugging in the values for L, α_1 and α_2

$$A(L) = 1 - \alpha_1 L - \alpha_2 L^2$$

$$A(1) = 1 - \alpha_1 - \alpha_2$$

$$A(1) = 1 - 1.335 + 0.401 = 0.066$$

(iv) Recall that whenever you are required to find the roots of the characteristic equation, the first step is to express your equation in terms of the lag polynomial. In this case, let us re-write

$$\Delta y_t = 0.003 + 0.369 \Delta y_{t-1} + v_t$$

as

$$y_t - y_{t-1} = 0.003 + 0.369 (y_{t-1} - y_{t-2}) + v_t$$

and then re-write this in terms of the lag operator as

$$y_t - 1.369 y_{t-1} + 0.369 y_{t-2} = 0.003 + v_t$$

$$(1 - 1.369L + 0.369L^2) y_t = 0.003 + v_t.$$

Following the same reasoning as in point(ii), the characteristic equation is:

$$1 - 1.369z + 0.369z^2 = 0$$

The roots of the equation $1 - 1.369z + 0.369z^2 = 0$ are exactly 1.000 and approximately 2.710.

TEACHING NOTES

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1 Matrix Notation in Ordinary Least Squares Regression and Its Interchangeability with Summarization Notation

In the realm of regression analysis, the Ordinary Least Squares (OLS) method is a cornerstone. This technique aims to find the line (or hyperplane) that minimizes the sum of the squared deviations (residuals) from the observed values. Traditionally, this is illustrated using summarization notation, which is more explicit and can be beneficial for a granular understanding. For instance, in a simple regression model with one independent variable, the equation is typically represented as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, where y_i denotes the dependent variable, x_i is the independent variable, β_0 and β_1 are the intercept and slope respectively, and ϵ_i represents the error term. The OLS estimators for these coefficients are derived using summation formulas which are intrinsically tied to the mean values of the variables.

However, as the complexity of models grows, especially in multiple regression scenarios, the matrix notation offers a more compact and generalized representation. In matrix terms, the regression equation can be succinctly expressed as $\mathbf{y} = \mathbf{X}\beta + \epsilon$, where $\mathbf{y}$ is a vector of the dependent variable, $\mathbf{X}$ is the matrix of independent variables (augmented with a column of ones for the intercept), and β is the vector of regression coefficients. The OLS estimator, in this case, is given by $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$.

Interestingly, there exists an underlying harmony between the matrix and summarization notations. They are essentially two sides of the same coin, each echoing the same mathematical relationships and computations. The former proves particularly handy in efficiently handling multiple predictors, sidestepping the need for extensive summation expressions. It's worth noting, however, that regardless of the chosen notation, the core principles and objectives of OLS remain consistent.

1.1 Summarization Notation for OLS

Given a dataset with n observations on a dependent variable y and a single independent variable x , the OLS regression equation is:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Where:

- y_i is the value of the dependent variable for the i -th observation.
- x_i is the value of the independent variable for the i -th observation.
- ϵ_i is the error term for the i -th observation.
- β_0 and β_1 are the intercept and slope of the regression line, respectively.

The OLS estimates for β_0 and β_1 can be derived from the following summation formulas:

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$
$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Where:

- $\bar{x}$ is the mean of x
- $\bar{y}$ is the mean of y

1.2 Matrix Notation for OLS

In matrix notation, the OLS regression equation for multiple regression (let's consider k predictors) can be written as:

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

Where:

- $\mathbf{y}$ is an $n \times 1$ vector of the dependent variable.
- $\mathbf{X}$ is an $n \times k$ matrix of the independent variables (with the first column being a column of ones for the intercept).
- β is a $k \times 1$ vector of regression coefficients.
- ϵ is an $n \times 1$ vector of error terms.

The OLS estimate for β is:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Where:

- $\mathbf{X}^T$ is the transpose of $\mathbf{X}$
- $(\mathbf{X}^T \mathbf{X})^{-1}$ is the inverse of $\mathbf{X}^T \mathbf{X}$

1.3 Matrix Representation of Summation of Deviations

The process of capturing the summation of deviations of individual data points from their mean can be elegantly expressed using matrix notation. Given a set of data points x_i , where each x_i denotes an individual observation, their deviation from the mean $\bar{x}$ is traditionally expressed as $\sum_{i=1}^n (x_i - \bar{x})$.

To represent this in matrix notation, one can consider a column vector $\mathbf{x}$ that encompasses all the x_i values:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Additionally, to aid in our computations, let's introduce a column vector $\mathbf{1}$ of length n , where each element is unity:

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Utilizing this vector, the mean $\bar{x}$ can be formulated as:

$$\bar{x} = \frac{1}{n} \mathbf{1}^T \mathbf{x}$$

The deviation of each data point x_i from the mean $\bar{x}$ can then be represented in matrix form as:

$$\mathbf{x} - \bar{x} \mathbf{1} = \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{bmatrix}$$

Summing up these deviations yields:

$$\sum_{i=1}^n (x_i - \bar{x}) = \mathbf{1}^T (\mathbf{x} - \bar{x} \mathbf{1})$$

However, it is crucial to recognize that this summation will invariably equate to zero. This is grounded in the fundamental property stating that the summation of deviations of data points from their average is always zero.

1.4 Matrix Representation of Sum of Products of Deviations

The expression denoted by $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ encapsulates the sum of the products of deviations of individual observations x_i from their mean $\bar{x}$ and the deviations of y_i values from their respective mean $\bar{y}$. To elucidate this relationship using matrix notation, we undertake the following steps:

1. First and foremost, we represent the observations x_i and y_i as distinct column vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

2. We introduce a column vector $\mathbf{1}$ of length n , populated solely with the element 1:

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

3. Using the aforementioned vector, the means $\bar{x}$ and $\bar{y}$ can be articulated as:

$$\bar{x} = \frac{1}{n} \mathbf{1}^T \mathbf{x} \quad \text{and} \quad \bar{y} = \frac{1}{n} \mathbf{1}^T \mathbf{y}$$

4. The deviations of each data point x_i and y_i from their respective means can be depicted in matrix form as:

$$\mathbf{x} - \bar{x}\mathbf{1} = \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{bmatrix} \quad \text{and} \quad \mathbf{y} - \bar{y}\mathbf{1} = \begin{bmatrix} y_1 - \bar{y} \\ y_2 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{bmatrix}$$

5. To deduce the sum of the products of these deviations, we transpose one of these deviation vectors and perform matrix multiplication with the other:

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = (\mathbf{x} - \bar{x}\mathbf{1})^T (\mathbf{y} - \bar{y}\mathbf{1})$$

The resultant scalar epitomizes the sum of the products of the deviations.

1.5 Demonstration Question: Deriving the OLS Estimator in Matrix Notation

Consider the simple linear regression model defined by:

$$Y_i = \alpha + u_i$$

Our goal is to derive the OLS estimator, $\hat{\alpha}$, utilizing matrix notation. For a general linear regression, the matrix form is given by:

$$\mathbf{Y} = \mathbf{X}\mathbf{b} + \mathbf{u}$$

In the context of our model, the dependent variable vector is $\mathbf{Y}$, while the design matrix $\mathbf{X}$ is solely a column of ones, as there isn't any predictor variable apart from the constant term. Consequently, the parameter vector $\mathbf{b}$ contains merely one element, α . Given:

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \quad \text{and} \quad \mathbf{b} = [\alpha]$$

The OLS estimator in matrix notation is articulated as:

$$\hat{\mathbf{b}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Substituting in the aforementioned matrices:

$$\hat{\alpha} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Simplifying further, we get:

$$\hat{\alpha} = \left(\sum_{i=1}^n 1 \right)^{-1} \sum_{i=1}^n Y_i$$

Given that the summation of ones is equivalent to n , it can be concisely represented as:

$$\hat{\alpha} = \frac{\sum_{i=1}^n Y_i}{n}$$

This unambiguously verifies that $\hat{\alpha} = \bar{Y}$, signifying the sample mean of the Y_i values.

As for R^2 in this model, it perpetually remains 0. This is attributed to the fact that no variance in Y is elucidated by predictor variables. The model's primary function is to estimate the mean of Y .

3.17. Prove that if a regression is fitted *without* a constant term, the residuals will not necessarily sum to zero, and R^2 , if calculated as $1 - e'e/(y'y - n\bar{Y}^2)$, may be negative.

When you fit a linear regression model with a constant term (intercept), the least squares algorithm sets the sum of the residuals to zero. This is part of the derivation of the ordinary least squares (OLS) estimators where the intercept term is specifically calculated to ensure that the average of the residuals is zero.

However, if you fit a linear regression model without a constant term, this property no longer holds. Here's why:

The sum of the residuals is given by:

$$\sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - \hat{y}_i)$$

When you have an intercept in your model, the equation of the line is:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

The least squares fitting procedure will choose $\hat{\beta}_0$ such that $\sum e_i = 0$.

However, if there's no constant term, then the equation is simply:

$$\hat{y}_i = \hat{\beta}_1 x_i$$

And there's no guarantee that $\sum e_i = 0$, because there's no intercept term to adjust the height of the line. The line will simply be forced through the origin (0,0), and the sum of the residuals can be any number.

As for the R^2 statistic, when you include a constant term, R^2 is guaranteed to be between 0 and 1. This is because R^2 is defined as the proportion of the variance in the dependent variable that is predictable from the independent variable(s), which is computed as:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

where SS_{res} is the sum of squares of residuals and SS_{tot} is the total sum of squares (a measure of the total variance in the dependent variable).

The formula you provided for R^2 , $1 - e'e/(y'y - n\bar{Y}^2)$, assumes that there is an intercept term since $n\bar{Y}^2$ is part of the total sum of squares under the assumption that the mean of the residuals is zero. Here's why:

The R^2 formula provided:

$$R^2 = 1 - \frac{e'e}{y'y - n\bar{Y}^2}$$

is typically derived under the assumption that there is an intercept term in the regression model. Here's why $n\bar{Y}^2$ is subtracted in the total sum of squares when an intercept is included:

Total Sum of Squares (SS_{tot}): The total sum of squares measures the total variance in the dependent variable, Y , and is defined as the sum of the squared differences between each observed value y_i and the mean of Y , $\bar{Y}$:

$$SS_{tot} = \sum (y_i - \bar{Y})^2$$

This can also be written in matrix notation as:

$$SS_{tot} = y'y - n\bar{Y}^2$$

where $y'y$ is the sum of squares of the observed values, and $n\bar{Y}^2$ is the squared mean multiplied by the number of observations, which represents the part of the variation that is explained simply by the mean of the data (i.e., the variance if we had a model with only an intercept).

- 1 Mean Centering: When an intercept is included, the regression line is allowed to shift up or down to best fit the data, which includes capturing the mean of the dependent variable, Y . The mean of Y is part of the total variation we want to explain with our model.
- 2 Interpretation: With an intercept, the total sum of squares is decomposed into the regression sum of squares (SS_{reg}) plus the sum of squares of residuals (SS_{res}), which together equal SS_{tot} . Subtracting $n\bar{Y}^2$ from $y'y$ gives us the part of the total variation that is "leftover" to be explained by the model after accounting for the mean.
However, without a constant term, the model does not include this mean adjustment. The line is forced to go through the origin (0,0), and the only variation it can explain is the variation around the origin, not around the mean. Therefore, the total sum of squares without a constant term is simply $y'y$, without subtracting $n\bar{Y}^2$, because the mean is not being considered or captured by the model.

Consequently, if you were to use the formula with $n\bar{Y}^2$ included when there's no intercept, you would be incorrectly accounting for the mean twice: once implicitly by the model not including an intercept, and once explicitly by the subtraction in the SS_{tot} formula. This could lead to incorrect calculations for R^2 , such as potentially negative values, which do not make sense within the standard interpretation of R^2 as the proportion of the variance explained by the model.

Without the constant term, SS_{tot} should not subtract $n\bar{Y}^2$, and as such, the formula for R^2 should be:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

where now $SS_{tot} = y'y$ without any adjustment for the mean of y , because the model does not ensure that the mean of the predicted values matches the mean of the actual values.

If you use the incorrect formula for R^2 when there's no constant term, you may end up with a negative value for R^2 , which doesn't make sense within the usual interpretation of R^2 as the proportion of variance explained. This happens because the sum of squares due to the regression may be greater than the total sum of squares if the line does not fitted well with the data.

Therefore, without a constant term, it's not appropriate to use the formula $1 - e'e/(y'y - n\bar{y}^2)$ for R^2 , and it's not correct to interpret R^2 as a proportion of variance explained in the usual way.

TEACHING NOTES

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DEPARTMENT OF FINANCE

Risk Management & Financial Engineering 2023

Financial Statistics

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1 Exploring the Delta Method - Real-Life Example: Return on Investment in Stock Market

The Delta Method is a powerful tool in statistics, especially when we want to determine the asymptotic distribution of a function of an estimator. Here's an application of the Delta Method derived from **Week 3 question 2** in a real-life scenario.

1.1 Background

Imagine you're an economist working for a financial firm. You're interested in understanding how the return on investment (ROI) of a stock depends on the overall market performance and other macroeconomic indicators.

Suppose, after much research, you arrive at the following linear regression model:

$$ROI_t = \alpha + \beta \times \text{MarketIndex } x_t + u_t$$

Here:

1. ROI_t is the return on investment of the stock at time t .
2. $\text{MarketIndex } x_t$ is the value of a significant market index (like the S&P 500) at time t .

From this model, you're especially interested in the "bang for your buck" measure, which is defined as the ratio of the intercept to the slope:

$$\text{Bang For Buck}_t = \frac{\alpha}{\beta}.$$

In essence, this measure indicates how much ROI (in percentage terms) a stockholder can expect for each unit increase in the market index.

1.2 Application of Delta Method

Step 1 Estimate the Model: Using historical stock prices and market index values, you estimate the model using OLS and get the estimates $\hat{\alpha}$ and $\hat{\beta}$.

Step 2 Compute the Nonlinear Transformation: Compute the estimated 'bang for your buck' as:

$$\text{Bang For Buck} = \frac{\hat{\alpha}}{\hat{\beta}}$$

Step 3 Apply the Delta Method: Using the principles outlined in the notes, you determine the asymptotic distribution of Bang For Buck . With this, you can compute confidence intervals and test hypotheses about the 'bang for your buck' measure.

Let's get into the specifics of applying the Delta Method to the "bang for your buck" measure, using the principles outlined:

1. **Compute the Gradient:** The gradient or the Jacobian matrix, F , is the first derivative of the nonlinear function with respect to the parameters. For the function $f(\beta) = \frac{\alpha}{\beta}$:

$$F(\beta) = \frac{\partial f(\beta)}{\partial \beta'} = \begin{bmatrix} \frac{1}{\beta} & -\frac{\alpha}{\beta^2} \end{bmatrix}$$

Plug in the OLS estimates $\hat{\alpha}$ and $\hat{\beta}$ into the above to get:

$$F(\hat{\beta}) = \begin{bmatrix} \frac{1}{\hat{\beta}} & -\frac{\hat{\alpha}}{\hat{\beta}^2} \end{bmatrix}$$

2. **Asymptotic Variance:** From the notes, the asymptotic variance of $f(\hat{\beta})$ is:

$$\text{Var}(f(\hat{\beta})) = \sigma^2 F Q^{-1} F'$$

Where:

- σ^2 is the consistent estimator of the error variance. In OLS, it can be estimated as $s^2 = \frac{1}{T-k} e'e$.
- Q is a limit of the matrix $\frac{X'X}{T}$ as $T \rightarrow \infty$.

In practice, Q is estimated by $\left(\frac{X'X}{T}\right)$, the sample analog. Thus, Q^{-1} is replaced by $\left(\frac{X'X}{T}\right)^{-1}$ in applications. So, plugging in our estimates, the asymptotic variance of Bang Fôr Buck becomes:

$$\text{Var}(\text{Bang Fôr Buck}) = s^2 F(\hat{\beta}) \left(\frac{X'X}{T}\right)^{-1} F(\hat{\beta})'$$

3. Confidence Interval: With the asymptotic variance computed, a 95% confidence interval for Bang Fôr Buck can be constructed as:

$$\text{Bang Fôr Buck} \pm 1.96 \times \sqrt{\text{Var}(\text{Bang Fôr Buck})}$$

4. Hypothesis Testing: If you want to test a hypothesis, e.g., $H_0 : \text{Bang Fôr Buck} = 2$, then the test statistic is:

$$t = \frac{\text{Bang Fôr Buck} - 2}{\sqrt{\text{Var}(\text{Bang Fôr Buck})}}$$

This test statistic will be approximately normally distributed under large samples due to the Central Limit Theorem.

Step 4 Interpret the Results: Suppose you determine that, on average, for each point increase in the market index, an investor can expect a 2% return on this particular stock, with a certain level of variability around this estimate. This information can be critical for investment decisions, portfolio management, and risk assessment.

1.3 Conclusion

This real-life example illustrates how the Delta Method can be applied to infer the statistical properties of nonlinear transformations of parameters estimated in a regression model. The "bang for your buck" measure is one of many possible applications in finance where such nonlinear transformations might be of interest.